

30 Years of
Oposa v. Factoran
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Mangrove Forests:
Suspend the Issuance of
Tenurial Instruments
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Lumad-Laban: An Exposition
Against Corporations on
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"Kontrabida Ang Bida":
An Authority Discrepant with
the Local Government Code of 1991
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Journal for Law Advocacy

**Enhancing Environmental Laws in the Philippines:
Strengths, Weaknesses, and Reforms
Toward a Balanced and Healthful Ecology**

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“ The right to a balanced and
healthful ecology carries with it
the correlative duty to refrain
from **impairing the**
environment. ”

*The Supreme Court through Justice Davide - Oposa v.
Factoran (G.R. No. 101083, July 30, 1993)*

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The preservation of our natural resources is a constitutional duty. The exploitation of these resources **must** always be done with a deep sense of responsibility for their conservation, ensuring that future generations may still enjoy the country’s **bountiful resources** and a **balanced environment**.

La Bugal-B'laan Tribal Association, Inc. v. Ramos (G.R. No. 127882, December 1, 2004)

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Luigi Bien. B. Villanueva
Executive Board of Editors



"The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature."¹

The enshrinement of environmental protection in the 1987 Constitution is evidence of our mutual recognition of the role of the environment in our national livelihood, development, and welfare. Section 16 of Article 2,² serves as an important foundation of morality and law, reaffirming the state's commitment to protecting ecological integrity. However, translating this into effective policy and practice in light of current ecological crises remain an ongoing challenge.

As the world faces a triple planetary crisis, the Philippines remains at a crossroad. In a country rich in natural resources but full of scarcity, the citizens have a special responsibility to strike a balance between development and environmental protection. Even with progressive legislation such as the Philippine Environmental Code (P.D. 1152), the Clean Air Act (R.A. No. 8749), and the Environmental Solid Waste Management Act (RA 9003), there are regulatory differences with respect to the gap between intent and implementation.

The triple planetary crisis³ pertains to the three main interlinked issues that humanity currently faces: climate change, pollution and biodiversity loss. Each of these issues having its own causes-and-effect as well as the need to be resolved in hopes of a viable future on this planet.⁴

This year's theme, *"Enhancing Environmental Laws in the Philippines: Strengths, Weaknesses, and Reforms towards a Balanced and Healthful Ecology,"* calls for critical evaluation on our legal framework and its capacity in addressing contemporary environmental problems. It also encourages forward-looking discussions on the policy and sustainability implications of a more flexible ecological paradigm.

¹ CONST. (1987), art. II, §16 (Phil.).

² *Id.*

³ What is the triple planetary crisis? (April 13, 2024), <https://unfccc.int/news/what-is-the-triple-planetary-crisis>

⁴ *Id.*

Going Back and Coming Through

The environmental legal framework of the Philippines circulates with the Philippine Environmental Code in its core. The system supporting such framework is being lifted by supporting statutes as well as landmark cases. In the first Article, **30 Years of Oposa v. Factoran: A Lasting Green Legacy for The Children of Tomorrow and Beyond**,⁵ looks back on the impact of intergenerational responsibility towards environmental protection imposed upon the present inhabitants in hopes of preservation for the future generation.

Two Trees for None

Trees are the lifeblood of the environment. It is evidenced by movements as well as legal placements ensuring that trees continue to provide as humans continue to exploit. In the second, third, and fourth articles, **Speak for the Trees: The Law as Voice for Defending Mangrove Forest Depletion, Tree-Transplanting: An Inter-Generational Operation for a Balanced and Healthful Ecology Using the Advancement of Technology**, and **Mangrove Forest: Suspend the Issuance of Tenurial Instruments** exhibits the relationship built surrounding trees and the environment as a whole as well as the legal framework for preservation and the consequences in scenarios of waste and destruction.

Environmental States and Stateless Environments

*“Humanity is waging war on nature.
This is suicidal.”*⁶

— UN Secretary-General António Guterres

An environmental state is one possessing a significant set of institutions and practices dedicated to environmental and societal-environmental interactions.⁷ It is one which the Philippines should aspire to be as a country blessed with rich and diverse resources.

⁵ Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.).

⁶ A Human Destroying or a Human Creating – One Climate Story (November 5, 2021), <https://www.undp.org/kyrgyzstan/blog/human-destroying-or-human-creating-one-climate-story>

⁷ Hausknot, D., & Hammond, M. (2019). Beyond the environmental state? The political prospects of a sustainability transformation. *Environmental Politics*, 29(1), 1–16, <https://doi.org/10.1080/09644016.2020.1686204>

In contrast, stateless environments can be denoted as those where environmental governance is weak or non-existent, resulting in degradation and often marked by unregulated industrial activity and illegal resource exploitation.

In the path towards environmental statehood, the Philippines is known to enact progressive and inclusive environmental legislations such as the Wildlife Resources Conservation and Protection Act (R.A. No. 9174).⁸ In the sixth article, **Mangayon Shot Dead** discusses the protection of species and their habitats in the fight against climate change, a fight where both wildlife and humans are comrades.

Other movements are also geared towards the support of indigenous population for economic development. The fifth article, **Lumad-Laban: An Exposition Against Corporations on Environmental Destruction** explains how Lumads are uprooted from their native lands as a result of rapid modernization and industrialization which makes them vulnerable to exploitation by strong interests.⁹

The Clean Air Act (R.A. No. 8749),¹⁰ as discussed in the tenth article, **25 Years Later: Is the Clean Air Act Of 1999 Still Relevant?** still proves to be ever needed now compared to the time when it was enacted. Climate change as well as global pollution fails to manifest signs of slowing down or stoppage anytime sooner.

The concept of environmental states must evolve to encompass not only national governance but also international cooperation. The Paris Agreement¹¹ represents such effort, which aims to unite nations in limiting global temperature increases. In the twelfth article, **Paris Agreement on Climate Change: Rising Sea Levels Threatening Philippine Cities** extensively widens the scope of legally binding obligation of the Philippines, as signatory, towards mitigation and adapting towards a series of scenarios involving climate change as a catalyst for global downfall.

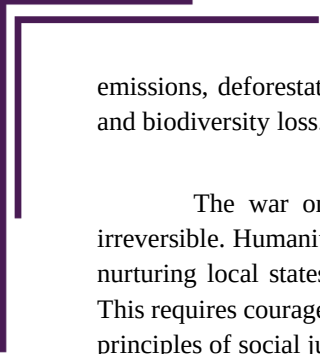
Antonio Guterres' dire warning underscores the urgent need to address these differences. Humanity's "war on nature" is reflected in increased greenhouse gas

⁸ Wildlife Resources Conservation and Protection Act, Rep. Act No. 9147 (July 30, 2001) (Phil.).

⁹ Espina-Varona, Inday, Lumad Children : War Targets in Ph's Last Frontiers (October 24, 2015), <https://indayvarona.wordpress.com/2015/10/24/lumad-children-war-targets-in-phs-last-frontiers/>

¹⁰ Philippine Clean Air Act, Rep. Act No. 8749, (June 23, 1999) (Phil.).

¹¹ United Nations Framework Convention on Climate Change (UNFCCC), *The Paris Agreement*, (n.d.), <https://unfccc.int/process-and-meetings/the-paris-agreement>



emissions, deforestation and pollution, all of which are worsening global warming and biodiversity loss.

The war on nature is definitely suicidal but nevertheless, such is not irreversible. Humanity can pave the way for a harmonious and sustainable future by nurturing local states and tackling the root causes of environmental statelessness. This requires courageous action, cooperation, and an unwavering commitment to the principles of social justice.¹²

A Call to Action

As legal scholars, practitioners and advocates, we are on a mission to ensure that environmental law evolves to meet the challenges of our time. Volume 4 of the *Journal for Law Advocacy* seeks to support this effort by offering interdisciplinary perspectives and practical notes for upholding environmental laws in the Philippines.

The articles presented here explore the interplay between legal theory and practice. It examines how our legal framework can serve as both a shield against environmental degradation and a sword for demanding accountability. This volume aims to inspire meaningful narrative and action and to achieve a truly balanced and healthful ecology.

This collection of scholarly works is not only informative but also persuasive in defending the cause of environmental justice because protecting our environment is tantamount to honoring our intergenerational responsibility to protect the future of the generations to come.

Yours in advocacy,

Luigi Bien B. Villanueva
Executive Board of Editors
Journal for Law Advocacy

¹² Calalang v. Williams, G.R. No. 47800, (December 2, 1940) (Phil.).

Bibliography

Constitution

1987 Constitution of the Philippines art. II, §16 (Phil.).

Statutes

Wildlife Resources Conservation and Protection Act, Rep. Act No. 9147, (July 30, 2001) (Phil.).

Philippine Clean Air Act, Rep. Act No. 8749, (June 23, 1999) (Phil.).

Jurisprudence

Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.).

Calalang v. Williams, G.R. No. 47800, (December 2, 1940) (Phil.).

Online Sources

A Human Destroying or a Human Creating – One Climate Story (November 5, 2021),
<https://www.undp.org/kyrgyzstan/blog/human-destroying-or-human-creating-one-climate-story>

Espina-Varona, Inday, Lumad Children: War Targets in Ph's Last Frontiers (October 24, 2015),
<https://indayvarona.wordpress.com/2015/10/24/lumad-children-war-targets-in-phs-last-frontiers/>

Hausknost, D., & Hammond, M. (2019). Beyond the environmental state? The political prospects of a sustainability transformation. *Environmental Politics*, 29(1), 1–16,
<https://doi.org/10.1080/09644016.2020.1686204>

What is the triple planetary crisis? (April 13, 2024),
<https://unfccc.int/news/what-is-the-triple-planetary-crisis>

United Nations Framework Convention on Climate Change (UNFCCC), *The Paris Agreement*, (n.d.),
<https://unfccc.int/process-and-meetings/the-paris-agreement>



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30 YEARS OF OPOSA V. FACTORAN: A LASTING GREEN LEGACY FOR THE CHILDREN OF TOMORROW AND BEYOND

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“...unless the rights to a balanced and healthful ecology and to health are mandated as state policies by the Constitution itself, thereby highlighting their continuing importance and imposing upon the state a solemn obligation to preserve the first and protect and advance the second, the day would not be too far when all else would be lost not only for the present generation, but also for those to come — generations which stand to inherit nothing but parched earth incapable of sustaining life.”

— Chief Justice Hilario Davide, Jr.

ABSTRACT

In 1993, the Supreme Court of the Philippines, through the landmark case of *Oposa v. Factoran*,¹ pioneered the inclusion of intergenerational responsibility as a basis for acquiring legal standing in environmental litigation revolutionizing the environmental advocacy and movement towards attaining climate justice across corners of the globe. This article is a legal analysis that aims to explore and revisit the case of the Minors Oposa, and trace back the laws and jurisprudence both in the Philippines and in other countries it has influenced over the decades since its promulgation. An in-depth and comprehensive case analysis and research has been conducted to selected local and international law and jurisprudence in relation to the case of Minors Oposa and its foundational and revolutionary concept of intergenerational responsibility. It was revealed that the case of *Oposa v. Factoran* has made a direct and indirect impact and influence to the favorable decision of cases concerning environmental protection. Further, its influence in the formation of the 2010 Rules of Procedure for Environmental Cases in the Philippines was highlighted, particularly the inclusion of the provisions on citizen suit that underscored the rights of the people and generations yet unborn to a balanced and healthful ecology. 30 years after its historic promulgation, *Oposa v. Factoran* still remains to be a trailblazing environmental jurisprudence, paving the way and leaving a lasting legacy for the continued advancement of rights to a balanced and healthful ecology of the people today and those of the children of tomorrow.

¹ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId.=27868&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

Introduction

The legal system in the Philippines holds a cradle of historic and revolutionary environmental jurisprudence, producing ground-breaking landmark court decisions that are also being celebrated across international legal landscapes since their promulgation up until today. Among these jurisprudential cases include Metro Manila Development Authority v. Concerned Residents of Manila Bay,² Resident Marine Mammals of the Tañon Strait Protected Seascape v. Reyes,³ Mosqueda v. Pilipino Banana Growers and Exporters Association, Inc.⁴ to name a few. But one environmental case that made a prominent mark in the history of the environmental movement and legal system not just of the Philippines, but also those of other countries across the globe, is the infamous Oposa v. Factoran.⁵ Regarded as one of the most important cases ever decided by the Supreme Court of the Philippines, Oposa v. Factoran, paved the way as the forefront for the success of other environmental cases, with the twin concept of “intergenerational justice and responsibility”⁶ as its major key points. Three decades after its promulgation in 1993, Oposa v. Factoran, as a landmark case have come so far already in terms of legal influence, and up until the present contemporary times, it still remains to hold a major significance and lasting impact as a subject and center of environmental discussions, basis of court decisions, policy reformations, and international legal foundation in the aspect of environmental law and climate justice.

Oposa v. Factoran in Summary

The case of *Oposa v. Factoran* stemmed from the initiated civil case by minors Juan Antonio, Anna Rosario, and Jose Alfonso Oposa, along with other minors who were joined and represented by their respective parents, against the Secretary of the Department of Environment and Natural Resources (DENR), Fulgencio S. Factoran, Jr., and the Honorable Eriberto U. Rosario, Presiding Judge of the RTC, Makati, Branch 66. The minors further assert that they do not just simply

² Metro Manila Development Authority v. Concerned Residents of Manila Bay, G.R. Nos. 171947-48, (December 18, 2008) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/48335>

³ Resident Marine Mammals of the Tañon Strait Protected Seascape v. Reyes, G.R. No. 180771, (April 21, 2015) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/60478>

⁴ Mosqueda v. Pilipino Banana Growers and Exporters Association, Inc., G.R. No. 189185, (August 16, 2016) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/62290>

⁵ Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId=27868&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

⁶ *Id.*

represent themselves and their interests; because more than that, they "represent their generation as well as generations yet unborn."

The petition seeks to cancel all existing Timber License Agreements (TLAs) and cease and desist from receiving, accepting, processing, and renewing or approving new TLAs in the country in the ambitious hopes of preventing further environmental diminution, which have been the consequence of massive deforestation over the years subsequently leading to the disturbance of the biodiversity and the delicate ecological balance – a balance that have been flourishing and withstanding the changing phases of the planet since time immemorial. The adverse consequences of continued and massive deforestation are so detrimental to the point that it becomes capable of unquestionable demonstration and that the same may be submitted as a matter of judicial notice.

They argued that such actions contravene the Filipinos' constitutional right to a balanced and healthful ecology and the state's obligation to protect such rights for present and future generations.

The Regional Trial Court (RTC) of Makati dismissed the case on the grounds that the complaints state no cause of action against the defendant and raises a political question incapable of judicial review. The respondent judge further reasoned that the grant of the reliefs sought would violate the established constitutional principle of non-impairment of contracts. Hence, the special civil action for certiorari before the Supreme Court praying for the rescission and to set aside the dismissal order.⁷

Historic Ruling of the Supreme Court

Although it was not raised as an issue, the Supreme Court classified the present case as a class suit, mainly because of the fact that the subject matter being tackled is of common and general interest to all citizens of the Philippines. In its historic ruling in 1993, Justice Davide, as the ponente, underscored and upheld the revolutionary concepts of intergenerational responsibility and justice as its primary basis for its favorable decision not just to the plaintiffs, but to the generations to come. It is worth giving emphasis in the ruling that the personality to sue in behalf of the succeeding generations can only be based on the concept of intergenerational responsibility insofar as the right to a balanced and healthful ecology is concerned. Further, the court ruled that the non-impairment clause cannot as yet be invoked, for a timber license is not a contract within the purview of the due process clause; it is only a privilege, which can be validly withdrawn by the

⁷ Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId.=27868&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

executive whenever dictated by public interest or welfare. Otherwise stated, the freedom of contract is not meant to be absolute; hence, the constitutional guarantee of non-impairment of obligations of contract is limited by the exercise of police power of the State, in the interest of the public health, safety, moral and general welfare.⁸

Intergenerational Responsibility clothed with Legal Standing

It has been established that the petitioners are not just representing themselves and their interest, but also encompasses the interest of those that are yet to be born in the future through the concept of intergenerational responsibility. It was probably quite a risk for the petitioners and their counsel to brave the unknown outcome of this novel case given the fact that intergenerational responsibility has rarely or even never been traced as a legal basis in the history of the legal system of any country in any part of the world as a valid reference for anyone to acquire a legal standing upon seeking a legal relief. Well settled is the fundamental principle that for someone to be considered to have a legal standing in a case, one must sustain or is in immediate danger of sustaining an injury as a result of the act complained of.⁹ Such actual or potential injury must first be established, otherwise a complainant cannot have the legal personality to sue nor raise a constitutional question before the courts of law. In simple terms, one cannot seek relief for an injury he/she never sustained in the first place. At first glance, one who has basic knowledge on the requisites of a judicial review might think that there is no way for the petitioners to possess a legal standing given that they did not sustain a direct, potential, or immediate injury upon the granting of Timber License Agreements (TLAs) of the DENR on anyone who they deem to be worthy and compliant of the requirements of such grant. However, in a careful and deep circumspection, there are no other proper entities that could better understand, empathize, and represent the future generation than the human kind of the present themselves. Logically speaking, the planet Earth, although holds life, continually being abused and therefore sustains an actual injury which is already tantamount to a cause of action, cannot represent itself in front of the jury, nor the future generations could represent themselves at the present time.

This right was already inherent under the 1987 Constitution, but it took so much keen attention for it to sprout, be interpreted and recognized as it is now. People were too busy making fortunes by exploiting the natural resources that they overlooked the fact that they are already committing a mortal sin to the planet, to

⁸ Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId.=27868&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

⁹ Ex Parte Levitt, 303 U.S. 633 (1938), in Philippine Political Law 506, (Cruz & Cruz, 2014).

the people of tomorrow, and even to themselves. The right exists, but it seemed dead. It is there, yet it does not seem like it. It is there, yet no one calls out... until Atty. Tony Oposa.

The cost of not noticing was expensive. If only it was discovered sooner, the long litigation process could have been prevented, and thousands of trees could have been saved from exploitation. But the real bottom line of all this is that: if only humans were kind and mindful enough from the very beginning, thousands of trees, if not millions, could have been saved and thus have continuously thrived and flourished.

Thus, intergenerational responsibility must have been rather a new and extraordinary specie of legal standing as a concept in the eyes of the courts of justice in the Philippines back in 1993. Despite its novel character, it is worth noting that no risk is too high for a conviction that is so strong – a conviction that is anchored in the strong pursuit of attaining climate justice due to the “continued rape of mother Earth”¹⁰ and noble intention of preserving natural resources for the future generations to witness and make use of. It is indeed true that “justice must be realized along a continuum of time”¹¹ hence it must not be confined neither in the past, present, nor future; to attain its holistic form, it must also take into account the phases in between.

It must have been a leap of faith on the part of the ponente to reconsider the petitioners’ contention, upon the liberal interpretation as to the aspect of peculiar form of legal standing and the exception from the constitutional provision of non-impairment of contracts when public interest so demands. Besides, the court is vested with power to apply a liberal interpretation of laws for the genuine purpose of meeting the interest of justice and in the pursuit of public welfare.

The Oposa Effect

Oposa v. Factoran, since its promulgation, set the stage for the developments of environmental laws, procedures, legal instruments, or other strategic litigation cases on the basis of intergenerational equity and justice, and has

¹⁰ Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId.=27868&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

¹¹ Singh Margaretha; et. al (August 3, 2023) <https://academic.oup.com/ejil/article/34/3/651/7236581>

since opened up opportunities for future child rights litigation particularly in the area of climate litigation.¹²

1. Oposa's Footprints on the Rules of Procedure for Environmental Cases

The fundamental doctrines and principles laid down under the *Oposa case* left a remarkable footprint, setting a guiding path towards the formulation and development of laws, rules, and procedures that lobby the right and provide relief for the people to realize their constitutionally vested right to a balanced and healthful ecology. The jurisprudential triumph of the *Oposa case* has become a major consideration, influence, and forms part as a rationale of the Supreme Court of the Philippines in the framework and promulgation of the 2010 Rules and Procedure for Environmental Cases.¹³ Under this rule, the concept of intergenerational responsibility as established in the *Oposa case* was explicitly manifested in the Rule 2, Section 5 on Citizen Suit, and Rule 5, Section 1 on the Reliefs in a Citizen Suit:

Rule 2, Sec. 5. *Citizen Suit*.— Any Filipino citizen in representation of others, including minors or generations yet unborn, may file an action to enforce rights or obligations under environmental laws...

Rule 5, Sec. 1. *Reliefs in a Citizen Suit*.— If warranted, the court may grant to the plaintiff proper reliefs which shall include the protection, preservation or rehabilitation of the environment and the payment of attorney's fees, costs of suit and other litigation expenses. It may also require the violator to submit a program of rehabilitation or restoration of the environment, the costs of which shall be borne by the violator, or to contribute to a special trust fund for that purpose subject to the control of the court.

Citizen suit seeks to liberalize the traditional rule on standing by relaxing the requirement of personal and direct interest for all cases filed to enforce environmental laws. This is intended to further encourage the protection of the environment as stewards of nature.¹⁴

¹² *Minors Oposa v. Secretary of the Department of Environment and Natural Resources* 33 I.L.M. 173 (1994). — Philippines, <https://www.acrisl.org/casenotes/m2l18m8skjpglk8-83mk2>

¹³ 2010 Rules and Procedure for Environmental Cases, <https://sc.judiciary.gov.ph/rules-of-procedure-for-environmental-cases/>

¹⁴ *Spouses Maliones et al. v. Mario Timario Jr. et al.* G.R. No. 252834, (February 6, 2023), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/69143>

These two pertinent provisions of the 2010 Rules and Procedure for Environmental Case underscores the value of intergenerational responsibility, emphasizing that citizens as legal representative can sue in behalf of the succeeding generations. Not only under this rule citizen suit is present, because it can also be traced even under some provisions of some revolutionary Philippine Environmental laws, such as the Clean Air Act¹⁵ and Ecological Solid Waste Management Act.¹⁶ Again, this goes to show the *Oposa case* has truly made a radical influence in the shaping and progress of the country's present and existing Environmental laws and procedures.

2. From one landmark case to another, Oposa Case Delivered

One notable landmark case that affirmed and drew inspiration from the *Oposa v. Factoran* case is the Metropolitan Manila Development Authority v. Concerned Residents of Manila Bay.¹⁷ The case revolved around the complaint of concerned residents of Manila Bay demanding the petitioners, MMDA and other government agencies for the clean-up, rehabilitation, and protection of the Manila Bay due to the drastic decline of water quality of the said body of water, which apparently endangers marine life and disturbs public health. Petitioners argue that the pertinent provisions of the Environment Code (P.D. 1152) relate only to the cleaning of specific pollution incidents and do not cover cleaning in general, and that the cleaning of the Manila Bay is not a ministerial act which can be compelled by mandamus.

The Court in its decision emphasized the extreme necessity for all concerned executive departments and agencies to immediately act and discharge their respective official duties and obligations. But the tasks ahead could only be accomplished if those mandated, with the cooperation of all civic-minded individuals, would put their minds to these tasks and take responsibility. Hence, the State, through petitioners, has to take the lead in the preservation and protection of the Manila Bay. The Court further reiterated that different government agencies and instrumentalities cannot shirk from their mandates; they must perform their basic functions in cleaning up and rehabilitating the Manila Bay.

So, it was that in *Oposa v. Factoran* the Court stated that the right to a balanced and healthful ecology need not even be written in the Constitution for it is assumed, like other civil and political rights guaranteed in the Bill of Rights, to exist from the inception of mankind” and it is an issue of transcendental importance with intergenerational implications. Even assuming the absence of a categorical legal

¹⁵ Philippine Clean Air Act of 1999, Rep. Act. 8749, (June 23, 1999), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/6028>

¹⁶ Ecological Solid Waste Management Act of 2000, REP. ACT 9003 (January 26, 2001), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/4800>

¹⁷ Metro Manila Development Authority v. Concerned Residents of Manila Bay, G.R. Nos. 171947-48, (December 18, 2008), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/48335>

provision specifically prodding petitioners to clean up the bay, they and the men and women representing them cannot escape their obligation to future generations of Filipinos to keep the waters of the Manila Bay clean and clear as humanly as possible.¹⁸

Similarly, in a recently decided case by the Supreme Court of the Philippines on February 6, 2023 *Spouses Maliones v. Timario, Jr.*,¹⁹ it was made evident how valuable the principle of intergenerational equity in *Oposa v. Factoran* is, in assailing people's right to a balanced and healthful ecology and demand for the protection and preservation of the environment, which is in accordance with a particular provision on citizen suit under the 2010 Rules of Procedure for Environmental Cases.

The case commenced through a citizen suit instituted by Timario et al.,²⁰ through a petition for enforcement of rights and obligations under environmental laws, cancellation of tax declarations, issuance of temporary environmental protection order and permanent environmental protection order, *mandamus*, cost of suit, and attorney's fees, against the Spouses Maliones, alleging that the subject land where the Datanians, the actual residents and individuals who trace their roots in Brgy. Data, freely pastured their farm animals and hold barangay picnics and scout jamborees of students in Mountain Province because tax declarations were issued in the names of the petitioners, *Spouses Maliones et al.*; they introduced improvements to the area that led to the exclusion of the public from its use and enjoyment equating to alleged violations of environmental laws that would lead to unimaginable environmental damage not only to the prejudice of their own constitutional right to a balanced and healthful ecology, but also to that of their children, the present generation, and the generations yet unborn.

Specifically, *Timario, et al.* alleged that Spouses Maliones were doing earth-moving and bulldozing activities that have destroyed the natural view and beauty of nature as the subject land had been turned into a wide vegetable plantation. They also claimed and alleged that Spouses Maliones are engaging in kaingin activities, and that alnus trees were cut, and pine trees were girdled and threatened to be cut which all apparently contributed to the destruction of the environment.²¹

They sought to enjoin activities which threatened to irreversibly damage and destroy a part of the public forest to which every Filipino citizen and even more

¹⁸ Metro Manila Development Authority v. Concerned Residents of Manila Bay, G.R. Nos. 171947-48, (December 18, 2008) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/48335>

¹⁹ *Spouses Maliones et al. v. Mario Timario Jr. et al.*, G.R. No. 252834, (February 6, 2023) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/6914>

²⁰ *Id.*

²¹ *Id.*

so the residents of the Municipality of Sabangan, Mountain Province, have a right to enjoy as part of their enforceable environmental rights. While they did not claim ownership over the subject land, they prayed for the cancellation of the tax declarations issued to their respective holders.²²

Timario, et al. prayed that *Spouses Maliones et al.* be ordered to cease from introducing improvements immediately and pleaded that the Department of Environment and Natural Resources - Community Environment and Natural Resources Office (DENR-CENRO) be compelled to zealously enforce and implement the environmental laws. They also prayed that the Provincial and Municipal Assessors of Mountain Province be compelled to release certified true copies of the tax declarations in the name of *Spouses Maliones et al.*; and mandated the Punong Barangay of Data to actively participate in the environmental management and protection programs of the government – all to which the Regional Trial Court favorably so ordered. Hence the appeal of *Spouses Maliones et al.* to the Court of Appeals. The appellate court denied the appeal, hence the elevation of the case to the Supreme Court.²³

The Supreme Court in its decision, reiterated that: while it has to refrain from resolving the underlying issues on the ownership of the subject land and the recognition of the parties as indigenous cultural communities/indigenous peoples, since the same must be addressed in the proper case and in the correct forum; it is nonetheless not precluded from granting reliefs available to *Timario, et al.*, in accordance with Section 1, Rule 5 of the Rules of Procedure for Environmental Cases. It has declared as an affirmation to RTC and CA's finding that there exist an actual or imminent threat that can be attributed to *Spouses Maliones et al.* and the activities they are conducting on the subject area that violate environmental statutes and cause prejudice the life, health, or property of residents of Barangay Data, Sabangan, Mountain Province. Hence, The DENR-CAR and CENRO in Sabangan, Mountain Province are ordered to prevent *Spouses Maliones et al.* from converting forest zones into vegetable farms; engaging in illegal activities; cause the rehabilitation of the area affected by the bulldozing activities; and ensuring environmental protection. The Punong Barangay of Data, Sabangan is also required to actively participate in government environmental management programs and assist in enforcing environmental laws.²⁴ This decision was primarily anchored from the pertinent provision of the 2010 Rules on Environmental Cases on Citizen Suit which was originally grounded from the concept of intergenerational responsibility as laid down in *Oposa v. Factoran*.

These cases apparently have more than a decade gap from the time of their respective promulgation, which might indicate that environmental cases in the Philippines are not after all as progressive as it seems based only from the accounts

²² *Spouses Maliones et al. v. Mario Timario Jr. et al.*, G.R. No. 252834, (February 6, 2023) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/6914>

²³ *Id.*

²⁴ *Id.*

of the cases of Oposa to Timario. Oposa did not even have a direct implication as to the result of the case of *Spouses Maliones v. Timario, Jr.*; in fact, there was no single word nor citation mentioning the name of Oposa indicated in the decision. Nonetheless, if we were to read between the lines, one would realize that the spirit of the Oposa case lingered around and served as a driving force in advancing the power and significance of the concept of intergenerational responsibility as adapted in the 2010 Rules of Procedure on Environmental Cases towards the protection and preservation of the natural environment.

Hence, it is of basic and common understanding that the right to a balanced and healthful ecology is a right inherently enshrined in each of dwellers of the planet, but it must also be instilled in each of us that this constitutional right carries with it a correlative duty to refrain from impairing the environment.²⁵

Oposa v. Factoran as a Global Jurisprudential Icon

After its historic win in 1993 in the Supreme Court of the Philippines, *Oposa v. Factoran* became a jurisprudential icon, prompting it to be featured as a subject in countless scholarly works in the aspect of environmental law and climate justice across the globe. The Oposa Doctrine has been cited and replicated all over the world,²⁶ it got even cited as a precedent in some Bangladeshi and Indian courts;²⁷ more recently in on-going climate litigation cases such as the Juliana case in the United States.²⁸ The concept of intergenerational responsibility as a form of legal standing as highlighted in the *Oposa v. Factoran* became an inspiration to several courts worldwide especially in the global South, and it signals that many more courts

²⁵ Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId.=27868&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

²⁶ Bueta, Gregorio Rafael (June 21, 2019) Environmental Jurisprudence from the Philippines: Are Climate Litigation Cases Just Around the Corner?, <https://www.iucn.org/news/world-commission-environmental-law/201906/environmental-jurisprudence-philippines-are-climate-litigation-cases-just-around-corner>

²⁷ Philippines: Minors Oposa v. Sec'y of Dep't of Env't & Natural Res., (December 27, 1993) (Phil.), https://www.hurights.or.jp/english/human_rights_and_jurisprudence/1993/12/philippines-minors-oposa-v-secy-of-dept-of-envt-natural-res-sup-ct-1993-33-ilm-173-1994.html

²⁸ Juliana v. United States, Ninth Circuit Holds that Developing and Supervising Plan to Mitigate Anthropogenic Climate Change Would Exceed Remedial Powers of Article III Court. Comment on: 947 F.3d 1159 (9th Cir. 2020), <https://harvardlawreview.org/print/vol-134/juliana-v-united-states/>

and other actors in the litigation process are about to join, diversify and enrich this global discussion.²⁹

30 Years after *Oposa v. Factoran*

The framing of the 1987 Constitution of the Philippines covered a self-executing provision that underscores the paramount importance of the promotion, protection, and preservation of the environment. “The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.”³⁰ This provision paved the way for *Oposa v. Factoran* to create a radical precedent for other cases with common interest of protecting and preserving the environment. 30 years later, it still remains as a revolutionary precedent, cementing itself in the international pedestal as a solid legal instrument that has a potential of reshaping and transforming international law that would address environmental crisis in the ambitious hope of attaining climate justice. It may even be considered as jurisprudential messiah in the aspect of environmental litigation, for it has assisted in saving natural forests or at least mitigating the drastic and rapid effects of environmental degradation.

Ironically, 30 years after *Oposa v. Factoran*, while it became a cornerstone, a global jurisprudential icon for environmental activism and climate justice, we are still confronted with the sad truth that it did not diminish the harsh reality that the planet still remains to be in constant peril. We, the present dwellers of the planet are fortunate enough that we can still be able to witness and experience the vestiges and wonders of nature up until this very moment; this was made possible because of our ancestors from yesterday, and so, we owe a home no less than of what we have today to the children of tomorrow.

²⁹ Singh Margaretha; et. Al, (August 3, 2023) <https://academic.oup.com/ejil/article/34/3/651/7236581>

³⁰ CONST. (1987), art. II, §16 (Phil.).

Bibliography

Constitution

1987 Constitution of the Philippines, art. II, §16 (Phil.).

Statutes

2010 Rules and Procedure for Environmental Cases, <https://sc.judiciary.gov.ph/rules-of-procedure-for-environmental-cases/>

Ecological Solid Waste Management Act of 2000, Rep. Act 9003, (January 26, 2001) <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/4800>

Philippine Clean Air Act of 1999, Rep. Act 8749, (June 23, 1999) <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/6028>

Jurisprudence

Ex Parte Levitt, 303 U.S. 633 (1938), in Philippine Political Law 506, (Cruz & Cruz, 2014).

Metro Manila Development Authority v. Concerned Residents of Manila Bay, G.R. Nos. 171947-48, (December 18, 2008) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/48335>

Mosqueda v. Pilipino Banana Growers and Exporters Association, Inc., G.R. No. 189185, (August 16, 2016) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/62290>

Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId=27868&Index=%2a4aeb4dbdcccda9b59b85ae3fb22cec0&HitCount=3&hits=4+10+1be4+&SearchForm=C%3a%5ce librev2%5csearch%5csearch%5fform

Resident Marine Mammals of the Tañon Strait Protected Seascape v. Reyes, G.R. No. 180771, (April 21, 2015) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/60478>

Spouses Maliones et al. v. Mario Timario Jr. et al., G.R. No. 252834, (February 6, 2023) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/69143>

Online Sources

Bueta, Gregorio Rafael (June 21, 2019) Environmental Jurisprudence from the Philippines: Are Climate Litigation Cases Just Around the Corner?, <https://www.iucn.org/news/world-commission-environmental-law/201906/environmental-jurisprudence-philippines-are-climate-litigation-cases-just-around-corner>

Juliana v. United States, Ninth Circuit Holds that Developing and Supervising Plan to Mitigate Anthropogenic Climate Change Would Exceed Remedial Powers of Article III Court. Comment on: 947 F.3d 1159 (9th Cir. 2020), <https://harvardlawreview.org/print/vol-134/juliana-v-united-states/>

Minors Oposa v. Secretary of the Department of Environment and Natural Resources 33 I.L.M. 173 (1994). – Philippines, <https://www.acrisl.org/casenotes/m2ll8m8skjpglk8-83mk2>

Philippines: Minors Oposa v. Sec'y of Dep't of Env't & Natural Res. (December 27, 1993) (Phil.), https://www.hurights.or.jp/english/human_rights_and_jurisprudence/1993/12/philippines-minors-oposa-v-secy-of-dept-of-envt-natural-res-sup-ct-1993-33-ilm-173-1994.html

Singh Margaretha; et. al., (August 3, 2023) <https://academic.oup.com/ejil/article/34/3/651/7236581>



Photo by Nandhu Kumar from Pexels: <https://www.pexels.com/photo/people-on-a-boat-near-mangroves-13791361/>



Image Source: Canva for Education

SPEAK FOR THE TREES: THE LAW AS VOICE FOR DEFENDING MANGROVE FOREST DEPLETION

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Introduction

The Philippines is a nation endowed not only with abundant natural resources but also with laws governing a wide range of issues of the country. Our environment is protected by a number of laws, including one that incorporates the right to a healthy and balanced ecology into the 1987 Constitution.

Laws are important for without it, everything will be in chaos. This is also true for our environment; how would we be able to protect it in the absence of laws? How will we be able to fight against the harm that humans are causing to our environment?

One of the most pressing challenges in our country now is the preservation of our mangrove regions. These mangrove areas must be safeguarded and preserved since they are home to one of our primary food sources, fish and other animals that flourish in these environments. As a result, laws were enacted to assure the conservation of the aforementioned mangrove areas throughout the country.

Thus, in this way, the laws speak for us and our environment.

Mangroves in Panay Island

With a total land area of over 1,239,400 hectares, Panay is the sixth largest of the more than 7,000 islands in the Philippines. Iloilo Province, which makes up 42% of the island, is a plain with vast estuaries where its rivers flow into the straits of Iloilo and Guimaras. Its tallest mountains divide it from the provinces of Antique and Capiz. Many offshore islands shield the erratic Iloilo coastline from the intense *habagat* (southeast monsoon) winds; the largest being the island-province of Guimaras. In 1951, the 12,400 acres of this broad, protected lowland area with winding tidal inlets was home to rich mangrove swamps. Now, endless box-like fishponds cover the landscape and only a few scattered mangrove trees can be seen along the creeks and riverbanks (300 hectares remaining in 1988).¹

Steep mountain ranges that nearly reach the coast, separating the province of Antique from the rest of the island, provide a rocky shoreline with minimal coastal

¹ Primavera, J.H., et. al., *Handbook of Mangroves in the Philippines – Panay*, (2004), <https://repository.seafdec.org.ph/handle/10862/3053>

plain. There is a restricted distribution of mangroves in Western Panay due to its narrow plain and extended dry season. Capiz is located on the northeastern side. Its huge valley basin, created by four major interconnected rivers, and wide protected plains make it an ideal place for mangrove habitation. One of these rivers is known by its native Sulod-Bukidnon name, Mapanhay, which means tranquil. After the first Spanish town was constructed near the river's mouth in Banica, Capiz, the entire island came to be known by this name, which is today abbreviated to Panay.²

There are five main rivers in the rugged, mountainous province of Aklan to the northwest. One of the largest is the Aklan River, which drains into a delta and is site to the successful mangrove reforestation project at Buswang, Kalibo. The 72 hectares Bugtong Bato-Naisud primary patch mangroves in Ibajay, which span the Naisud River, served as our primary collecting site. The comparatively long age of Panay and its previous mainland ties to Luzon and the rest of the archipelago can be attributed to the large diversity of mangroves (33 species) on the island. Some forests, notably mangroves, have remained undeveloped in Aklan due to the slow rate of "development," and they must be protected from being converted to ponds used for aquaculture and other illegal cutting.³

Importance of Mangrove Forests

The Philippines has about 7,100 islands surrounding the mainland of Luzon in the north, Visayas in the middle and Mindanao in the south. The country has about 18,000 kilometers (km) of shorelines and vast areas of mangroves totaling 500,000 hectares.⁴

Mangrove habitats are critical to many coastal populations because they provide livelihood opportunities, food sources, and protection from coastal hazards. Given the country's longest coastlines, which act as pathways for tropical cyclones from the Pacific Ocean, mangroves and beach forests provide fishers and rural communities with sustainable and cost-effective ways to protect themselves, their properties, and communities from the devastating effects of natural disasters. Mangroves also play an important role in climate stabilization on a global scale

² Primavera, J.H., et. al., *Handbook of Mangroves in the Philippines – Panay*, (2004), <https://repository.seafdec.org.ph/handle/10862/3053>

³ *Id.*

⁴ Brown, W.H. & A.F. Fisher, *Philippine Mangrove Swamps. Minor Products of Philippine Forests Vol. I*, 22, DENR, Bureau, (1920), Cited in: Mangroves Management and Development, https://repository.seafdec.org.ph/bitstream/handle/10862/712/RTCCode_p39-47.pdf

because they have high carbon storage and sequestration capability, which is thought to be greater than that of tropical forests.⁵

Mangroves are often misunderstood and undervalued ecosystems. These coastal forests are sometimes perceived as “dirty” or “dead areas”, a wasteland that could be cleared in favor of sandy beaches, swanky resorts or other developments.⁶

More than 1,500 plant and animal species rely on mangroves. This includes fish and birds that use the shallow waters beneath mangrove plants as nursery. According to recent research, mangroves are also important for larger mammals such as monkeys, sloths, tigers, hyenas, and wild dogs.⁷ Fifteen percent of the more than 1,500 species that rely on mangroves for their survival are in danger of going extinct. When it comes to mammals, this number is rising. In the next few years, almost half of the species that live in or feed on mangrove forests may become extinct, and the situation is expected to get worse for the majority of them. Extreme weather events like storms and flood surges are growing more common and intense as global temperatures rise. Mangroves serve as an excellent first line of defense that serves to protect higher ground because their trunks absorb the impact of waves. The resilience of coastal communities and national economies is increased by valuing, restoring, and safeguarding mangroves as a natural solution.⁸

Protecting and restoring mangroves also aids in climate change mitigation through carbon sequestration, as they are among the most carbon-rich ecosystems on the globe, storing an average of 1,000 tons of carbon per hectare in biomass and underlying soils. However, mangroves are threatened. A fifth of them have already disappeared globally. Coastal development, when mangrove forests are cleared to make way for buildings and fish or shrimp farms, is the main driver of mangrove loss.⁹

For fish and shellfish, the mangrove root systems below the water provide safe nesting and breeding habitats. Fish species that flourish in mangrove forests can support local communities' diets with protein and provide jobs for fishermen in the area. In fact, it has been estimated that 80% of the global fish catch relies directly or indirectly on mangrove forests.¹⁰

⁵ Oceana Campaign Overview, <https://ph.oceana.org/our-campaigns/restore-mangrove-forests/#news-and-reports>

⁶ UN Environment Programme, *An Inside Look at the Beauty and Benefits of Mangroves*, (July 25, 2023), <https://www.unep.org/news-and-stories/story/inside-look-beauty-and-benefits-mangroves>

⁷ *Id.*

⁸ *Id.*

⁹ *Id.*

¹⁰ Goodman, Ella. *Benefits of Mangroves—Socio-economic*, (December 12, 2021), <https://www.theleafcharity.com/blog/benefits-of-mangroves-socio-economic>

Philippine Laws and Jurisprudence on Mangrove Forests

In the Philippines, we have several laws that protect our very own Mangrove Forests, among them are: Expanded National Integrated Protected Areas System (eNIPAS) Act of 2018 (R.A. No. 11038), National Integrated Protected Area System Act of 1992 (R.A. No. 7586), Strategic Environmental Plan (SEP) for Palawan Act (R.A. No. 7611), An Act Incorporating Certain Sections of the National Internal Revenue Code Of 1977, as amended, to Presidential Decree No. 705, as amended (R.A. No. 7161), The Revised Forestry Code of the Philippines (P.D. No. 705), Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and injuring of Certain Trees, Plants and Vegetation (P.D. No. 953).

Republic Act No. 11038: Expanded National Integrated Protected Areas System (eNIPAS) Act of 2018

The National Integrated Protected Areas System (NIPAS)¹¹ encompass ecologically rich and unique areas and biologically important public lands that are habitats of rare and threatened species of plants and animals, biogeographic zones and related ecosystems, whether terrestrial, wetland or marine, all of which shall be designated as 'protected areas'. The Law provides for the environmental and natural resources offices and protected area management offices where protected areas are located. The Environmental Management Bureau shall consider the environmental impact assessment. The Act further provides for heritage preservation, development, protection and conservation of biologically significant areas with their indigenous knowledge systems and practices. Violation of laws, rules and regulations in the protected areas are investigated by the Department of Justice. In Section 20, the prohibited acts are nominated such as killing, destroying and disturbing any wildlife, hunting, defacing boundaries, damaging roads and trails, using blasting caps and occupying any public land. All property and private rights within the protected area and its buffer zones already existing and/or vested upon the effectivity of this Act shall be protected and respected.¹² This act protects mangrove forests that are classified as protected areas.

Republic Act No. 7611: Strategic Environmental Plan (SEP) for Palawan Act

An Act adopting the strategic environmental plan for Palawan, creating the administrative machinery for its implementation, converting the palawan integrated area development project office to its support staff, providing funds therefor, and for other purposes.

¹¹ National Integrated Protected Areas System (NIPAS) Act of 1992, Rep. Act No. 7586, (June 1, 1992) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/3508>

¹² eNIPAS Act Summary, <https://www.informea.org/en/content/legislation/republic-act-11038-expanded-national-integrated-protected-areas-system-act-2018>

A comprehensive plan for the Palawan province is adopted based on a philosophy of sustainable development for Palawan and the conservation of life supporting ecosystems. An integrated approach to problems and issues shall be adopted. The Plan shall serve as a framework for government agencies concerned with the formulation and implementation of plans. The Plan shall establish a "graded system of protection and development control over Palawan including tribal lands, mangroves, seabed, etc. This system shall be known as the Environmentally Critical Areas Network (ECAN).¹³ Areas covered by ECAN shall be classified into 3 components: terrestrial, Coastal /marine areas, and tribal ancestral lands (sect. 8). The second part of the Act concerns management of resources outside of ecologically critical areas. Other provisions deal with the Palawan Council for Sustainable Development created under section 16 and conversion of an existing office to the Council.¹⁴

Republic Act No. 7161 Sec. 4: An Act Incorporating Certain Sections of the National Internal Revenue Code Of 1977, as amended, to Presidential Decree No. 705, as amended, otherwise known as "The Revised Forestry Code of the Philippines," And Providing Amendments Thereto By Increasing the Forest Charges on Timber and Other Forest Products

SEC. 4. Section 71 of Presidential Decree No. 705, as amended, (formerly Section 233 of the National Internal Revenue Code) is hereby amended to read as follows:

"SEC. 71. Charges on Firewood, Branches and Other Recoverable Wood Wastes of Timber. - Except for all mangrove species whose cutting shall be banned, there shall be collected forest charges on each cubic meter of firewood cut in forestland, branches and other recoverable wood wastes of timber, such as timber ends, tops and stumps, when used as raw materials for the manufacture of finished products, Ten pesos (P10.00).

"Only third or fourth group wood can be taken for firewood. However, if jointly authorized by the Secretaries of both the Departments of Environment and Natural Resources, and Agriculture, first and second group woods may be removed for firewood purposes from land which is more valuable for agriculture than for forest purposes."¹⁵

¹³ PCSDS. 2006. ECAN Zones Management Plan for El Nido Municipality Palawan Council for Sustainable Development, Puerto Princesa City, Philippines, <https://pkp-new.pcsd.gov.ph/environmentally-critical-areas-network-ecan-zones-management-plan-for-el-nido-municipality-2006/>

¹⁴ Republic Act No. 7611: Strategic Environmental Plan (SEP) for Palawan Act Summary, <https://www.fao.org/faolex/results/details/en/c/LEX-FAOC019797/>

¹⁵ Revised Forestry Code of the Philippines, Rep. Act No. 7161 §4, (October 10, 1991) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/2885>

Presidential Decree No. 705, s. 1975: Revised Forestry Code of the Philippines

The Revised Forestry Code of the Philippines, also known as Presidential Decree No. 705, was strengthened by Republic Act 7161, which made cutting down mangroves illegal. In 1998, the Philippine Fisheries Code¹⁶ was introduced, specifying in Sections 46 and 49 that all abandoned fishponds must be reverted to their original mangrove state.

It must also be noted that in the case of *Director of Forestry v. Villareal*, the Supreme Court ruled that conformably to the legislative definition embodied in Section 1820 of the Revised Administrative Code of 1917, which remains unamended up to now, mangrove swamps or manglares form part of the public forests of the Philippines. As such, they are not alienable under the Constitution and may not be the subject of private ownership until and unless they are first released as forest land and classified as alienable agricultural land.¹⁷

Presidential Decree No. 953, s. 1976: Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and injuring of Certain Trees, Plants and Vegetation

Section 3 states that “any person who cuts, destroys, damages or injures, naturally growing or planted trees of any kind, flowering or ornamental plants and shrubs, or plants of scenic, aesthetic and ecological values, along public roads, in plazas, parks other than national parks, school premises or in any other public ground or place, or on banks of rivers or creeks, or along roads in land subdivisions or areas therein for the common use of the owners of lots therein, or any species of vegetation or forest cover found therein shall, be punished with imprisonment for not less than six months and not more than two years, or a fine of not less than five hundred pesos and not more than five thousand pesos, or with both such imprisonment and fine at the discretion of the court, except when the cutting, destroying, damaging or injuring is necessary for public safety or the pruning thereof is necessary to enhance beauty, and only upon the approval of the duly authorized representative of the head of agency or political subdivision having jurisdiction therein, or of the Director of Forest Development in the case of trees on banks of rivers and creeks, or of the owner of the land subdivision in the case of trees along roads and in other areas therein for the common use of owners of lots therein. If the offender is a corporation, partnership or association, the penalty shall be imposed upon the officer or officers thereof

¹⁶ Philippine Fisheries Code of 1998, Rep. Act No. 8550, (February 25, 1998) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/4002>

¹⁷ *Director of Forestry v. Villareal*, G.R. No. L-32266, (February 27, 1989) (Phil.), https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId.=37189&Index=%2a4aeb4dbdcceeda9b59b85ae3fb22cec0&HitCount=4&hits=4+5+11+12+&SearchForm=C%3a%5celibrev2%5csearch%5csearch%5fform

responsible for the offense, and if such officer or officers are aliens, in addition to the penalty herein prescribed, he or they shall be deported without further proceedings before the Commission on Immigration and Deportation. Nothing in this Decree shall prevent the cancellation of a license agreement, lease, license or permit from the Government, if such cancellation is prescribed therein or in Government regulations for such offense.”¹⁸

Related Cases

The Court of Appeals (CA) upheld a lower court's ruling in favor of the issuance of a permanent protection order (PPO) against the illegal occupation and destruction of five hectares of protected land within a mangrove reserve in Puerto Princesa City, Palawan.¹⁹

In a decision dated February 21, the appellate court's Special Seventh Division dismissed the appeal filed by businessman Rafael Cervantes and affirmed the Puerto Princesa Regional Trial Court's (RTC) decision which required him to post a Php 10 million initial trust fund for the rehabilitation and restoration of the damaged areas at the Mangrove Swamp Forest Reserve, in Sitio Barimbing in Barangay San Manuel.²⁰

Under a number of laws including Presidential Proclamation 2152²¹ signed in 1981, the entire province of Palawan is a mangrove swamp forest reserve and cutting of mangroves is prohibited.

In *Leynes v. People*, Petitioner Efren R. Reyes was convicted for the offense of conversion of mangroves as punishable under Section 94 of Republic Act No. 8550, otherwise known as the “Philippine Fisheries Code of 1998.” The Supreme Court said that “it is high time and therefore, to avoid confusion, that mangrove forests do not consist of the typical mangrove trees only. As defined, mangroves are “a community of intertidal plants including all species of trees, shrubs, vines and herbs found on coasts, swamps, or borders of swamps.” Contrary to Efren's belief, “the word ‘mangroves’ refers to a group of plants which may actually belong to several families (species that distinctly belong to their own

¹⁸ Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and Injuring of Certain Trees, Plants and Vegetation, P.D. No. 953, §3, (July, 6, 1976) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327>

¹⁹ Benjamin Pulta, *Palawan Court's Protection Order on Mangrove Reserve Affirmed*, (Feb. 25, 2020), <https://www.pna.gov.ph/articles/1094701>

²⁰ *Id.*

²¹ Presidential Proclamation 2152, s. 1981, <https://www.officialgazette.gov.ph/1981/12/29/proclamation-no-2152-s-1981/>

evolutionary group)." By cutting a tree in the mangrove forest, regardless of its species, Efren caused conversion of the same.²²

House Bill No. 2275 also known as the “Mangrove Forest Protection and Reservation Act”

On July 22, 2022, House Bill No. 2275 was introduced in the 19th Congress of the Philippines, wherein it proposes the establishment of Mangrove Reservation Areas in all local government units with coastal areas within the Philippines portions of land solely for the preservation, protection, reforestation, and sustainable management of mangrove forests; establishment and composition of the National Council for the Preservation of Mangrove Forests which shall be attached agency of the Department of Environment and Natural Resources (DENR); and establishment of Local Councils for the Preservation of Mangrove Forests in each of the local government units where a mangrove reservation area has been identified. It also discusses the powers and functions of the National and Local Council.²³

The bill also requires mandatory prior consultations which states that no project of program that may cause pollution, deduction or loss of mangrove forests or extinction of marine and estuarine plant and animal species shall be implemented by the National or Local Council without prior consultations with the local government unit concerned, non-governmental or people’s organizations, and other sectors concerned.²⁴ House Bill No. 2275 is geared towards utilizing Local Government Units to implement further protection and preservation of Mangrove Forests in the Philippines, thus the passage of this bill into law would be a great step to a more effective instrument in safeguarding Mangrove Forests, as it is a specific law that goes deeper into the local level.

Conclusion

Our country is indeed a nation that has numerous laws on environmental protection, however, there is still a need to make the public aware of the value of mangrove areas and the laws protecting them. Thus, to protect and preserve the mangrove forests, local government units should take the lead in encouraging their constituents to plant more mangrove trees. They should also make sure that the laws, rules, and regulations pertaining to these forests are strictly followed and that they are implemented appropriately, not only to protect our environment but also to support the livelihood of fishermen who depend entirely on the mangroves for a plentiful harvest of seafood.

²² Leynes v. People, G.R. No. 224804, (September 21, 2016) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/62395>

²³ House Bill No. 2275 - An Act Establishing the National Council for the Preservation of Mangrove Forests, https://docs.congress.hrep.online/legisdocs/basic_19/HB02275.pdf

²⁴ *Id.*

Bibliography

Statutes

eNIPAS Act Summary,

<https://www.informea.org/en/content/legislation/republic-act-11038-expanded-national-integrated-protected-areas-system-act-2018>

Philippine Fisheries Code of 1998, Rep. Act No. 8550, (February 25, 1998) (Phil.),
<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/4002>

Presidential Proclamation 2152, s. 1981,
<https://www.officialgazette.gov.ph/1981/12/29/proclamation-no-2152-s-1981/>

National Integrated Protected Areas System (NIPAS) Act of 1992, Rep. Act No. 7586, (June 1, 1992) (Phil.),
<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/3508>

Republic Act No. 7611: Strategic Environmental Plan (SEP) for Palawan Act Summary,
<https://www.fao.org/faolex/results/details/en/c/LEX-FAOC019797/>

Revised Forestry Code of the Philippines, Rep. Act No. 7161 §4, (October 10, 1991) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/2885>

Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and Injuring of Certain Trees, Plants and Vegetation, P.D. No. 953, §3, (July, 6, 1976) (Phil.),
<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327>

House Bill

House Bill No. 2275 - An Act Establishing the National Council for the Preservation of Mangrove Forests,
https://docs.congress.hrep.online/legisdocs/basic_19/HB02275.pdf

Jurisprudence

Director of Forestry v. Villareal, G.R. No. L-32266, (February 27, 1989) (Phil.),
https://elibrary.judiciary.gov.ph/assets/dtSearch/dtSearch_system_files/dtisapi6.dll?cmd=getdoc&DocId=37189&Index=%2a4aeb4dbdceeda9b59b85ae3fb22cec0&HitCount=4&hits=4+5+11+12+&SearchForm=C%3a%5ce librev2%5csearch%5csearch%5fform

Leynes v. People, G.R. No. 224804, (September 21, 2016) (Phil.),
<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/62395>

Online Sources

Benjamin Pulta, Palawan Court's Protection Order on Mangrove Reserve Affirmed, (February 25, 2020),
<https://www.pna.gov.ph/articles/1094701>

Brown, W.H. & A.F. Fisher, Philippine Mangrove Swamps. Minor Products of Philippine Forests Vol. I, 22, DENR, Bureau, (1920), Cited in: Mangroves Management and Development,
https://repository.seafdec.org.ph/bitstream/handle/10862/712/RTCCode_p_39-47.pdf

Goodman, Ella. Benefits of Mangroves—Socio-economic, (December 12, 2021),
<https://www.theleafcharity.com/blog/benefits-of-mangroves-socio-economic>

Oceana Campaign Overview,
<https://ph.oceana.org/our-campaigns/restore-mangrove-forests/#news-and-reports>

PCSDS. 2006. ECAN Zones Management Plan for El Nido Municipality Palawan Council for Sustainable Development, Puerto Princesa City, Philippines,
<https://pkp-new.pcsd.gov.ph/environmentally-critical-areas-network-ecan-zones-management-plan-for-el-nido-municipality-2006/>

Primavera, J.H., et. al., Handbook of Mangroves in the Philippines – Panay, (2004).,
<https://repository.seafdec.org.ph/handle/10862/3053>

UN Environment Programme, An Inside Look at the Beauty and Benefits of Mangroves, (July 25, 2023),
<https://www.unep.org/news-and-stories/story/inside-look-beauty-and-benefits-mangroves>



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TREE-TRANSPLANTING: AN INTER-GENERATIONAL OPERATION FOR A BALANCED AND HEALTHFUL ECOLOGY USING THE ADVANCEMENT OF TECHNOLOGY

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Grow

To grow takes time. To grow is uncertain. When a person plants a tree, he plants hope. A hope that there will be a better future. Meanwhile, to hope for a better future is to believe that growth will go with it. Unfortunately, hope, growth and belief sometimes do not turn up.

When we cut trees and plant new ones, for instance in the town of Pototan wherein an affidavit stated that 100 tree seedlings should be planted in other areas as replacement for every tree that Department of Public Works and Highways (DPWH) felled¹, did we plant hope or did we wager our growth?

Move out the way

Republic Act No. 3571 (R.A. No. 3571), enacted on June 21, 1963, prohibits the cutting, destroying or injuring of planted or growing trees, flowering plants and shrubs or plants of scenic value along public roads, in plazas, parks, school premises or in any other public pleasure ground. The exceptions for the cutting, destroying or injuring of same apply when it is necessary for public safety, or such pruning of same is necessary to enhance its beauty and only upon the recommendation of the committee and approved by the Director of Parks and Wildlife.² The law aims to prohibit the cutting, destroying, or injuring of planted or growing trees, flowering plants, shrubs, or plants of scenic value in public places. R.A. No. 3571 was amended by Presidential Decree No. 953 on July 06, 1976 providing for additional authorization on exceptions of cutting, destroying, damaging or injuring trees when it is necessary for public safety or the pruning thereof is necessary to enhance beauty, and upon the approval of the duly authorized representative of the head of agency or political subdivision having jurisdiction therein.³ These laws are as ancient as some

¹ Sornito, I. (2018, June 27), Cutting of trees in Pototan continues, Panay News, <https://www.panaynews.net/cutting-of-trees-in-pototan-continues/>

² Republic Act No. 3571 (June 21, 1963), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/25793>

³ Pres. Dec. No. 953 - Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and Injuring of Certain Trees, Plants and Vegetation. - Supreme Court E-Library. (1976, July 6), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327><https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327>

of the affected trees along the road during road widening. Thus, it is imperative that these legislations be reviewed to address current environmental issues.

Our Supreme Court upheld the concept of inter-generational responsibility in the case of *Oposa vs. Factoran*.⁴ The Court emphasized that every generation has a responsibility to the next to preserve that rhythm and harmony for the full enjoyment of a balanced and healthful ecology.⁵ In the field of Public International Law, 1972 Stockholm Declaration⁶ as further affirmed by 1992 Rio Declaration⁷, respectively established the inter-generational responsibility of States. The capacity of the earth to produce vital renewable resources should be maintained, and where practicable, be restored, or improved in order to guard such resources against the danger of their future exhaustion.⁸

DPWH was given permit by the Department of Environment and Natural Resources (DENR) to cut over 5,000 trees to give way to the project that covers thirteen (13) barangays in Pototan, requiring DPWH to replace each tree with at least 100 seedlings.⁹ More than 5,000 trees would be cut down to widen the two-lane national highway that links Iloilo province to the provinces of Aklan and Capiz.¹⁰ This not only happens in developing provinces but to developed provinces like Cebu Province. In cutting of roadside trees in the cities of Naga and Carcar in the province

⁴ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.).

⁵ *Id.*

⁶ United Nations. (1972). United Nations Conference on the Human Environment, Stockholm 1972 | United Nations, <https://www.un.org/en/conferences/environment/stockholm1972> [Principle 2: The natural resources of the earth, including the air, water, land, flora and fauna and especially representative samples of natural ecosystems, must be safeguarded for the benefit of present and future generations through careful planning or management, as appropriate.].

⁷ United Nations. (1992). United Nations Conference on Environment and Development, Rio de Janeiro, Brazil, 3-14 June 1992 | United Nations, <https://www.un.org/en/conferences/environment/rio1992> [Principle 3: The right to development must be fulfilled so as to equitably meet developmental and environmental needs of present and future generations.].

⁸ Nachura, A. E. B., & Gatdula, J. Outline Introduction to Public International Law (2022 Edition, p. 198). REX Book Store.

⁹ The Department of Public Works and Highways (DPWH) 2nd Iloilo District Engineering Office and the Department of Environment and Natural Resources (DENR) Region 6, media briefing on the replacement of trees affected by the road widening project in Pototan, Iloilo. (May 17, 2018), <https://www.dpwh.gov.ph/dpwh/news/13277#:~:text=DPWH%20was%20given%20permit%20by,With%20at%20least%20100%20seedlings>

¹⁰ Burgos, N. P., Jr. (2016, December 1). Not 300, but 5,000 trees to be cut for road | Inquirer News. INQUIRER.net, <https://newsinfo.inquirer.net/849500/not-300-but-5000-trees-to-be-cut-for-road>

of Cebu, DENR stressed that public safety is a major consideration in granting tree-cutting permits.¹¹

Cutting trees will continue as long as there is a need to widen highways for economic transportation purposes. In the United States of America, the benefits of a transportation investment are typically estimated by comparing the amount of travel time, vehicle miles traveled and expected number of crashes for the Alternative to the Base Case.¹² As applied in Philippines highways, national agencies also aim for travel-time savings, vehicle cost operating savings, and safety benefits. Further, in the Philippine Development Plan, investments in infrastructure are linked to job creation, inclusive growth, the wider business enabling environment, and many government priorities.¹³ This shows that the government is concerned about the traffic in roads and even in highways thus they see that expansion and building new ones are the solutions available in addressing them.

Technology has advanced with development and discovery of new machineries, methods, and strategies. Hence, we should ask ourselves whether cutting trees and planting new ones is still ideal. When we cut trees, it is as if we are cutting our source of life. With the idea of planting new trees for every tree we felled, how long can we hold our breath?

Survival

The Department of Environment and Natural Resources (DENR) former chief, Secretary Roy Cimatu, sought ways to raise NGP seedlings' survival rate.¹⁴ The survival rate of its National Greening Program (NGP) seedlings in respective regions way back his tenure ranges between 50 percent to 80 percent.¹⁵ The DENR reported that there was a 78% survival rate of the trees planted under NGP from 2011 to 2016.¹⁶ An experiment showed results with loss of 20% of saplings in the first few

¹¹ Department of Environment and Natural Resources (DENR). (2024).. DENR: Public safety is a major consideration in granting tree-cutting permit, <https://denr.gov.ph/news-events/denr-public-safety-is-a-major-cons/d.eration-in-granting-tree-cutting-permit/>

¹² Benefit-Cost Analysis - MNDOT. <https://www.dot.state.mn.us/planning/program/benefitcost.html>

¹³ Patinio, F. (2018). DPWH to build more roads to ease Metro Manila traffic. <https://www.pna.gov.ph/articles/1045213>

¹⁴ Teves, C. (2018). DENR chief seeks ways to raise NGP seedlings' survival rate. <https://www.pna.gov.ph/articles/1056443>

¹⁵ *Id.*

¹⁶ Gozum, I. (2023, September 26). Lawmaker seeks bigger maintenance, compensation budget for reforestation, <https://pid.s.gov.ph/details/news/in-the-news/lawmaker-seeks-bigger-maintenance-compensation-budget-for-reforestation#:~:text=The%20Department%20of%20Environment%20and,NGP%20from%202011%20to%202016>

months amounted to approximately 4800 plants, and approximately 12,000 plants in the first 20 months.¹⁷

While the result shows a promising rate, it is uncertain whether the seedling will prosper into a full-grown tree. It is highly likely that the rate will go down considering rapid changes in our climate. In Calasiao, Pangasinan, at least 2,500 saplings and seedlings of native trees in the mountains of Mangatarem town in this province did not survive the recent heat and have withered from extremely high temperatures.¹⁸

Tree of spades

As a tool, tree spades have been around in some form since as early as 1961, and make moving trees only possible but far less laborious.¹⁹ It consists of a series of sharp blades that are attached to a hydraulic system which are used to dig around the roots of a tree, creating a root ball.²⁰ The root ball is then lifted out of the ground and moved to a new location.²¹ With the advancement of mechanical engineering, a tree spade now comes with a variety of sizes attached to heavy-duty trucks or construction vehicles.

Transplanting is the term used to describe the digging and replanting of trees from one location to a new location hence, due to the wide extent of morphology of the tree root system, transplanting of trees involves substantial removal of roots.²² Transplanting or moving trees or shrubs from one location to another site is a major operation from which most plants recover slowly.²³ There are numerous benefits to using a tree spade than a chainsaw. Aside from saving time and efforts, using tree spade enables us to transport mature trees at a higher survival rate that would be hard to move otherwise, preserve trees that would otherwise be destroyed, save labor costs

¹⁷ Preece, N. D., Van Oosterzee, P., & Lawes, M. J. (2023). Reforestation success can be enhanced by improving tree planting methods. *Journal of Environmental Management*, 336, 117645, <https://doi.org/10.1016/j.jenvman.2023.117645>

¹⁸ Sotelo, Y. (2024, April 19). Intense heat takes toll on young native trees in Pangasinan. INQUIRER.net, <https://newsinfo.inquirer.net/1931222/intense-heat-takes-toll-on-young-native-trees-in-pangasinan>

¹⁹ Campbell, R. (2023, December 28). Using a tree spade to preserve your favorite tree. Miller's Tree Service, <https://www.millertreesrv.com/how-to-use-a-tree-spade/>

²⁰ Insta-Shade, O. (n.d.). Tree Spades - Oxford Insta-Shade. Oxford Insta-Shade. <https://www.oxfordinstashade.com/tree-spades>

²¹ *Id.*

²² Greening, Landscape and Tree Management Section Development Bureau. (2014). Guidelines on Tree Transplanting, https://www.greening.gov.hk/filemanager/greening/en/content_28/Guidelines_on_Tree_Transplanting_e.pdf

²³ Transplanting trees and shrubs. (n.d.). Natural Resource Stewardship, <https://naturalresources.extension.iastate.edu/encyclopedia/transplanting-trees-and-shrubs>

required for hand-digging and removal, and by giving us the ability to move established trees to landscape an area, rather than waiting for saplings to grow.²⁴

Planting an Idea

We should invest in tree spade as it has proven to offer mutual benefits between us and the environment. During the replacing of trees felled to widen the national highway in Pototan town, DPWH acknowledged it has been slow in replacing the said felled trees. According to Edna Mae Cañete, assistant engineer of DPWH-6's 2nd Engineering District during the duration of the project, DPWH did not have enough funds to procure tree seedlings as the budget for the highway widening did not include the needed money for tree planting.²⁵ Applying the most efficient strategy and logistics, several tree spades are enough for an entire region as it is not always on demand unlike other heavy construction machinery. Another obstacle encountered during the cutting down of trees was the lack of expertise to plant tree seedlings requiring DENR's rigorous assistance.

There is no guarantee that a tree planted today will be alive in the upcoming years. Hence, the best practice is to allow fully grown trees which are thriving at the moment to continue with its purpose. A study of tree growth, mortality and harvest at plantations in the Philippines shows that the growth rate of individual trees differed widely in the mixed plantations, ranging from 0.001 to 271.44 cm²/year.²⁶ This is not to discourage planting trees but to prioritize an idea that it is better to transplant than to cut then plant. There is no harm in pursuing this policy of using the advancement of technology to save a living thing that is not yet possible to recreate. We can always buy seedlings, but nowhere will we be able to buy the time it takes for trees to grow.

²⁴ Heritage. (2024, April 4). What's a tree spade – and can you actually dig up a tree with it? - Heritage Tree Care. Heritage Tree Care, <https://www.heritagetreecare.com.au/whats-a-tree-spade-and-can-you-actually-dig-up-a-tree-with-it/>

²⁵ Sornito, I. (2018, May 14).. DPWH slow to replace felled trees in Pototan. Panay News, <https://www.panaynews.net/dpwh-slow-to-replace-felled-trees-in-pototan/>

²⁶ Nguyen, H., Vanclay, J., Herbohn, J., & Firn, J. (2016). Drivers of Tree Growth, Mortality and Harvest Preferences in Species-Rich Plantations for Smallholders and Communities in the Tropics, <https://doi.org/10.1371/journal.pone.0164957>

Bibliography

Statutes

Republic Act No. 3571 (June 21, 1963),
<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/25793>

Pres. Dec. No. 953 - Requiring the Planting of Trees in Certain Places and Penalizing Unauthorized Cutting, Destruction, Damaging and Injuring of Certain Trees, Plants and Vegetation. - Supreme Court E-Library. (1976, July 6),
<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327><https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/26/18327>

Jurisprudence

Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.).

Treaties and Declarations

United Nations. (1972). United Nations Conference on the Human Environment, Stockholm 1972 | United Nations,
<https://www.un.org/en/conferences/environment/stockholm1972>

United Nations. (1992). United Nations Conference on Environment and Development, Rio de Janeiro, Brazil, 3-14 June 1992 | United Nations,
<https://www.un.org/en/conferences/environment/rio1992>

Publications

Nachura, A. E. B., & Gatdula, J. Outline Introduction to Public International Law (2022 Edition, p. 198). REX Book Store.

Online Sources

Benefit-Cost Analysis – MNDOT,
<https://www.dot.state.mn.us/planning/program/benefitcost.html>

Burgos, N. P., Jr. (2016, December 1). Not 300, but 5,000 trees to be cut for road | Inquirer News. INQUIRER.net, <https://newsinfo.inquirer.net/849500/not-300-but-5000-trees-to-be-cut-for-road>

Campbell, R. (2023, December 28). Using a tree spade to preserve your favorite tree. Miller's Tree Service, <https://www.millertreesrv.com/how-to-use-a-tree-spade/>

-
- Department of Environment and Natural Resources (DENR). (2024). DENR: Public safety is a major consideration in granting tree-cutting permit, <https://denr.gov.ph/news-events/denr-public-safety-is-a-major-consideration-in-granting-tree-cutting-permit/>
- Gozum, I. (2023, September 26). Lawmaker seeks bigger maintenance, compensation budget for reforestation, <https://pids.gov.ph/details/news/in-the-news/lawmaker-seeks-bigger-maintenance-compensation-budget-for-reforestation#:~:text=The%20Department%20of%20Environment%20and,NGP%20from%202011%20to%202016>
- Greening, Landscape and Tree Management Section Development Bureau. (2014). Guidelines on Tree Transplanting, https://www.greening.gov.hk/filemanager/greening/en/content_28/Guidelines_on_Tree_Transplanting_e.pdf
- Heritage. (2024, April 4).. What's a tree spade – and can you actually dig up a tree with it? - Heritage Tree Care. Heritage Tree Care, <https://www.heritagetreecare.com.au/whats-a-tree-spade-and-can-you-actually-dig-up-a-tree-with-it/>
- Insta-Shade, O. (n.d.). Tree Spades - Oxford Insta-Shade. Oxford Insta-Shade, <https://www.oxfordinstashade.com/tree-spades>
- Nguyen, H., Vanclay, J., Herbohn, J., & Firn, J. (2016). Drivers of Tree Growth, Mortality and Harvest Preferences in Species-Rich Plantations for Smallholders and Communities in the Tropics, <https://doi.org/10.1371/journal.pone.0164957>
- Patinio, F. (2018). DPWH to build more roads to ease Metro Manila traffic, <https://www.pna.gov.ph/articles/1045213>
- Preece, N. D., Van Oosterzee, P., & Lawes, M. J. (2023). Reforestation success can be enhanced by improving tree planting methods. *Journal of Environmental Management*, 336, 117645, <https://doi.org/10.1016/j.jenvman.2023.117645>
- Sornito, I. (2018, June 27). Cutting of trees in Pototan continues. Panay News, <https://www.panaynews.net/cutting-of-trees-in-pototan-continues/>
- Sornito, I. (2018, May 14).. DPWH slow to replace felled trees in Pototan. Panay News, <https://www.panaynews.net/dpwh-slow-to-replace-felled-trees-in-pototan/>

Sotelo, Y. (2024, April 19). Intense heat takes toll on young native trees in Pangasinan. INQUIRER.net, <https://newsinfo.inquirer.net/1931222/intense-heat-takes-toll-on-young-native-trees-in-pangasinan>

Teves, C. (2018). DENR chief seeks ways to raise NGP seedlings' survival rate, <https://www.pna.gov.ph/articles/1056443>

The Department of Public Works and Highways (DPWH) 2nd Iloilo District Engineering Office and the Department of Environment and Natural Resources (DENR) Region 6, media briefing on the replacement of trees affected by the road widening project in Pototan, Iloilo. (May 17, 2018), <https://www.dpwh.gov.ph/dpwh/news/13277#:~:text=DPWH%20was%20given%20permit%20by,With%20at%20least%20100%20seedlings>

Transplanting trees and shrubs. (n.d.). Natural Resource Stewardship, <https://naturalresources.extension.iastate.edu/encyclopedia/transplanting-trees-and-shrubs>



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Photos by: Luigi Bien B. Villanueva

MANGROVE FORESTS: SUSPEND THE ISSUANCE OF TENURIAL INSTRUMENTS

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Introduction

Mangrove or “Bakawan” is a tree that breeds in sea and river waters. It thrives along the complex conditions of tropical and subtropical coastlines. They link crucial support to marine and terrestrial ecosystems. From their leaves to their branches, trunks, and roots, Mangroves support the multiple intricacies of human needs.¹ To wit, they act as carbon sinks, biodiversity hotspots, improve and maintain local water quality, double as coastal protection, and are essential to the livelihood resources of the people.²

In the Philippines, mangrove trees can be seen anywhere along the coastlines and river waters of the archipelago, generously multiplying. It has served significant protection from the typhoons. However, the International Union for Conservation of Nature (IUCN) reported that due to climate change, half of the world’s mangrove ecosystems are at risk. While the Philippines experienced a significant mangrove decline, losing 10.5 percent of its mangrove forests between 1990 and 2010, because of the rapid expansion of aquaculture ponds for fish farming.³

Unfolding the economic and scientific importance of Mangrove Trees marks the milestone of the preservation and breeding of Mangrove forests and timberlands. This, however, did not warrant protection from commercial and industrial human exploitation. Its existence radiated so much potential financial gain and profit that it invited several business perpetrators, aqua farmers, and private investors, taking advantage of the gaps of the law to exploit mangrove timberlands in the guise of eco-tourism development purposes.

¹ The Philippines Today, What is a Bakawan? (July 4, 2019), <https://thephilippinestoday.com/what-is-a-bakawan/>

² Dhaliwal, Bani The Canopy Project, The Importance of Mangrove Forests (February 14, 2023), <https://www.earthday.org/the-importance-of-mangroveforests/#:~:text=Mangroves%20improve%20and%20maintain%20local,aland%20the%20species%20within%20them>

³ Lu, Brian James J. Fresh Viewpoints: A New Perspective, The Importance of Philippine Mangroves (May 31, 2024), <https://www.pna.gov.ph/opinion/pieces/888-the-importance-of-philippine-mangroves>

The Law Provides

DENR Administrative Order No. 15-90 (AO No. 15-90),⁴ or otherwise known as the Regulations Governing the Utilization, Development and Management of Mangrove Resources; established the policy to conserve, protect, rehabilitate and develop the remaining mangrove resources; regulate the utilization and development of the mangrove resources; stop the wanton exploitation of the mangrove resources; and enhance the replenishment of the denuded areas through natural or artificial means. In addition to the policy, paragraph (l), Sec. 2, of DENR AO No. 15-90, defined that “Mangrove Swamp Forest Reserves are mangrove areas of the public domain which are declared as such under Presidential Proclamation 2152 and are determined to be needed for conservation and protection purposes.”⁵

On the other hand, the Republic Act No. 11038 (R.A. No. 11038),⁶ or otherwise known as the Expanded National Integrated Protected Areas System Act of 2018, established a comprehensive system of protected areas in the country to protect and conserve the environment, biodiversity, and threatened species. Furthermore, in paragraphs (n) and (o), Sec. 18 of RA No. 11038, provides that, “Except as may be allowed by the nature of their categories and pursuant to rules and regulations governing the same, the following acts are PROHIBITED within Protected Areas:

(n) Occupying or dwelling in any public land within the protected area without clearance from the Protected Management Area (PAMB);

(o) Constructing, erecting, or maintaining any kind of structure, fence or enclosure, conducting any business enterprise within the protected area without prior clearance from the PAMB and permit from the DENR, or conducting these activities in a manner that is inconsistent with the management plan duly approved by the PAMB;⁷

The existing policies extend its protection to not just with the mangrove reserves, forests, and timberlands, but to all extent of the wildlife, terrestrial, and marine ecosystem and resources. With the armor of the law, these environmental reserves are warranted against human exploitations. However, such policies are limited when it prescribes otherwise, providing for special exceptions. And in most cases, the exceptions to the rules open the floodgates to abuse.

⁴ DENR Administrative Order No. 15-90 (AO No. 15-90), <http://lis.dar.gov.ph/documents/9231#:~:text=To%20sustain%20optimum%20productivity%2C%20it,development%20of%20the%20mangrove%20resources%3B>

⁵ Regulations Governing the Utilization, Development and Management of Mangrove Resources, DENR Administrative Order No. 15-90 (February 1, 1990), <http://lis.dar.gov.ph/documents/9231>

⁶ Expanded National Integrated Protected Areas System Act of 2018, Republic Act No. 11038 (June 22, 2018), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/84999>

⁷ Republic Act No. 11038 (June 22, 2018) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/84999>

Detriments of Issuance of Tenurial Instruments

Tenurial instruments are leases, permits, agreements, joint venture or production sharing agreements, and licenses that are granted by the Department of Environment and Natural resources (DENR) for various reasons: conservation, exploration, development, and utilization of natural resources.⁸

The gradual decline of mangrove forest areas is rooted in the exploitation of permits or tenurial instruments. It is clear that the law is keen on protecting and conserving mangrove resources and reserves, however, due to the exceptions provided by the existing laws, such preservation policies are violated. Take for instance the report from Palawan News notifying the public to take safeguards and perform due diligence when entering into tenurial instruments or license agreements, especially those involving forest and other government lands. This implies that although possessing or occupying forest reserves are prohibited, the same is not absolute when such permits or licenses are granted and approved.⁹

Another alarming report coming from the DENR, announcing the eviction notice of the illegal occupants in Palawan protected areas for violating the provisions of R.A. No. 11038 or the Expanded National Integrated Protected Areas System or E-NIPAS Act of 2018. The law prohibits occupation or dwelling in any public land within a protected area without clearance from the Protected Management Area (PAMB). Furthermore, DENR-MIMAROPA Regional Executive Director Henry Adornado said “The PAMB has neither accepted from the respondents any application for tenurial instrument nor approved any construction of structures in the said area.”¹⁰ Hence, if it weren't for the permits unapproved, the illegal occupants in

⁸ Gozum, Iya DENR to Stop Issuing Permits, Agreements for Use of Protected Areas (October 18, 2023), [https://www.rappler.com/philippines/denr-stop-issuing-permits-instruments-protected-areas/#:~:text=DENR%20to%20stop%20issuing%20permits%2C%20agreements%20for%20use%20of%20protected%20areas,-Oct%2018%2C%202023&text=MANILA%2C%20Philippines%20%E2%80%93%20The%20Department%20of,said.%20on%20Wednesday%2C%20October%](https://www.rappler.com/philippines/denr-stop-issuing-permits-instruments-protected-areas/#:~:text=DENR%20to%20stop%20issuing%20permits%2C%20agreements%20for%20use%20of%20protected%20areas,-Oct%2018%2C%202023&text=MANILA%2C%20Philippines%20%E2%80%93%20The%20Department%20of,said.%20on%20Wednesday%2C%20October%20)

⁹ Palawan News, DENR Warns Public Against Fraudulent Land Transactions in MIMAROPA (May 20, 2022), <https://palawan-news.com/denr-warns-public-against-fraudulent-land-transactions-in-mimaropa/#:~:text=DENR%20warns%20public%20against%20fraudulent%20land%20transactions%20in%20MIMAROPA,-By&text=The%20environment%20department%20in%20MIMAROPA,forest%20and%20other%20government%20lands>

¹⁰ DENR, DENR: Eviction Notice v. Illegal Occupants in Palawan Protected Area Issued ‘as early as March’ (June 17, 2020), <https://denr.gov.ph/news-events/denr-eviction-notice-vs-illegal-occupants-in-palawan-protected-area-issued-as-early-as-march/#:~:text=As%20early%20as%20March%20this,in%20Taytay%20town%2C%20Palawan%20province>

the “*protected areas*” of Mangrove forests and timberlands could have gotten away with the law.

To give credence to those issues, I had a personal interview with one of the retired employees of the City Agriculture Office of Palawan, named Delia B. Martinez. She attested that the pressing issues are indeed happening in Palawan. She said that our environmental salvage zones are all occupied and titled lots, even the timberland “mangrove areas.” She further said that these areas are, in fact, cannot be sold nor can be possessed by any private investors without certification from the DENR. She explained that these buyers apply for Forest Landuse Agreement for Tourism Purposes (FLAGT) or Foreshore Lease Agreement to acquire the said protected areas, and the moment DENR approves such agreement, the buyers can then covertly generate the lot title for the said subject property and gradually destroy the Mangrove Trees. And when there are no more Mangrove trees, the buyers can now then apply for a tenurial instrument, and pay for the taxes of said property so that they can acquire tax declaration for the processing of land titling. And she concluded her statement that only those well-off individuals who have actually unlawfully acquired those protected properties.

Conclusion

“Good people do not need laws to tell them to act responsibly, while bad people will find a way around the laws.” –Plato

It is a well settled rule that for every general rule there will always be an exception. In light of this, DENR AO No. 15-90 and R.A. No. 11038 established the rules for the protection and preservation of Mangrove Forests and Timberlands. However, violators are able to go around the law by taking advantage of the exception provided therefor. Hence, the occupying or dwelling of mangrove forests, timberlands, or even protected zones are prohibited until clearance, permit, or tenurial instruments are granted by the proper authorities. The guarantee of the protection of Mangrove Forests and Timberlands are open to the abuse of the perpetrators and may cause its further destruction. And we can do nothing about it except through lobbying such issues to the attention of the legislators. As Martinez said, “The law is made firm and polished, however, the law always has a loophole that is money talks.”

It is alarming that tenurial instruments are granted even to the protected areas of marine, timberland, and coastal forests. Because, although the process therefor is extensive and closely evaluated by the proper authorities, perpetrators are still able to go around the law by means of promised considerations, which could potentially influence the judgment of the assigned authorities. Therefore, we appeal to suspend the issuance of tenurial instruments, not just on mangrove protected areas, but likewise to all extent where Mangrove Trees and Timberlands are already existing and thriving.

Bibliography

Statutes

Expanded National Integrated Protected Areas System Act of 2018, Republic Act No. 11038 (June 22, 2018), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/84999>

Regulations Governing the Utilization, Development and Management of Mangrove Resources, DENR Administrative Order No. 15-90 (February 1, 1990), <http://lis.dar.gov.ph/documents/9231>

Online Sources

DENR Administrative Order No. 15-90 (AO No. 15-90), <http://lis.dar.gov.ph/documents/9231#:~:text=To%20sustain%20optimum%20productivity%2C%20it,development%20of%20the%20mangrove%20resources%3B>

DENR, DENR: Eviction Notice v. Illegal Occupants in Palawan Protected Area Issued 'as early as March' (June 17, 2020), <https://denr.gov.ph/news-events/denr-eviction-notice-vs-illegal-occupants-in-palawan-protected-area-issued-as-early-as-march/#:~:text=As%20early%20as%20March%20this,in%20Taytay%20to wn%2C%20Palawan%20province.>

Dhaliwal, Bani The Canopy Project, The Importance of Mangrove Forests (February 14, 2023), <https://www.earthday.org/the-importance-of-mangrove-forests/#:~:text=Mangroves%20improve%20and%20maintain%20local,an d%20the%20species%20within%20them>

Gozum, Iya DENR to Stop Issuing Permits, Agreements for Use of Protected Areas (October 18, 2023), <https://www.rappler.com/philippines/denr-stop-issuing-permits-instruments-protected-areas/#:~:text=DENR%20to%20stop%20issuing%20permits%2C%20agreements%20for%20use%20of%20protected%20areas,-Oct%2018%2C%202023&text=MANILA%2C%20Philippines%20%E2%80%93%20The%20Department%20of,said%20on%20Wednesday%2C%20October%2018>

Lu, Brian James J. Fresh Viewpoints: A New Perspective, The Importance of Philippine Mangroves (May 31, 2024), <https://www.pna.gov.ph/opinion/pieces/888-the-importance-of-philippine-mangroves>

Palawan News, DENR Warns Public Against Fraudulent Land Transactions in MIMAROPA (May 20, 2022), <https://palawan-news.com/denr-warns-public-against-fraudulent-land-transactions-in-mimaropa/#:~:text=DENR%20warns%20public%20against%20fraudulent%20land%20transactions%20in%20MIMAROPA,By&text=The%20environment%20department%20in%20MIMAROPA,forest%20and%20other%20government%20lands>

The Philippines Today, What is a Bakawan? (July 4, 2019), <https://thephilippinestoday.com/what-is-a-bakawan/>



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LUMAD-LABAN: AN EXPOSITION AGAINST CORPORATIONS ON ENVIRONMENTAL DESTRUCTION

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Introduction

The term 'Lumad,' meaning 'born of the earth' or 'homegrown,' refers to the indigenous peoples of Mindanao, a region in the southern Philippines. This term collectively describes 15 to 18 distinct ethnolinguistic groups that have maintained their indigenous beliefs, remaining outside the influences of Islam and Christianity. Since 2010, military and paramilitary organizations in Mindanao have extrajudicially assassinated over 60 indigenous leaders.¹ Over 40,000 Lumad have been uprooted by the attacks, which nearly invariably occur before plantations and mining operations are established, according to the human rights organization Karapatan.²

The Lumads face a dire crisis where violence, displacement, and political marginalization converge to threaten their very existence. Violence from the Philippine military, corporate armies, and rebel groups results in brutal extrajudicial killings, fostering a culture of pervasive fear and forcing many Lumad to flee their ancestral lands. This displacement shatters their livelihoods, disrupts their education, and severs their connection to deeply rooted cultural practices. Further compounding their plight, the Lumad are excluded from political processes, leaving them voiceless and vulnerable. Most tragically, this relentless cycle culminates in the destruction of their environment. Large-scale mining and logging operations, often accompanied by heavy militarization, ravage their forests, rivers, and natural resources. This environmental annihilation is the most profound blow, erasing not just their means of survival but the very essence of their cultural identity, leaving their lives and heritage hanging by a thread.

Statement of the Problem

The operations of mining companies and enterprises pose a serious threat to the Lumad people's ancestral lands. Furthermore, environmental laws and regulatory frameworks that do not adequately safeguard these lands and the indigenous communities that live on them worsen this problem. The Lumad people's livelihoods and cultural legacy are under grave danger due to the displacement, which also poses serious environmental hazards that compromise local ecosystems' resilience and biodiversity.

¹ Espina-Varona, Inday, Lumad Children: War Targets in Ph's Last Frontiers, indayvarona (October 24, 2015), <https://indayvarona.wordpress.com/2015/10/24/lumad-children-war-targets-in-phs-last-frontier>

² *Id.*

Lumads and Ancestral Domain

Lumads face loss of land due to land grabbing and exploitation. Some communities have been forced out of their lands for resisting encroachment by mining, logging, and energy companies.¹ The Lumad people have a very strong sense of ownership over their ancestral lands, which they view as private community property with the right of community members to exploit any unclaimed area. These ancestral lands include fields under cultivation, forests, rivers, hunting places, and the mineral resources underneath.²

The Indigenous Peoples Rights Act (IPRA)³ was a legislative step towards protecting their rights, yet its implementation faces challenges due to the prevailing power dynamics and economic interests at play. The conflict is not merely a matter of land rights; it is also a clash of cultural values and a question of how to reconcile economic development with the preservation of indigenous ways of life. The Lumad struggle reflects a broader issue faced by indigenous communities globally: how to navigate the pressures of modernization and globalization while retaining a connection to their heritage and land.

Land, Laws and Loss

The foundation of Moro customary law, or *adat*, is the idea that total land ownership is unattainable. Islamic doctrine holds that God is the owner of the entire earth and that mankind are only His stewards. Therefore, rather than being based on ownership, the Moros' concept of land ownership was the right to its produce.

Numerous land regulations enacted by the American colonial authority allowed for the eviction of Moro and other indigenous people from all across the Philippines.⁴ These rules limited the amount of land that people and businesses could purchase and mandated land registration through titles. Unregistered land became

¹ Rina Chandran, Thomson Reuters Foundation. Driven from Home, Philippine Indigenous People Long for Their Land. ABS (June 2, 2024), <https://news.abs-cbn.com/focus/04/19/18/driven-from-home-philippine%20indigenous-people-long-for-their-land>

² Rodil, B. R., The Lumad and Moro of Mindanao. London: Minority Rights Group (1993), <https://minorityrights.org/app/uploads/2024/04/the-lumad-and-moro-of-mindanao.pdf>

³ An Act to Recognize, Protect and Promote The Rights Of Indigenous Cultural Communities/Indigenous Peoples, Creating A National Commission On Indigenous Peoples, Establishing Implementing Mechanisms, Appropriating Funds Therefor, And For Other Purposes, Republic Act No. 8371 (October 29, 1997) (Phil.).

⁴ Abaya-Ulindang, Faina C., Land Resettlement Policies in Colonial and Post Colonial Philippines: Key to Current Insurgencies and Climate Disasters in its Southern Mindanao Island (May 2015), https://www.iss.nl/sites/corporate/files/CMCP_54-Abaya-Ulindang.pdf

available for American and Filipino citizens to explore, inhabit, and buy.⁵

Many Moro people either chose not to register their lands or refused to do so because at first, few of them had the English literacy necessary to understand the land registration procedure. However, a few Moro leaders took advantage of the new legislation to register large areas under their own names, laying the groundwork for the Moro landed elite of today.⁶

The Lumad people are excluded from the previous political structure. They are unable to obtain the assistance they require to defend their ancestral lands due to language problems. The legislature ought to guarantee that every person has a voice and be more inclusive. Perhaps one of the reasons why educated indigenous people are marginalized is that they are afraid of standing out for their rights. This is an academic onslaught.

The large arable lands of Mindanao occupied by the Moro and Lumad people, which were sold or leased to settlers and plantation firms, were ignored by U.S. authorities, who acknowledged land claims from the Spanish era in the lowlands of Luzon and Visayas. Seven agricultural colonies were established by the colonial government between 1913 and 1917, with the intention of fostering "good working relations" between Christian settlers and indigenous Muslims. However, the true objective was to disperse peasant unrest in Luzon and drive out troublemakers from the northern and central Philippines. The native residents had ten (later reduced to eight) hectares of land, while Christian newcomers obtained bigger tracts—16 hectares. Any Moro opposition was put down by the Philippine Constabulary, which was primarily Christian.⁷ As a result, the Moros and Lumads on their own ancestral lands were reduced to destitute squatters.⁸

Whereas, this is incongruent with the court decision in the landmark case of *Mateo Cariño v. Insular Govt.*,⁹ Mateo Cariño, an Igorot from Benguet, adamantly defended his family's ownership of property they had farmed and lived on for more than fifty years. The court recognized Cariño's long-standing possession, albeit in the absence of official registration, and based its ruling on Spanish law. It made clear that ownership alone might be sufficient in certain cases, especially if it was supported by continuous use for long stretches of time.

⁵ *Id.*

⁶ *Id.*

⁷ Abaya-Ulindang, Faina C., Land Resettlement Policies in Colonial and Post Colonial Philippines: Key to Current Insurgencies and Climate Disasters in Its Southern Mindanao Island (May 2015), https://www.iss.nl/sites/corporate/files/CMCP_54-Abaya-Ulindang.pdf

⁸ Ocampo, Satur C., The Moro Land Problem., Philstar.com., <https://www.philstar.com/opinion/2011/03/12/664993/moro-land-problem>

⁹ *Carino v. Insular Government*, G.R. No. 2869, (March 25, 1907) (Phil.).

However, mining firms have aggressively expanded their activities in the current economic environment, frequently at the expense of vulnerable populations. In "Costly Dirty Money-Making Schemes,"¹⁰ Docena details the serious encroachment of mining firms on the Lumad people's traditional lands in the Philippines. According to the book, up to half of the places that these corporations have claimed coincide with regions that are thought to be ancestral grounds. Notwithstanding regulatory safeguards that demand approval from native populations prior to commencing mining operations, corporations frequently exploit their assets and clout to circumvent these laws.

One prominent offender named Nickel Asia Corporation is the wealthy Zamora family.¹¹ Through its four subsidiaries, which include operating interests in Taganito, Cagdianao, Tagana-an, and South Dinagat in Surigao del Norte—some of which are portions of the Lumad ancestral lands—this corporation dominates the nickel mining sector in the area. Philip T. Ang, the vice chairman of one of these subsidiaries, is one of the wealthiest people in the nation. Furthermore, a Japanese consortium led by Sumitomo Metal Mining Corporation, the country's top nickel producer, owns the majority of Coral Bay Nickel Corporation, in which Nickel Asia also has minority interests.¹² These subsidiaries exceeded the Department of Environment and Natural Resources' (DENR) budget in 2007 with a total net income of around ₱15 billion. Moreover, a sizeable share of the projected ₱0.3 billion to ₱1.7 billion in annual profits from their Mindanao project is expected to be earned by the Zamoras and their CDM venture partners.¹³ This significant financial impact highlights their power and demonstrates how the intricate network of interests within the mining industry exacerbates the Lumad people's predicament by causing environmental damage, community violence, and relocation.

The question is: How do they pervert the law?

Despite the legal protections enshrined in the Indigenous Peoples' Rights Act (R.A. No. 8371)¹⁴ and the Philippine Mining Act (R.A. No. 7942),¹⁵ companies

¹⁰ Docena, Herbert, *Costly dirty money-making schemes: The Clean Development Mechanism Projects in the Philippines*. Quezon City, Philippines: Focus on the Global South, Philippines Program (June 2010), <https://focusweb.org/wp-content/uploads/2017/04/CDM-Web-version-lowres.pdf>

¹¹ Lopez, Antonio, *The Troubleshooters*, Asiaweek (March 19, 1999), <http://www.cgi.cnn.com/ASIANOW/asiaweek/99/0319/nat1.html>

¹² *Id.*

¹³ *Id.*

¹⁴ An Act to Recognize, Protect And Promote The Rights Of Indigenous Cultural Communities/Indigenous Peoples, Creating A National Commission On Indigenous Peoples, Establishing Implementing Mechanisms, Appropriating Funds Therefor, And For Other Purposes, Republic Act No. 8371 (October 29, 1997) (Phil.).

¹⁵ An Act Instituting A New System of Mineral Resources Exploration, Development, Utilization, And Conservation, Republic Act No. 7942, (March 3, 1995) (Phil.).

like Nickel Asia Corporation often exploit loopholes and exert undue influence to bypass these regulations. R.A. No. 8371 mandates that any project affecting indigenous peoples' ancestral domains must secure their free, prior, and informed consent (FPIC). However, these companies frequently circumvent this requirement by leveraging their substantial financial resources and “political connections.”¹⁶ They manipulate the consent process, often securing agreements through coercion, misinformation, or by negotiating with illegitimate representatives not recognized by the community.

Additionally, the Philippine Mining Act (R.A. No. 7942), while promoting the rational exploration, development, and utilization of mineral resources, includes provisions that are supposed to ensure environmental protection and the equitable sharing of benefits with affected communities. Yet, enforcement is weak, and regulatory capture is rampant. Mining companies often neglect environmental safeguards, leading to severe ecological degradation, such as deforestation, water contamination, and habitat destruction, all of which are expressly against the protective measures outlined in RA 8371.¹⁷

The landmark decision in *La Bugal-B'laan Tribal Association, Inc. v. Ramos*, highlight these tensions. The court ruling interprets the constitutional provision regarding agreements with foreign-owned corporations for mineral exploration, development, and utilization (EDU). It rejects the notion that such agreements are restricted solely to technical or financial assistance, emphasizing that the term “involving” suggests a broader scope. The ruling underscores the State's ownership of mineral resources and its mandate for full control and supervision over their EDU. While foreign assistance in financial and technical aspects is allowed, the State must retain ultimate control to protect its interests. Upholding this control standard, the court deems the Philippine Mining Law and related regulations constitutional insofar as they pertain to financial and technical agreements.¹⁸ Despite the legal frameworks intended to protect them, the Lumad and other indigenous groups remain vulnerable to exploitation and environmental destruction due to the pervasive influence of powerful mining interests. The intricate web of corporate power and political leverage underscores the urgent need for robust enforcement of existing laws and genuine respect for the rights and welfare of indigenous peoples.

Conclusion

The suffering of the Lumad people in Mindanao serves as evidence that the

¹⁶ FPIC: A Shield or threat to Indigenous Peoples' Rights? (Accessed June 11, 2024), <https://minorityrights.org/app/uploads/2024/04/the-lumad-and-moro-of-mindanao.pdf>

¹⁷ *Id.*

¹⁸ *La Bugal-B'laan Tribal Association, Inc. v. Ramos*. G.R. No. 127882, <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/46804>

exploitation of indigenous populations for economic development is a global issue that also affects the Philippines. They are now uprooted from their native lands due to rapid modernization and industrialization, making them vulnerable to exploitation by strong interests. Respecting the rights of indigenous peoples who have lived on the land for many years is just as vital as economic advancement.

While it is impossible to stop Mindanao's development, there are things that can be done to help the Lumad population. Giving them access to tools like livelihood programs, legal assistance, and scholarships can lessen their difficulties and keep them safe from exploitation. Their claims to the remaining ancestral lands must also be honored in order to prevent foreign interests from further encroaching.

Furthermore, given the extensive ecological harm that has been done to Mindanao over time, protecting the ecosystem is of the utmost importance. Excessive structural development could worsen the state of the environment. An act of reconciliation that preserves the environment and pays tribute to the island's indigenous inhabitants is giving back what is left of the land to the Lumad people. In the interest of economic advancement, we respect the values of justice and sustainability by putting the rights and welfare of the Lumad community first and protecting their natural environment and cultural legacy.

Bibliography

Statutes

An Act Instituting A New System of Mineral Resources Exploration, Development, Utilization, And Conservation, Republic Act No. 7942, (March 3, 1995) (Phil.).

An Act to Recognize, Protect and Promote the Rights of Indigenous Cultural Communities/Indigenous Peoples, Creating A National Commission on Indigenous Peoples, Establishing Implementing Mechanisms, Appropriating Funds Therefor, And For Other Purposes, Republic Act No. 8371 (October 29, 1997) (Phil.).

Jurisprudence

Carino v. Insular Government, G.R. No. 2869, (March 25, 1907) (Phil.).

La Bugal-B'laan Tribal Association, Inc. v. Ramos. G.R. No. 127882, (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/46804>

Online Sources

Abaya-Ulindang, Faina C., Land Resettlement Policies in Colonial and Post Colonial Philippines: Key to Current Insurgencies and Climate Disasters in its Southern Mindanao Island (May 2015), https://www.iss.nl/sites/corporate/files/CMCP_54-Abaya-Ulindang.pdf

Docena, Herbert, Costly dirty money-making schemes: The Clean Development Mechanism Projects in the Philippines. Quezon City, Philippines: Focus on the Global South, Philippines Program (June 2010), <https://focusweb.org/wp-content/uploads/2017/04/CDM-Web-version-lowres.pdf>

Espina-Varona, Inday, Lumad Children : War Targets in Ph's Last Frontiers, indayvarona (October 24, 2015), <https://indayvarona.wordpress.com/2015/10/24/lumad-children-war-targets-in-phs-last-frontier>

FPIC: A Shield or threat to Indigenous Peoples' Rights? (Accessed June 11, 2024), <https://minorityrights.org/app/uploads/2024/04/the-lumad-and-moro-of-mindanao.pdf>

Lopez, Antonio, The Troubleshooters, Asiaweek (March 19, 1999), <http://www.cgi.cnn.com/ASIANOW/asiaweek/99/0319/nat1.html>

Ocampo, Satur C., The Moro Land Problem., Philstar.com., <https://www.philstar.com/opinion/2011/03/12/664993/moro-land-problem>

Rina Chandran, Thomson Reuters Foundation. Driven from Home, Philippine Indigenous People Long for Their Land. ABS (June 2, 2024)., <https://news.abs-cbn.com/focus/04/19/18/driven-from-home-philippine%20indigenous-people-long-for-their-land>

Rodil, B. R., The Lumad and Moro of Mindanao. London: Minority Rights Group (1993), <https://minorityrights.org/app/uploads/2024/04/the-lumad-and-moro-of-mindanao.pdf>



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MANGAYON SHOT DEAD: R.A. NO. 9147 AND WILDLIFE JUSTICE

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The Philippines is one of 18 mega-biodiverse countries of the world, containing two-thirds of the earth's biodiversity and between 70% and 80% of the world's plant and animal species.¹ With five percent of the world's plant species, the Philippines is home to a fifth of all plant species. The nation ranks fourth in the world for bird endemism, but its species endemism is quite high, encompassing at least 25 genera of plants and 49% of terrestrial fauna.²

With at least 700 vulnerable species, the Philippines is among the top worldwide conservation regions and one of the world's biodiversity hotspots. 42 species of land mammals, 127 species of birds, 24 species of reptiles, and 14 species of amphibians are included in the national list of vulnerable faunal species, which was created in 2004.³

In terms of marine life, there are at least 3,214 species of fish in the Philippines, with roughly 121 being endemic and 76 being threatened. A nationwide list of threatened plant species was established in 2007 by an administrative order from the Department of Environment and Natural Resources. This list included 64 additional threatened species, 187 endangered species, 99 severely endangered species, and 176 vulnerable species.⁴ Considering these data, it is arguably that the Philippines is a God-gifted country not only by culture but by nature.

However, despite these God-given gifts, man tends to stand by its nature of being abusive. Recently, a Philippine eagle was struck down from the heights. The Philippine Eagle Foundation (PEF) reported that on the afternoon of July 8, 2024, PEF received a report from the DENR-PENRO Davao de Oro about an injured Philippine Eagle. The eagle was under the custody of the Philippine Army patrol base in Barangay Mangayon, Compostela. According to Sgt. Badon, the patrol led by 2nd Lt. Cabajar discovered the weakened eagle with an apparent left-wing injury on the ground in Bermuda, a part of Barangay Mangayon.⁵ The eagle was named "Mangayon."

¹ Convention on Biological Diversity. (n.d.), <https://www.cbd.int/countries/profile?country=ph>

² *Id.*

³ *Id.*

⁴ *Id.*

⁵ Philippine Eagle Foundation, Another Philippine Eagle killed in Mangayon, Compostela Valley, Davao de Oro (2024),

Mangayon underwent physical and medical examinations, revealing several injuries, including shattered bones in the left tarsal joints. The attending veterinarian, Dr. Bayani Vandenbroeck, discovered an entry wound in the left tarsal area of the eagle's wing, which exited through the opposite side and extended to the left keel area of Mangayon's wing.⁶

The procedure also revealed that Mangayon began discharging foul fluids and solid materials from his mouth, which were identified as an undigested half-cooked chick. The primary cause of death was severe blood loss from the wounds in the left wing, with a high probability of sepsis stemming from the injuries.⁷ Clearly, the bird was intentionally shot as a product of poaching. As per reports, the identity of the suspect is still unknown.

Mangayon is the fourth rescued eagle in 2024 to have fallen victim to deliberate shooting apparent in their gunshot wounds. Apart from this, there are other cases of deliberate shootings of these endangered species. The number of the shootings may indicate of the lax implementation of environmental laws in the Philippines.

Wildlife Resources Conservation and Protection Act (R.A. No. 9147)

The Wildlife Resources Conservation and Protection Act of 2001, also commonly known as R.A. No. 9147, provides for the conservation, preservation and protection of wildlife species and their habitats, in order to preserve and encourage ecological balance and biological diversity.⁸ It provides, furthermore, for the control and supervision of wildlife capture, hunting and trade; finally, it provides for supporting and promote scientific research on the protection of biodiversity.⁹

Some of the key provisions of R.A. No. 9147 include the (1) Regulation of Wildlife Trade, to control the trade of wildlife species, both domestically and internationally, in preventing illegal trafficking and over-exploitation of animal species; (2) Protection of Habitat, to designate critical habitats and areas which are to be protected in ensuring the survival of wildlife species in their natural environments; (3) Education and research, the act encourages scientific research on

<https://www.philippineeaglefoundation.org/post/philippine-eagle-rescued-in-mangayon-compostela-valley-davao-de-oro-declared-dead>

⁶ Patumbon, Rojean Grace G., Another PH eagle dies of gunshot wounds (July 14, 2024), [https://www.sunstar.com.ph/davao/another-ph-eagle-dies-of-gunshot-wounds#:~:text=THE%20Philippine%20Eagle%20Foundation%20\(PEF,%2C%20on%20July%208%2C%202024](https://www.sunstar.com.ph/davao/another-ph-eagle-dies-of-gunshot-wounds#:~:text=THE%20Philippine%20Eagle%20Foundation%20(PEF,%2C%20on%20July%208%2C%202024)

⁷ *Id.*

⁸ Wildlife Resources Conservation and Protection Act, Rep. Act No. 9147 (July 30, 2001) (Phil.), <https://www.officialgazette.gov.ph/2001/07/30/republic-act-no-9417/>

⁹ *Id.*

wildlife, promote public awareness and education about wildlife conservation; and (4). the Penalties for the violations and the violators of the act.

Research Studies on Implementation of R.A. No. 9147

Implementation of R.A. No. 9147 Among the Coastal Communities of the Municipality of Manay, Davao Oriental

The research was conducted on the Office of the MENRO in the Municipality of Manay. The researcher did additional research and focus group discussion on the different activities being made by the agency for the implementation of the law specifically Article 3 Sec. 27 of the R.A. No. 9147 along the coastal community namely; Central, San Ignacio, Holy Cross, and Zaragoza all in the Municipality of Manay, Davao Oriental.¹⁰

The summary of the findings shows important problems that need to be addressed in implementing R.A. No. 9147. Specifically, it showed that:

- 1) There are limited activities such as advocacies in the implementation of R.A. No. 9147.¹¹ The result was that the Information Education Campaign (IEC) was not enough as it was done only on a quarterly basis. The researcher asserts that the community's awareness of such laws can contribute to the conservation and protection of marine wildlife;
- 2) There are low-observation checkpoints that are done in the municipality which, according to the researcher, “can be very alarming and that violators can escape the municipality discreetly with the wildlife with them.”¹²
- 3) There are less attendees in the IEC programs. As per the study, there were 63 or 74.12% said that they have attended the IEC done especially on the threatened species.¹³

Implementation of R.A. No. 9147 “Wildlife Resources Conservation and Protection Act

The research was to evaluate the implementation of the R.A. No. 9147 “Wildlife Resources Conservation Act” in the jurisdiction of DENR-CENRO

¹⁰ Belena, Jan Michael L., The Implementation of RA 9147 Among the Coastal Communities of The Municipality of Manay (n.d.), <https://ejournals.ph/article.php?Id=15473>

¹¹ *Id.*

¹² *Id.*

¹³ *Id.*

Guiguinto, Bulacan. The study shows issues hindering R.A. No. 9147 its effective implementation. According to the study, these are education, engineering, and enforcement.¹⁴

Education

“The lack of information materials regarding R.A. No. 9147 such as what are the ban species for trading and what are the prohibited acts is a significant information gap... The primary goal of IEC materials is to inspire and to educate people about the importance of the conservation and protection of wildlife and for better understanding of what the law prohibits in a more comprehensive way.”¹⁵

Engineering

“...the lack of wildlife rescue centers, and lack of wildlife traffic monitoring units. At present, only the airports of Davao, Zamboanga, Manila, Cebu and Subic have monitoring units, while only the port of Manila has a monitoring unit. For this reason, the law cannot be fully implemented because rescue centers and monitoring units are essential to perform the process of achieving the objectives of the law.”¹⁶

Enforcement

“When it comes to the aspect of enforcing the law, there are significant gaps that we need to look upon. One is the lack of resources for wildlife protection... Another is the absence or lack of institutional arrangements with other countries to combat entry and exit of wildlife and / or exotic wildlife species... the lack of checkpoints and monitoring units at airports and ports hinders the effective implementation of the law.”¹⁷

Challenges and Opportunities in Philippine Wildlife Conservation

Despite the efforts of the government in implementing R.A. No. 9147, the studies, and the recent incidents prove that these efforts are not enough. The full implementation of the law still remains to be challenged. Indeed, public awareness campaigns and Community-Based Conservation Programs have been launched to raise public awareness about the importance of wildlife conservation. However, they

¹⁴ Cruz, Angelique G., Implementation of Republic Act 9147 “Wildlife Resources Conservation and Protection Act. (n.d.), <https://www.studocu.com/ph/document/san-beda-university/natural-resources-and-environmental-laws/implementation-of-republic-act-9147-wild/71046612>

¹⁵ *Id.*

¹⁶ *Id.*

¹⁷ *Id.*

are proven to be insufficient. The Philippines lacks the resources for the full implementation of conservation programs and funds for the enforcing agencies.

Nevertheless, there are still rooms for improvements. The Philippine government may opt to (1) strengthen institutional capacity, that is, to enhance the capabilities of institutions responsible for wildlife conservation such as the Department of Environment and Natural Resources (DENR); (2) engage local communities, that is by promoting community-based conservation initiatives; (3) leveraging technology, that is by utilizing technology such as the internet for monitoring, enforcement, and public awareness campaigns; and (3) enhance international collaboration, engage with global partners for technical and financial support.

Likewise, according to Environment Secretary Roy A. Cimatu, a more stringent law is needed to curb illegal wildlife trade as culprits are not deterred by the punishments provided under R.A. No. 9147 or the Wildlife Resources Conservation and Protection Act of 2001.¹⁸ This calls for a strict implementation of the penalties as provided by the republic act.

The survival of human societies and the health of our planet depend on the preservation of wildlife. Every species in our intricately interrelated natural environment contributes to the preservation of equilibrium. By preserving wildlife, we help ecosystems continue to function efficiently and continue to supply us with necessities like clean water, air, and fertile soil. The extinction of a single species can have a cascading effect on ecosystem resilience by upsetting food systems. In addition to preserving plants and animals, wildlife conservation ensures a stable ecosystem for present and future generations. Additionally, there are significant cultural, aesthetic, and financial benefits to wildlife preservation. Numerous groups, particularly those of indigenous descent, have long-standing relationships to the local fauna and natural environments. Their cultural identities and customs are shaped by these interactions. Furthermore, wildlife tourism contributes significantly to the economy by creating jobs and advancing sustainable development. The wonder and diversity of animals arouse admiration and respect for the natural world, which strengthens one's sense of duty to preserve it.

It is essential to protect species and their habitats in the fight against climate change. As carbon sinks, healthy ecosystems contribute to reducing global warming. Protecting these habitats and stopping actions like illegal trading and deforestation that worsen climate change require laws like the Wildlife Resources Conservation and Protection Act (R.A. No. 9147). Our dedication to a sustainable future in which people and the natural world coexist peacefully is reflected in our efforts to safeguard

¹⁸ DENR, Cimatu: Stricter law needed to curb illegal wildlife trade (July 19, 2020), <https://denr.gov.ph/news-events/cimatu-stricter-law-needed-to-curb-illegal-wildlife-trade/>

wildlife. We make an investment in a resilient, thriving planet for all living things when we value and conserve wildlife.

Bibliography

Statutes

Wildlife Resources Conservation and Protection Act, Rep. Act No . 9147 (Jul. 30, 2001) (Phil.),
<https://www.officialgazette.gov.ph/2001/07/30/republic-act-no-9417/>

Online Sources

Belena, Jan Michael L., The Implementation of RA 9147 Among the Coastal Communities of The Municipality of Manay (n.d.),
<https://ejournals.ph/article.php?id=15473>

Convention on Biological Diversity (n.d.),
<https://www.cbd.int/countries/profile?country=ph>

Cruz, Angelique G., Implementation of Republic Act 9147 “Wildlife Resources Conservation And Protection Act (n.d.),
<https://www.studocu.com/ph/document/san-beda-university/natural-resources-and-environmental-laws/implementation-of-republic-act-9147-wild/71046612>

DENR Cimat: Stricter law needed to curb illegal wildlife trade. (July 19, 2020),
<https://denr.gov.ph/news-events/cimatu-stricter-law-needed-to-curb-illegal-wildlife-trade/>

Patumbon, Rojean Grace G. Another PH eagle dies of gunshot wounds. (July 14, 2024),
[https://www.sunstar.com.ph/davao/another-ph-eagle-dies-of-gunshot-wounds#:~:text=THE%20Philippine%20Eagle%20Foundation%20\(PEF,%2C%20on%20July%208%2C%202024](https://www.sunstar.com.ph/davao/another-ph-eagle-dies-of-gunshot-wounds#:~:text=THE%20Philippine%20Eagle%20Foundation%20(PEF,%2C%20on%20July%208%2C%202024)

Philippine Eagle Foundation. (2024),
<https://www.philippineeaglefoundation.org/post/philippine-eagle-rescued-in-mangayon-compostela-valley-davao-de-oro-declared-dead>



Photos by: Mary Gemzel M. Alcazar

“KONTRABIDA ANG BIDA”: AN AUTHORITY DISCREPANT WITH THE LOCAL GOVERNMENT CODE OF 1991

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The environment and ecological health play a pivotal role in the Philippines, being significant to other facets of the country in terms of disaster mitigation and climate action, but most especially in economic growth through sustainable tourism. Branded as an archipelagic country that boasts more than 7,000 islands, the best beaches, and breathtaking mountains with rare endemic wildlife, it is an understatement to claim the imperative role of the country's environmental condition.

Many parts of the Philippines have been developed through sustainable tourism in relation to the environment. The promotion of the regional economy through creating an alternative source of income has generated foreign exchange, hence benefiting the primary industry. The Philippine national government, local authorities, and private capital in various sectors are involved in tourism development¹ and thus render several legislations to endeavor its protection for the national interest. Nevertheless, these statutes are not far from being imperfect, hampering other laws and bringing socio-political implications. This includes the proposal for the establishment of the Boracay Island Development Authority.

House Bill No. 9826

Among the many legislations proposed in the Philippines that have the environment as its focal point is House Bill No. 9826 – An Act Creating the Boracay Island Development Authority (BIDA)², Defining its Powers and Functions, and Providing Funds Therefor. The said bill was proposed before the House of Representatives, co-authored by the likes of Davao Representative Paolo Duterte, Anti-Crime and Terrorism Community Involvement and Support Partylist (ACT-CIS) Representative Eric Go Yap, and Marino Representative Sandro Gonzales among many others.

First introduced to Congress in 2021, the proposed BIDA Act was a priority bill by the former President, Rodrigo Duterte, aimed at developing responsible tourism through the participation of local communities, including

¹ Torii, T., Masaaki, S., Mizuho, K., & Kitolelei, J., Tourism Development in Boracay Island and its Impacts on the Local Communities. ResearchGate (July 2011), https://www.researchgate.net/publication/279861756_Tourism_Development_in_Boracay_Island_and_its_Impacts_on_the_Local_Communities

² House Bill No. 9826 – An Act Creating the Boracay Island Development Authority (BIDA), <https://legacy.senate.gov.ph/lisdata/36300327311.pdf>

indigenous peoples in conserving biophysical and cultural diversity and through the creation of the proposed government-owned and controlled corporation to be known as BIDA.³ The said bill is positioned to carry out the declared policy for the sustainable development of Boracay Island, with the utilization of its natural resources and all development activities therein to be in strict conformity with all the laws on the utilization, conservation, preservation, and protection of natural resources and the environment.⁴

Section 3 of House Bill No. 9826 enumerates its development framework which includes:

1. Sustainable and balanced development;
2. Sustainable domestic and foreign investments;
3. Sustainable employment; and
4. Sustainable growth with environmental protection.

The said bill also emphasized the formulation of a comprehensive long-term Boracay Action Plan (BAP) for adoption and implementation in coordination with concerned national government agencies, as well as the assurance of prompt and accurate collection, analysis, and dissemination of all data that will provide for environmental protection and management.

Boracay Island's Struggle for Sustainable Tourism

Unlike many environmental laws in the Philippines that have a broader scope, the bill creating the Boracay Island Development Authority (BIDA) is, by all accounts, concentrated on Boracay Island - a small island that is part of the Municipality of Malay, Aklan, located at the northwest portion of Panay in Western Visayas.

Boracay Island is one of the famous local tourist destinations in the Philippines, with white sandy beaches stretching for four kilometers. The island is internationally famed for its beaches, thus bolstering the country's tourism. In 2022 alone, at least 1.75 million people visited Boracay according to data from the Department of Tourism (DOT). Domestic tourists, including foreigners living in the

³ Cervantes, F.M., Bill creating Boracay Island Dev't Authority gets House nod. Philippine National Authority (August 4, 2021), <https://www.pna.gov.ph/articles/1149377>

⁴ 18th Congress of the Philippines, 3rd Regular Session, H. No. 9826, <https://legacy.senate.gov.ph/lisdata/36300327311.pdf>

country, comprised the bulk of visitors at 1.59 million. Foreign tourists, on the other hand, reached 132,687 with most coming from South Korea and the United States.⁵

According to the Philippine Statistics Authority (PSA), data from the same year showed a spike in Aklan's significant economic growth due to Boracay's tourism aptitude. The PSA showed Aklan posted a growth of 22.5 percent, proving its imperative role not only in its municipality but also in the province as a whole.

Even with data supporting the island as a tourism hotspot, Boracay is also paralleled with serious environmental degradation issues due to the influx of tourists and poor implementation of policies for the protection of the island's natural resources. With tourism development, changes are evident in Boracay Island in terms of sand erosion, coastal pollution, and environmental changes such as the sporadic felling of timber. A wastewater treatment facility was built as a joint venture with Japan, and this facility was intended to keep pace with waste disposed of from various tourism operations. However, wastewater was not properly treated, and evidence of sulfide spilled over to the shorelines. This has resulted in the sand having a strange smell and implications for the future to remain negative.⁶

The Dire Enactment of Proclamation No. 475 and Its Drastic Changes

Due to the apparent environmental woes the island has undergone, several policies and interventions have already been made. Among the significant interventions by far that the government has imposed was Proclamation No. 475, issued on April 26, 2018, which declared a State of Calamity and temporary closure of Boracay Island to address environmental concerns, protect public health, and rehabilitate the island due to various environmental violations and degradation, including a high concentration of fecal coliform, insufficient sewer lines, illegal discharge of untreated wastewater, degradation of coral reefs, excessive solid waste generation, and illegal encroachment of structures on wetlands.⁷ The said proclamation is in pursuant to Republic Act No. 10121 (R.A. No. 10121) or the Philippine Disaster Risk Reduction and Management Act of 2010.

⁵ Inquirer.Net, DOT: Boracay welcomed 1.75 million tourists in 2022 (Jan. 27, 2023), <https://newsinfo.inquirer.net/1721727/dot-boracay-welcomed-1-75-million-Tourists-in-2022>

⁶ Torii, T., Masaaki, S., Mizuho, K., & Kitolelei, J., Tourism Development in Boracay Island and its Impacts on the Local Communities. ResearchGate (July 2011) https://www.researchgate.net/publication/279861756_Tourism_Development_in_Boracay_Island_and_its_Impacts_on_the_Local_Communities

⁷ Office of the President, Declaring A State Of Calamity In The Barangays Of Balabag, Manoc-Manoc And Yapak (Island Of Boracay) In The Municipality Of Malay, Aklan, And Temporary Closure of The Island As A Tourist Destination, Proclamation No. 475 s. 2018 (Apr. 26, 2018) (Phil.), <https://www.officialgazette.gov.ph/2018/04/26/proclamation-no-475-s-2018/>

Proclamation No. 475 was faced with several political and legal challenges, primarily on matters of the constitutional powers of the president to impose the said policy.

In the case of *Zabal v. Duterte*,⁸ the proclamation was argued to be an invalid exercise of legislative powers. The petitioners in the case posited that its issuance is, in truth, a law-making exercise since the proclamation imposed a restriction on the right to travel and therefore substantially altered the relationship between the state and its people by increasing the former's power over the latter. Simply stated, petitioners posited that Proclamation No. 475 partook of a law, the issuance of which is not vested in the President. As such, Proclamation No. 475 must be struck down for being the product of an invalid exercise of legislative power.

Furthermore, petitioners argued that Proclamation No. 475 is unconstitutional for infringing on the constitutional rights to travel in reference to the Philippine 1987 Constitution's right to travel provision and due process, which was averred to cover property rights. These rights include the right to work and earn a living. With the proclamation restricting the entry of tourists and non-residents into the island, petitioners claim that they, as well as all others who work, do business, or earn a living on the island, were deprived of the source of their livelihood as a result thereof.⁹

Despite the challenges, the Supreme Court held in favor of the respondent, declaring the proclamation a valid exercise of police power by the executive branch of government, and that there is an absence of a clear showing of grave abuse of discretion, unreasonableness, arbitrariness, or oppressiveness. The Court, therefore, chose to not disturb the executive determination that the closure of Boracay Island was necessitated by the environmental circumstances.

Through the above-mentioned policy, the Boracay Inter-Agency Task Force (BIATF) was established to evaluate the environmental state of Boracay Island and investigate possible violations of existing environmental and health laws, rules, and regulations. The task force was composed of the DOT, the Department of Environment and Natural Resources (DENR), and the Department of the Interior and Local Government (DILG).

From their investigation, the BIATF found several violations of environmental laws that include the lack of proper waste disposal by most commercial establishments and residences on the island, and illegal encroachment of structures on wetlands, where 937 identified illegal structures constructed on

⁸ *Zabal v. Duterte*, G.R. No. 238467, (February 12, 2019) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/64872>

⁹ *Id.*

forestlands and wetlands and 102 illegal structures constructed on areas already classified as easements.

Implications on the Local Government Code of 1991

The bill proposing the establishment of BIDA is not categorically lodged as a statute for environmental protection and preservation alone but also involves taxation and regulation. This is where House Bill No. 9826 becomes a discrepant proposition to existing laws, specifically the Local Government Code of 1991.

Among the features of the BIDA bill is its provision with regard to its collection of taxes. According to Section 10, under the Taxes and Fees of the Boracay Island Development Zone:

(a) Any provision of existing law, rules or regulations to the contrary notwithstanding, all local and national taxes shall be imposed on business establishments operating and registered within the Development Zone: Provided, That the power to impose and collect real estate taxes shall be retained by the Provincial Government of Aklan and the Local Government of Malay, _____ respectively; _____ and _____

(b) The Authority may also impose fees, imposts, and assessments as may be deemed necessary for the sustainability of its operations. (HREP)

Through the aforementioned provision, it is clear that the bill imposes the right of the authority to collect taxes, specifically from business establishments. Meanwhile, in a further provision in Section 20, Relationship of the Authority and Local Government Units, the authority is granted to utilize the collected amount as per its administrative and rulemaking power.

Section 20 (a) provides that:

(a) The provisions of existing laws, rules and regulations to the contrary notwithstanding, the Authority shall exercise administrative powers, rulemaking powers, and shall be authorized to disburse funds, and to require the accreditation of enterprises to ensure compliance with the developmental, management, and protection plans in the Development Zone.

The above-mentioned provisions of the bill are in conflict with existing statutes in relation to taxation imposed upon businesses, as well as the involvement of local government units in utilizing their funds. The Local Government Code of 1991 provides in its Section 129 that local government units (LGUs) have the power to create their own sources of revenue and to levy taxes, fees, and charges subject to the provisions therein, consistent with the basic policy of local autonomy.¹⁰ The aforementioned sources of funds are expressly stated in the said code that they shall be accrued exclusively to the LGUs.

¹⁰ Local Government Code, Rep. Act No.7160 (Phil.).

The fact that a codified statute establishing the taxing power of LGUs is already operational renders the provisions of the BIDA bill, to actually acquire some of those taxing powers, hampering the Local Government Code provision. There is clear discord when a new bill intends to impart an exercise of the same taxing power of LGUs, taking the said units' capability to collect funds for their own. For local operations and policy implementation, the LGUs need the realization of the Local Government Code not only to supply for their own needs but also to empower them when it comes to governance and in providing social services.

Furthermore, based on the data previously stated in relation to the imperative role of Boracay as a source of economic growth, BIDA removes the province's opportunity to make use of its own funds from its own resources, supplied through the island's tourism aptitude. This, consequently, may impact not only the island's economic growth but also that of the province of Aklan.

Additional provisions that establish the LGUs to be the collectors and beneficiaries of taxes include Section 132 that stipulates the generation of revenue be exercised by the LGU and Section 137, which provides for the collection of franchise tax. The franchise tax provision delves into the imposition of taxes on businesses enjoying franchises.¹¹

Despite the scientific findings on the environmental condition of Boracay, there is still a need to ensure that environmental policies adhere to other statutes and observe already guaranteed rights and privileges, such as those designated by the Local Government Code of 1991.

Remedy: Strict Compliance of Environmental Laws

There are already several statutes intended for the protection of the environment as well as the assurance of ecological health, and many of them are very much applicable to Boracay and would aid in its preservation. The real problem is not the absence of laws but implementation.

Though noble as it may be, the BIDA bill is not the only solution to rectifying the environmental degradation the island is already experiencing. It's a matter of executing regulations and properly imposing penalties on environmental law offenders.

An example of a stronger implementation and compliance with a general mandate for environmental protection was Proclamation No. 475. There may have

¹¹ Local Government Code, Rep. Act No.7160 (Phil.).

been affairs surrounding its enactment and implementation, yet the closure became the model for sustainable tourism, and the success of the rehabilitation program was a potential springboard for establishing a culture of environmentally conscious tourism across the country.¹² It paved the way for other rehabilitation programs in other tourist destinations and presented stronger government intervention through upholding existing policies effectively in curtailing prevalent environmental problems. It clearly proves that the BIDA bill is not the only remedy for solving the island's condition.

Ultimately, it is instrumental to retain the strict observance of environmental laws in the province, and governments have already seen this opportunity to maintain compliance with the environment. The Environmental Management Bureau of Region 6 (EMB R6), for one, has already established its pursuit of the said cause, positioning itself to be relentless in its commitment to ensure that commercial establishments on the island comply with environmental laws and regulations.

As part of their ongoing efforts, EMB R6 recently conducted an on-site data validation initiative to update and harmonize its database of establishments on Boracay Island, which aligned with their pledge to collaborate with various stakeholders and government agencies, including the Department of Tourism and the Local Government of Malay, to ensure the protection of Boracay's unique environment.¹³ This process involves sending teams to physically visit each establishment on Boracay Island to collect essential information with the following multifaceted purpose:

- **Data Accuracy:** The primary goal is to ensure that the information on each establishment, including its environmental compliance, is accurate and up-to-date.
- **Collaboration with Tourism and Local Government:** As part of its commitment to harmonize data with other government agencies, EMB R6 is working closely with the Department of Tourism and the Local Government of Malay. By validating the data on commercial establishments, EMB R6 is actively contributing to a shared understanding of the island's environmental landscape.
- **Regulatory Oversight:** By having accurate data, EMB R6 can effectively monitor and regulate the activities of commercial establishments, ensuring that they adhere to environmental laws and regulations.
- **Enhanced Environmental Impact Assessment:** The results of this undertaking serve as a vital planning tool for assessing and mitigating the environmental effects of upcoming projects in Boracay.

¹² Oxford Business Group, The successful rehabilitation of Boracay Island in the Philippines leads the way for environmentally conscious tourism, (November 15, 2022), <https://oxfordbusinessgroup.com/reports/philippines/2019-report/economy/new-standard-the-success-of-the-rehabilitation-of-boracay-island-is-leading-the-way-for-environmentally-conscious-tourism>

¹³ *Id.*

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- Environmental Preservation: Ultimately, the On-Site Data Validation initiative serves the greater purpose of preserving Boracay Island's pristine environment. It allows for better control of pollution and sustainable development, reducing the environmental impact of commercial operations.

Conclusion

Legislation plays a key role in rectifying social ailments and resolving issues that not only concern socio-political conditions but also those of the environment. Among the unfortunate realities of it, however, is the lack of attention towards its implementation despite, in theory, they are already effective enough to solve environmental concerns. Due to the lack of outcomes from any of the existing environmental laws through the observation of the environment itself, many would see that it's the number of statutes enacted, not their implementation, as the cause of the problem. House Bill No. 9826 manifests this perspective.

Boracay Island is a pride of the Philippines' tourism, yet there is also no question in actuality that it has become a potluck of several environmental degradations and environmental law violations. There is no question about the necessity for interventions to solve its pending struggle for sustainable tourism, even after its closure through Proclamation No. 475. Thus, the momentum in taking direct action for its protection must not halt.

Regardless of the need, however, for intervention, it is imperative to guarantee that the intervention does not conflict with any rights and privileges already ascribed by existing and prevailing laws. Regrettably for the BIDA Bill, it fails to prevent this discrepant situation, specifically on the Local Government Code of 1991.¹⁴ The said bill's provision on stating its taxing power disputes numerous provisions already stipulated in the code; hence, Congress must not pursue its passage unless any stipulations for any possible disparities are resolved.

¹⁴ Local Government Code, Rep. Act No.7160 (Phil.).

Bibliography:

Statute

An Act Providing for A Local Government Code of 1991, Republic Act No. 7160 (October 10, 1991) (Phil.).

Office of the President, Declaring A State of Calamity In The Barangays Of Balabag, Manoc-Manoc And Yapak (Island Of Boracay) In The Municipality Of Malay, Aklan, And Temporary Closure Of The Island As A Tourist Destination, Proclamation No. 475 s. 2018 (Apr. 26, 2018) (Phil.), <https://www.officialgazette.gov.ph/2018/04/26/proclamation-no-475-s-2018/>

House Bill

House Bill No. 9826 – An Act Creating the Boracay Island Development Authority (BIDA), <https://legacy.senate.gov.ph/lisdata/36300327311.pdf>

Jurisprudence

Zabal v. Duterte. G.R. No. 238467, (February 12, 2019) (Phil.).

Online Sources

Admin, O. (2022, November 15). The successful rehabilitation of Boracay Island in the Philippines leads the way for environmentally conscious tourism - Asia 2019 - Oxford Business Group. Oxford Business Group. <https://oxfordbusinessgroup.com/reports/philippines/2019report/economy/new-standard-the-success-of-the-rehabilitation-of-boracay-island-is-leading-the-way-for-environmentally-conscious-tourism>

Cervantes, F.M. (2021). Bill creating Boracay Island Dev't Authority gets House nod. Philippine National Authority. <https://www.pna.gov.ph/articles/1149377>

Congress of the Philippines. 13th Congress. https://hrep-website.s3.ap-southeast1.amazonaws.com/legisdocs/third_18/HBT9826.pdf

Congress of the Philippines. 18th Congress. https://hrep-website.s3.ap-southeast1.amazonaws.com/legisdocs/third_18/HBT9826.pdf

Torii, T., Masaaki, S., Mizuho, K., & Kitolelei, J. (2011). Tourism Development in Boracay Island and its Impacts on the Local Communities. ResearchGate, https://www.researchgate.net/publication/279861756_Tourism_Development_in_Boracay_Island_and_its_Impacts_on_the_Local_Communities

Visayas, I. (2023, January 27). DOT: Boracay welcomed 1.75 million tourists in 2022 | Inquirer News. INQUIRER.net, <https://newsinfo.inquirer.net/1721727/dot-boracay-welcomed-1-75-million-tourists-in-2022>



Photo by: Christine Joy Y. Gabayoyo



Photo by: Mary Gemzel M. Alcazar



Photos by: Christine Joy Y. Gabayoyo



COASTAL AND MARINE RESOURCE MANAGEMENT LAWS: THE KEY TO SAFEGUARDING GUIMARAS' BODIES OF WATER

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Guimaras is an island province, formerly known as '*Himal-us*', that is best known for its sweetest mangoes, pristine beaches, and crystal-clear waters¹ which is famous for both local and foreign tourists. It lies southeast of Panay Island and northwest of Negros Island. Five (5) municipalities composed the province, namely: Buenavista, Jordan, Nueva Valencia, San Lorenzo, and Sibunag.

Beach resorts can be found along the coastline of the Five Municipalities, including a few island resorts. As Guimaras Province's tourism capital, with its enormous tourism potential, Nueva Valencia ports the largest number of beach resorts. Diving sites are identified in several spots throughout the coastal areas, which are noted for their richness, abundance, and diversity in marine resources.² Because of its wide range of seawater, locals venture for fishing as their primary employment and usually seafood is the main source of food.

Guimaras has one protected area under the National Integrated Protected Areas System (NIPAS) category, which is known as the Taklong Island National Marine Reserve located in the municipality of Nueva Valencia. Meanwhile, there are eleven (11) locally declared marine protected areas (MPAs) established within the five municipalities.³ The breathtaking view of the coasts and the marine areas captures the heart of every people that visit it. It is the duty of every person, may it be a local resident or a guest in the vicinity to protect, conserve, and preserve it for the betterment of our biodiversity.

To a greater degree of safeguarding the coasts and marine resources, here are some of the national and local laws to which the province of Guimaras conform.

1987 Philippine Constitution

The Constitution provides for the fundamental framework to which all laws of the land must abide. It is clearly mandated under Article II Section 16 that the State shall protect and advance the right of the people to a balanced and healthful ecology in accordance with the rhythm and harmony of nature.⁴ "The environment occupies

¹ Guimaras: Guimaras Past and Present from (Province of Guimaras, n.d.), <https://guimaras.gov.ph/featured-articles/>

² *Id.*

³ *Id.*

⁴ CONST. (1987), art. II, §16 (Phil.).

pride of place in the fundamental law of the land; we, the people, must defend it and ensure that it is truly honored” " as the Philippine Daily Inquirer quoted.⁵

In the landmark case of *Oposa v. Factoran*, the Court defines the phrase “in accord with the rhyme and rhythm of nature” (of Section 16) as the “judicious disposition, utilization, management, renewal and conservation of the country’s forests, minerals, land, waters, fisheries, wildlife, offshore areas, and other natural resources to the end that their exploitation, development, and utilization be equitably accessible to the present as well as future generations.”⁶

Also, Article XII Section 2 provides that the State shall protect the nation's marine wealth in its archipelagic waters, territorial sea, and exclusive economic zone, and reserve its use and enjoyment exclusively to Filipino citizens.⁷

These Constitutional provisions made it clear that it is the duty and responsibility of the State to create laws that will protect and sustain the Philippine environment and its natural resources.

National Laws

Coastal and marine resource management in the Philippines is primarily based on a number of national laws, including the Philippine Environmental Code (P.D. No. 1151 and 1152), the Fisheries Decree of 1975 (P.D. No. 704), the Local Government Code of the Philippines (R.A. No. 7160), the National Integrated Protected Areas System (NIPAS) Act of 1991 (R.A. No. 7586), the Philippine Fisheries Code of 1998 (R.A. No. 8550), the Extended Producer Responsibility Act of 2022 (R.A. No. 11898), and the Philippine Clean Water Act of 2004 (R.A. No. 9275).

The Fisheries Decree of 1975⁸ revised and unified all laws and decrees about fishing and fisheries to foster and expedite integrated development of the fishing sector and to maintain the nation's fisheries resources at their most productive state through appropriate preservation and safeguarding. It was able to support and boost exports.

⁵ PH Laws Ensure Protection of Ecosystem (Inquirer, 2018), <https://opinion.inquirer.net/110981/ph-laws-ensure-protection-ecosystem>

⁶ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.), <https://elibrary.judiciary.gov.ph/assets/pdf/philrep/2006/G.R.%20NO.%20158290.pdf>

⁷ CONST. (1987), art. XII, § 2 (Phil.).

⁸ Fisheries Decree of 1975, P.D. No. 704 (Phil.).

A comprehensive law was enacted in 1977 entitled “Philippine Environmental Code” to protect and preserve the environment of the Philippines.⁹ One of its aims is to improve the quality of Philippine water resources through the classification of Philippine waters, establishment of water quality standards, protection and improvement of the quality of the Philippine water resources, and responsibilities for surveillance and mitigation of pollution incidents.

The Philippines enacted a significant piece of national legislation known as the Local Government Code (R.A. No. 7160). This law went into force in January 1, 1992 marked the start of a new chapter in municipal development and citizen empowerment, opening the door for achieving community self-sufficiency and local autonomy.¹⁰ The Code's adoption caused a systemic power shift that prioritized coastal cities and municipal governments for managing resources sustainably and efficiently providing guidelines for environmental maintenance.

In 1992, the NIPAS Law made it the responsibility of the Department of Environment and Natural Resources to safeguard and sustainably manage natural habitats, like marine ecosystems, in protected areas across the Philippines. This led to the creation of both extensive and smaller marine protected areas. Management efforts focused on strengthening institutions and involving local communities. If a protected area falls within municipal waters, the local government is represented on the Protected Area Management Board.¹¹

The Philippine Fisheries Code of 1998 was anchored from the Constitutional Provision of the right of every Filipino citizen to access the fishery and aquatic resources. When it comes to environmental aspects, its main purpose is to ensure the rational and sustainable development, management, and conservation of the fishery and aquatic resources in Philippine waters including the Exclusive Economic Zone (EEZ) and the adjacent high seas, consistent with the primordial objective of maintaining a sound ecological balance, protecting and enhancing the quality of the environment.¹²

Under the Extended Producer Responsibility Act of 2022, Chapter III-A Article 1 (4). clean-up of waste leaked to coastal areas, was highlighted in adopting a systematic comprehensive, and ecological solid waste management program.

Lastly, the Philippine Clean Water Act of 2004 is a comprehensive law that applies to water quality management in all water bodies: fresh, brackish, and marine waters.¹³ It provides strategies and regulations on how to protect and conserve the

⁹ Philippine Environmental Code, P.D. 1152 (Phil.).

¹⁰ Local Government Code, Rep. Act No. 7160 (Phil.).

¹¹ Expanded National Integrated Protected Areas System Act of 2018, Rep. Act. No. 11038 (Phil.).

¹² Fisheries Code of 1998, Rep. Act No. 8550 (Phil.).

¹³ Philippine Clean Water Act of 2004, Rep. Act. No. 9275 (Phil.).

country's water resources. The main goal of this law is to ensure that every Filipino has access to clean and safe water while also safeguarding the environment, particularly the aquatic resources.

Local Ordinances

The province and municipalities of Guimaras have enacted several ordinances for environmental resources protection, conservation, and management, such as Provincial Ordinance No. 2, 2005 for the Guimaras Environment Code, and other sectoral ordinances covering fishery, extraction of natural resources, the establishment of protected areas, pollution, and solid waste management, the effectiveness of such legislations is difficult to discern.

The enforcement of these laws, including the commitment of concerned authorities in implementing them, needs to be considered and strengthened as highlighted in the State of the Coasts of Guimaras Province.¹⁴

Guimaras Coastal Strategy

The province of Guimaras adopted Integrated Coastal Management with the assistance of the Partnerships in Environmental Management for the Seas of Southeast Asia. It acts as a complement to the existing environmental strategies to bolster the coastal protection that is specifically essential to the province. It was mainly anchored in the international convention and agreements ratified by the Philippines, national legislation, national and local policies, plans, and programs. Consequently, Resolution No. 160, the "Resolution Adopting the Guimaras Coastal Strategy 2020 of this Province"¹⁵ was endorsed.

The strategic principles adopted are environment protection, mitigation of environmental risks, climate adaptation, sustainable development, and community involvement to achieve its mission and vision.¹⁶

National government agencies, local government units, non-governmental organizations, civic organizations, media, religious organizations, private sector, industry, academe, communities/households, and the youth are the partners to bring into action the strategic plans which clearly define the respective roles and commitment to the adoption of the coastal management plan.

¹⁴ State of the Coasts of Guimaras Province
https://seaknowledgebank.net/sites/default/files/state-of-the-coasts-of-guimaras_0.pdf

¹⁵ Resolution No. 160, Resolution Adopting the Guimaras Coastal Strategy 2020 of this Province.

¹⁶ Guimaras Coastal Strategy, [https://www.pemsea.org/sites/default/files/2023-12/2020 Guimaras Coastal Strategy compressed.pdf](https://www.pemsea.org/sites/default/files/2023-12/2020%20Guimaras%20Coastal%20Strategy%20compressed.pdf)

Risks and Challenges

Implementing coastal strategies requires extensive efforts to ensure sustainable development. Some of the possible risks are water pollution, overexploitation of natural resources, climate change, solid wastes, economic constraints, and community involvement.

Notably, in terms of water pollution, people living in coastal areas often use the bodies of water as their garbage dump to dispose of their wastes. Tourism is also a contributing factor, as tourists generate large amounts of plastic waste and their water activities pose significant risks to the coral reefs and marine ecosystems.

There are many cases of illegal fishing that depletes fish stocks and irreparable environmental damage due to the illicit use of commercial fishing methods, large-scale operations that wreak havoc on marine habitats, and the use of destructive techniques.¹⁷

Economic constraints and community involvement are also barriers to achieving these strategies as it requires a hefty amount of finances and the spirit of volunteerism of the barangay members. The struggle to secure funds to facilitate the strategic program is unavoidable because of the high-cost items that are imperative to the establishment of the facilities and for overseeing the developmental changes.

Recommendations

Allocating the budget to finance the enforcement of the strategy is the most primary significant action to be done. Financial capacity is crucial for the protection of coastal and marine resources for the construction of facilities, for acquiring tools and equipment necessary for its development, and to compensate enforcement officers who will be in charge of securing and protecting the areas.

Following immediately after is the information dissemination to coastal barangays to equip them with capacity development, knowledge, and practices to ensure their participation and collaboration in protecting the maritime areas. Monthly community-based programs regarding marine marine-protected area management and other similar maritime protection activities. A weekly clean-up drive is also commendable as it encourages public participation and environmental awareness.

The establishment of an advocacy group will be beneficial because the membership will be likely composed of the youth. In this manner, inhabitants of the

¹⁷ Strengthening Maritime Patrol to Combat Illegal Fishing, <https://www.panaynews.net/strengthening-maritime-patrols-to-combat-illegal-fishing/>

coastal areas will gain insights into how to manage, protect, and preserve their community.

Organization of a group of qualified persons to create a sector that performs oversight duties such as reviewing, monitoring, and ensuring that the policies, plans, programs, and projects implemented are fully functional and effectively operating. On the other hand, the strengthening of maritime patrols to keep an eye on maritime occurrences and combat illegal activities that are detrimental to the water environment.

Lastly, the imposition of a penalty to violators of the law who commit prejudicial activities that result in environmental harm. This will strengthen the implementation of the strategy and the strict compliance of the constituents.

Conclusion

The recent legislation enacted by the Province of Guimaras is known as the “Integrated Coastal Management Ordinance of the Province of Guimaras”, aims to ensure the sustainable development of coastal and marine resources and environment of the province as a basic service to the people who live in the island.¹⁸

The laws and jurisprudence tackled here are indispensable for safeguarding the coastal areas and marine life while combating environmental challenges. Proper management of natural resources is the ultimate determining factor that would increase the survivability and sustainability of our environment. How we can use these supplies has only two results and that is to either contribute to the environmental degradation or prolong its accessibility.

It is really important to be on track with these laws and declarations because these are the key factors essential in utilizing, maintaining, preserving, conserving, and sustaining the coastal and marine resources of Guimaras Province. Furthermore, the island has to strictly enforce its implementing rules and regulations for the law to efficiently and effectively be of purpose because it is the future’s generation who is at stake.

¹⁸ Integrated Coastal Management Ordinance of the Province of Guimaras, Ordinance No. 2018-06 Series of 2018, <https://guimaras.gov.ph/wp-content/uploads/2019/04/Ordinance-2018-06.pdf>

Bibliography

Constitution

1987 Constitution of the Philippines, art. II, §16 (Phil.).

1987 Constitution of the Philippines, art. XII, § 2 (Phil.).

National Legislation

Expanded National Integrated Protected Areas System Act of 2018, Rep. Act No. 11038 (Phil.).

Fisheries Code of 1998, Rep. Act No. 8550 (Phil.).

Local Government Code, Rep. Act No. 7160 (Phil.).

Philippine Environmental Code, P.D. 1152 (Phil.).

Philippine Clean Water Act of 2004, Rep. Act No. 9275 (Phil.).

Local Legislation

Integrated Coastal Management Ordinance of the Province of Guimaras, Ordinance No. 2018-06 Series of 2018,
<https://guimaras.gov.ph/wp-content/uploads/2019/04/Ordinance-2018-06.pdf>)

Jurisprudence

Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.),
<https://elibrary.judiciary.gov.ph/assets/pdf/philrep/2006/G.R.%20NO.%20158290.pdf>

Online Sources

Balgos, M. C. (2002, March 14-15). Provincial and Regional Institutions in the Philippines,
https://www.crc.uri.edu/download/Philippines_Background_Paper.pdf

Guimaras Coastal Strategy,
https://www.pemsea.org/sites/default/files/2023-12/2020_Guimaras_Coastal_Strategy_compressed.pdf

Guimaras: Guimaras Past and Present from (Province of Guimaras, n.d.),
<https://guimaras.gov.ph/featured-articles/>

State of the Coasts of Guimaras Province,

https://seaknowledgebank.net/sites/default/files/state-of-the-coasts-of-guimaras_0.pdf

Strengthening Maritime Patrol to Combat Illegal Fishing,

<https://www.panaynews.net/strengthening-maritime-patrols-to-combat-illegal-fishing/>

PH Laws Ensure Protection of Ecosystem (Inquirer, 2018),

<https://opinion.inquirer.net/110981/ph-laws-ensure-protection-ecosystem>

Province of Guimaras. (n.d.),

<https://guimaras.gov.ph/featured-articles/>



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BALANCING AUTHORITIES: CONFLICTS BETWEEN LOCAL ENVIRONMENTAL CODES AND NATIONAL GOVERNMENT AGENCIES

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Environmental governance in the Philippines is characterized by a complex interplay between local and national authorities. This dynamic often leads to jurisdictional conflicts and challenges in the implementation and enforcement of environmental laws. The Philippine Environment Code (Presidential Decree No. 1152) is the primary law in addressing the concerns of the country with respect to the environment on a broad spectrum.¹ The Local Government Code of 1991 (Republic Act No. 7160) empowers local government units (LGUs) to enact local environmental ordinances specifically tailored to address environmental issues affecting their area of jurisdiction.² Simultaneously, national agencies like the Department of Environment and Natural Resources (DENR) retain broad regulatory authority under various national statutes.

This paper explores the issues arising from this dual governance system, analyzing relevant cases and decisions. It also aims to understand resolutions between conflicting jurisdiction and authority between Local Environmental Codes and National Agencies empowered to implement environmental objectives.

Environmental Legal Framework of the Philippines

The Philippine environmental legal framework is rooted in both national statutes and local ordinances. The primary national laws include the Philippine Environmental Code (Presidential Decree No. 1152), the Clean Water Act (Republic Act No. 9275), and the Clean Air Act (Republic Act No. 8749), among others. The DENR is the principal agency responsible for enforcing these laws, overseeing environmental management, and regulating natural resources.

The enactment of the Local Government Code in 1991 marked a significant shift towards decentralization, granting LGUs substantial autonomy to manage local affairs, including environmental protection. Sections 3 and 16 of the Local Government Code emphasize the role of LGUs in promoting the general welfare, which encompasses environmental management. The Local Government Code of the Philippines empowers provinces and cities to enact their own Local Environment Code addressing areas and issues not specifically raised in national statutes. It hands the LGUs the reins to decide which direction to take in improving the conditions and

¹ Philippine Environment Code, Pres. Dec. No. 1152 s. 1977, (June 6, 1977) (Phil.), <https://www.officialgazette.gov.ph/1977/06/06/presId.ential-decree-no-1152-s-1977/>

² Local Government Code, Rep. Act No. 7160 (Phil.).

strengthening the protection of their areas. Each locality deals with various types of ecosystems, protected areas, and endangered species which require distinct initiatives formulated through first-hand analyses, observation, and calculation.

This decentralization has led to overlapping authorities and jurisdictional conflicts between LGUs in implementing their ordinances and Local Environment Codes and national agencies on the authority conferred upon them by national statutes.

Jurisdictional Conflicts

Jurisdictional conflicts between local environmental codes and national government agencies manifest in various ways—

1. Overlapping Regulations and Permits

LGUs often enact ordinances that require local permits for activities already regulated by national agencies. For instance, a local ordinance may require a municipal permit for quarrying activities, while the DENR also mandates a separate Environmental Compliance Certificate (ECC). This dual permitting process can cause delays, increased costs, and confusion among stakeholders. To illustrate, coastal development projects in Batangas such as resorts and marine sanctuaries must adhere to local zoning ordinances and obtain permits from the municipal government.³ At the same time, they need to comply with national regulations under the Philippine Fisheries Code and secure permits from the DENR for activities that affect coastal and marine environments. This dual regulation can cause significant delays and increase the complexity of project implementation.⁴

2. Contradictory Policies

In some cases, local ordinances may contradict national policies. These contradictions can undermine the effectiveness of both local and national environmental strategies. A subject illustrating this matter is the Baguio City Ordinance 36, series of 2017, or the “Plastic and Styrofoam-Free Baguio City

³ An Ordinance Adopting the City of Batangas Revised Zoning Ordinance and Providing for The Administration, Enforcement And Amendment Thereof And For The Repeal Of All Ordinances In Conflict Therewith, (March 17, 2016), <https://www.batangascity.gov.ph/web/images/Offices/LEIPO/Zoning-Ordinance.pdf>

⁴ The Philippine Fisheries Code Of 1998, Republic Act No. 8550, (February 25, 1998), <https://www.bfar.da.gov.ph/wp-content/uploads/2021/02/Philippine-Fisheries-Code-of-1998.pdf>

Ordinance”⁵ which represents an effort towards combating plastic pollution and advancing environmental conservation within the city. However, its implementation raises complex legal questions surrounding jurisdiction and alignment with national legislation. This conflict underscores a broader debate over the authority of local governments to enact environmental regulations that may intersect with national policies on waste management and packaging.

The case of *City of Manila v. Laguio*⁶ has established the autonomy of local government units in crafting ordinances that promote public welfare within their jurisdictions. It may come into conflict with national laws such as Republic Act No. 9003 (R.A. No. 9003),⁷ also known by its short title as the “Ecological Solid Waste Management Act of 2000.” As a legislation, R.A. No. 9003 provides a comprehensive framework for solid waste management at the national level, including provisions on segregation, recycling, composting, and the regulation of certain types of packaging materials. While Baguio’s ordinance aims to address plastic pollution by banning plastic bags, it may be perceived as inconsistent with the broader national strategy outlined in R.A. No. 9003, which emphasizes waste segregation, recycling, and proper disposal rather than outright bans.

3. Legal Disputes

Jurisdictional ambiguities frequently result in legal disputes between LGUs and national agencies. Courts are often called upon to interpret the extent of local autonomy versus national authority, leading to varying judicial interpretations and further complicating the legal landscape.

3.1 Boracay Island Rehabilitation

The rehabilitation of Boracay Island in 2018 highlighted jurisdictional issues between local and national authorities. In Proclamation No. 475, s. 2018,⁸ declaring a state of calamity in the island of Boracay and temporary closure of the island as a tourist destination and to facilitate its rehabilitation. The order, issued by the President, aimed to address the environmental issues on the island.

⁵ Philippine News Agency, <https://www.pna.gov.ph/articles/1058830#:~:text=Ordinance%2036%2C%20series%20of%202017,same%20in%20packaging%20their%20products>

⁶ *City of Manila v. Laguio*, G.R. No. 118127, (April 12, 2005) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/43608>

⁷ Republic Act No. 9003, (January 26, 2001) (Phil.), <https://www.officialgazette.gov.ph/2001/01/26/republic-act-no-9003/>

⁸ Office of the President, Declaring A State of Calamity in The Barangays Of Balabag, Manoc-Manoc And Yapak (Island Of Boracay) In The Municipality Of Malay, Aklan, And Temporary Closure Of The Island As A Tourist Destination, Proc. No. 475, s. 2018, (April 26, 2018) (Phil.), <https://www.officialgazette.gov.ph/2018/04/26/proclamation-no-475-s-2018/>

This decision was met with resistance from local officials who argued that such actions should involve more consultation with LGUs. In *Zabal v. Duterte*,⁹ the petitioners in their Supplemental Petition, argue that that Proclamation No. 475 unduly disturbs upon the local autonomy of affected LGUs since it orders the said LGUs to implement the closure of Boracay and the ban of tourists and non-residents therefrom. While petitioners acknowledge the President's power of supervision over LGUs, they nevertheless point out that he does not wield the power of control over them. The President can only call upon the attention of the LGUs with respect to rules which they fail to follow. They argue that the President may supervise but he cannot lay down the rules himself as this already constitutes control. The arguments laid down by the petitioners are grounded on the constitutional provision on the President's exercise of general supervision over local governments¹⁰ in relation to Republic Act No. 7160 (R.A. No. 7160),¹¹ restating such prerogative.

The Supreme Court in its decision sustained the constitutionality and validity of the proclamation and slammed down the issue of encroachment. The involvement of other government agencies in Boracay's rehabilitation does not imply that LGU powers are being encroached upon. Each agency's role is clearly defined in Executive Order No. 53¹² and aligns with their mandates. Boracay, a major tourist destination, has issues that affect the national level, not just local. Despite LGUs being aware of Boracay's problems, they have failed to effectively address them.¹³ The case underscored the need for clearer guidelines on the roles and responsibilities of local and national entities in environmental crises. It is evident how confusion between jurisdiction of local and national authorities affects legitimate objectives towards environmental protection.

3.2 Manila Bay Rehabilitation

The rehabilitation of Manila Bay has been a longstanding environmental issue in the Philippines. Manila Bay, one of the country's most important coastal resources, has suffered from severe pollution due to various anthropogenic activities, including improper waste disposal, industrial runoff, and urban encroachment. In 2008, the Supreme Court issued a continuing mandamus in response to petitions filed by concerned residents and environmental groups, directing various government

⁹ *Zabal v. Duterte*, G.R. No. 238467, (February 12, 2019) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/64872>

¹⁰ CONST. (1987), art. X (Phil.).

¹¹ Local Government Code, Rep. Act No. 7160 (Phil.).

¹² Office of the President, Creating a Boracay Inter-Agency Task Force, Providing for its Powers and Functions and Those of the Member-Agencies Thereof, and Other Measures to Reverse the Degradation of Boracay Island, Exec. Ord. No. 53, s. 2018 (May 8, 2018) (Phil.).

¹³ *Zabal v. Duterte*, G.R. No. 238467, (February 12, 2019) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/64872>

agencies, including the DENR and the Metropolitan Manila Development Authority (MMDA), to clean up and rehabilitate Manila Bay.¹⁴

The decision balanced the necessity of urban progress with the imperative of safeguarding Manila Bay's ecological integrity. While affirming the MMDA's authority in urban planning, the court also underscored the paramount importance of environmental preservation, directing collaborative efforts between the government agencies to ensure sustainable development practices and mitigate adverse environmental impacts, thereby striking a balance between progress and protection for the benefit of both present and future generations.

The duty of MMDA with respect to solid waste disposal is set forth not only in the Philippine Environmental Code, but in its charter as well. This duty of raising a proper waste disposal system cannot be characterized as discretionary. The government agencies enjoined, as a matter of statutory obligation, to perform certain functions relating directly or indirectly to the cleanup, rehabilitation, protection, and preservation of the Manila Bay. They are precluded from choosing not to perform these duties.¹⁵

The implementation of the Supreme Court's directive faced challenges due to overlapping responsibilities and coordination issues among various government agencies, LGUs, and stakeholders. Local governments along the Manila Bay coastline have jurisdiction over land use planning, waste management, and other environmental issues within their respective jurisdictions. However, the magnitude of the pollution problem in Manila Bay requires concerted efforts at the national and regional levels. Coordination among LGUs, the DENR, the MMDA, and other concerned agencies is essential for effective implementation, but differing priorities and resource constraints hinders its progress.

Local Environmental Codes vis-à-vis the Philippine Environmental Code

The interplay between local environmental codes and the Philippine Environmental Code is shaped by the constitutional framework of local autonomy and the overarching authority of national legislation. The Local Government Code of 1991 (R.A. No. 7160) empowers local government units (LGUs) to enact ordinances to protect and enhance the environment, fostering a system where local and national environmental policies coexist and occasionally intersect. *Laguna Lake Development Authority v. Court of Appeals*¹⁶ underscores that while LGUs have the right to regulate within their jurisdiction, their ordinances must not contravene national laws

¹⁴ MMDA v. Concerned Residents of Manila Bay, G.R. Nos. 171947-48, (December 18, 2008) (Phil.) <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/48335>

¹⁵ *Id.*

¹⁶ *Laguna Lake Development Authority v. Court of Appeals*, G.R. No. 120865-71, (December 7, 1995) (Phil.).

and policies. Furthermore, *Boracay Foundation, Inc. v. Province of Aklan*¹⁷ illustrates the need for local environmental initiatives to adhere to national legislation and procedural due process.

Local environmental codes must align with the Philippine Environmental Code's comprehensive framework, which addresses various aspects such as air and water quality, land use, and natural resource management. In practice, local ordinances often complement the national code by addressing specific regional environmental concerns, such as those seen in the Quezon City Environmental Code,¹⁸ which supplements national policies with local initiatives on waste management and pollution control. However, conflicts arise when local ordinances impose stricter regulations or diverge from the standards set by national laws, requiring judicial interpretation to reconcile these differences. The harmonization of local and national environmental codes thus relies on a balance between respecting local autonomy and maintaining the supremacy of national standards, ensuring that local initiatives contribute to the national environmental policy goals without overstepping legal boundaries. This delicate balance is crucial in the pursuit of sustainable development, where both local and national governments play vital roles in environmental protection.

Harmonizing Local and National Initiatives

The harmonization of local and national environmental initiatives in the Philippines is imperative to prevent overlapping authority and jurisdiction, ensuring efficient and effective environmental governance. The Local Government Code grants LGUs autonomy in environmental management, empowering them to enact ordinances and regulations for environmental protection within their territorial jurisdiction. However, to avoid conflicts and inconsistencies, LGU actions must align with national policies and laws such as the Philippine Environmental Code, Clean Air Act, Clean Water Act, Ecological Solid Waste Management Act, among others.

These laws provide the legal framework for environmental protection at the national level, establishing standards and guidelines that LGUs must adhere to in making their local environmental initiatives. Furthermore, mechanisms for coordination and cooperation between national agencies and LGUs promote balance and collaboration in addressing environmental challenges. By harmonizing local and national efforts, duplication of efforts is minimized, confusion in authority is reduced, and the collective goal of sustainable environmental management is pursued effectively.

¹⁷ *Boracay Foundation, Inc. v. Province of Aklan*, G.R. No. 196870, (June 26, 2012) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/54938>

¹⁸ An Ordinance Providing For The Environmental Protection And Waste Management Code Of Quezon City, (June 13, 2016), <https://quezoncitycouncil.ph/ordinance/SP/SP-2350,%20S-2014.pdf>

Conclusion

The interaction between local and national authorities in the Philippines presents a unique challenge in environmental governance. As examined, the overlapping jurisdictions of LGUs and national agencies often lead to conflicts that complicate the implementation and enforcement of environmental laws. This dual governance system, while designed to ensure comprehensive coverage and local adaptability, sometimes creates inefficiencies and legal ambiguities.

Local governments, empowered by the Local Government Code are positioned to address specific environmental issues within their jurisdictions through tailored ordinances. This local autonomy allows for more responsive and context-specific environmental management. However, when local ordinances conflict with national regulations and the broader mandates of national agencies and statutes, the result is often a contentious struggle over authority. The resolution of these conflicts requires a balanced approach that respects both the necessity of national oversight and the benefits of local governance. Legal frameworks and judicial decisions have shown that collaborative approaches, where both local and national bodies work in tandem rather than in opposition, are crucial. Mechanisms for dialogue and cooperation, such as joint task forces or inter-agency committees, can facilitate this balance by allowing for shared responsibility and coordinated action.

Moreover, there is a pressing need for clearer legislative guidance on the interplay between local and national authorities. Revisions to existing laws or new legislation could provide more precise definitions of jurisdiction and authority, reducing the scope for conflicts. This legislative clarity would empower LGUs to pursue effective environmental governance while ensuring that national standards and objectives are upheld. The effective governance of environmental issues in the Philippines hinges on the harmonization of local and national efforts. Addressing jurisdictional conflicts through clearer laws, cooperative mechanisms, and a commitment to mutual goals is essential. As the country faces increasing environmental challenges, the ability of LGUs and national agencies to work together will be critical in achieving sustainable and effective environmental management. The path forward lies in fostering a collaborative character that exhibits the strengths of both local responsiveness and national oversight, ensuring that environmental protection efforts are coherent, comprehensive, and capable of addressing the diverse needs of the Philippine environment.

Bibliography

Statute

1987 Constitution of the Philippines, art. X (Phil.).

An Ordinance Adopting The City Of Batangas Revised Zoning Ordinance And Providing For The Administration, Enforcement And Amendment Thereof And For The Repeal Of All Ordinances In Conflict Therewith, (March 17, 2016),

<https://www.batangascity.gov.ph/web/images/Offices/LEIPO/Zoning-Ordinance.pdf>

An Ordinance Providing For The Environmental Protection And Waste Management Code Of Quezon City, (June 13, 2016),

<https://quezoncitycouncil.ph/ordinance/SP/SP-2350,%20S-2014.pdf>

Local Government Code, Rep. Act No. 7160 (Phil.).

Office of the President, Creating a Boracay Inter-Agency Task Force, Providing for its Powers and Functions and Those of the Member-Agencies Thereof, and Other Measures to Reverse the Degradation of Boracay Island, Exec. Ord. No. 53, s. 2018 (May 8, 2018) (Phil.).

Office of the President, Declaring A State of Calamity In The Barangays Of Balabag, Manoc-Manoc And Yapak (Island Of Boracay) In The Municipality Of Malay, Aklan, And Temporary Closure Of The Island As A Tourist Destination, Proc. No. 475, s. 2018, (April 26, 2018) (Phil.),

<https://www.officialgazette.gov.ph/2018/04/26/proclamation-no-475-s-2018/>

Philippine Environment Code, Pres. Dec. No. 1152 s. 1977, (June 6, 1977) (Phil.),

<https://www.officialgazette.gov.ph/1977/06/06/presidential-decree-no-1152-s-1977/>

Philippine News Agency,

<https://www.pna.gov.ph/articles/1058830#:~:text=Ordinance%2036%2C%20series%20of%202017,same%20in%20packaging%20their%20products.>

Republic Act No. 9003, (January 26, 2001) (Phil.),

<https://www.officialgazette.gov.ph/2001/01/26/republic-act-no-9003/>

The Philippine Fisheries Code Of 1998, Republic Act No. 8550, (February 25, 1998),

<https://www.bfar.da.gov.ph/wp-content/uploads/2021/02/Philippine-Fisheries-Code-of-1998.pdf>

Jurisprudence

Boracay Foundation, Inc. v. Province of Aklan, G.R. No. 196870, (June 26, 2012) (Phil.),

<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/54938>

City of Manila v. Laguio, G.R. No. 118127, (April 12, 2005) (Phil.),

<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/43608>

Laguna Lake Development Authority v. Court of Appeals, G.R. No. 120865-71, (December 7, 1995) (Phil.).

MMDA v. Concerned Residents of Manila Bay, G.R. Nos. 171947-48, (December 18, 2008) (Phil.),

<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/48335>

Zabal v. Duterte, G.R. No. 238467, (February 12, 2019) (Phil.),

<https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/64872>



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25 YEARS LATER: IS THE CLEAN AIR ACT OF 1999 STILL RELEVANT?

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One of the most human and environmental health issues that the world has faced over the years is air pollution. In the Philippines, air pollution has been the top problem, especially in Metro Manila. The major contributors to the emission are motor vehicles, power plants, industrial processes, and the combustion of biomass, coal, and kerosene.

To prevent pollution, the Clean Air Act, also known as Republic Act No. 8749 (R.A. No. 8749), was signed by former president Joseph Estrada on June 23, 1999, becoming one of the major environmental laws in the Philippines.¹

With a focus on the "polluters must pay" idea, the Clean Air Act seeks to guarantee a "balanced and healthful ecology" for all Filipinos, in line with "the rhythm and harmony of nature." In carrying out the law, the Department of Environment and Natural Resources (DENR) serves as the "lead agency." Each airshed will have a common action plan and "a system of planning and coordination" to control the air quality more effectively. Pursuant to the law, the nation is divided into various airsheds based on "areas with similar climate, meteorology, and topology which affect the interchange and diffusion of pollutants in the atmosphere, or areas which share a common interest or face similar development programs, prospects, or problems."²

Legislative Framework

Key Provisions of the Act

The Clean Air Act of 1999 was legislated in order to ensure the proper implementation of government initiatives to lessen air pollution to protect public health and the environment. This act aims to reduce the harmful emissions coming from vehicles, industrial facilities, and other pollutants. The key provisions of the act focus on the Air Quality Monitoring and Information Network, National ambient air quality standards, and regulations for pollution From Stationary Sources and from Motor Vehicles.

¹ Alexa Basa, Clean Air Act after 23 years: Time to revisit the law <https://mb.com.ph/2022/11/12/clean-air-act-after-23-years-time-to-revisit-the-law>

² *Id.*

a. Air Quality Monitoring and Information Network

Section 6, Article 1 of the Clean Air Act of 1999 is the Air Quality Monitoring and Information Network which provides that the Department of Environment and Natural Resources Shall prepare an annual National Air Quality Status Report. Its purpose is to provide various measures and strategies to reduce air pollution and maintain healthy air quality levels, educate the public and provide awareness of the issues involved in maintaining and improving air quality, and have a systematic process to collect, analyze, and disseminate information about air quality levels across the country.³

The annual National Air Quality Status Report shall include, but shall not be limited to the following:

- a) Extent of pollution in the country, per type of pollutant and per type of source, based on reports of the Department's monitoring stations;
- b) Analysis and evaluation of the current state, trends and projections of air pollution at the various levels provided herein;
- c) Identification of critical areas, activities, or projects which will need closer monitoring or regulation;
- d) Recommendations for necessary executive and legislative action; and
- e) Other pertinent qualitative and quantitative information concerning the extent of air pollution and the air quality performance rating of industries in the country.⁴

Using this report, it can be ensured that the Department of Environment and Natural Resources will properly monitor data, provide accurate public information, and impose regulations that will ensure improvement for a healthier environment.⁵

b. National Ambient Air Quality Standards (NAAQS).

One of the fundamental parts of this legislation is the creation of Ambient Air Quality Guideline Values and Standards, which operate as the standard for evaluating and controlling the air quality to which the general public is exposed.

Section 12, Article 2 of the Clean Air Act of 1999 set forth that the Department of Environment and Natural Resource, in coordination with other concerned agencies, shall review and/or revise and publish annually a list of

³ Philippine Clean Air Act of 1999, Republic Act No. 8749 <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

⁴ *Id.*

⁵ *Id.*

hazardous air pollutants with corresponding ambient guideline values and/or standard necessary to protect public health and safety, and general welfare.⁶

These guidelines, values, and standards are designed to ensure the protection of the public against the detrimental effects of air pollution. This is based on the latest scientific knowledge and is based on the World Health Organization (WHO) standard. These include the information on:

- a) Variable factors, including atmospheric conditions, which of themselves or in combination with other factors may alter the effects on public health or welfare of such air pollutants;
- b) The other types of air pollutants that may interact with such pollutants to produce an adverse effect on public health or welfare; and
- c) The kind and extent of all identifiable effects on public health or welfare which may be expected from the presence of such pollutants in the ambient air, in varying quantities.

The National Ambient Air Quality Standards is a strong help for the Philippine government to achieve the sustainable development goals for the well-being of the Filipino people. This provision ensures that the standards set by the government are aligned with the international standard and it is carefully assessed and complied with.⁷

c. Regulation of Pollution from Stationary Sources

Section 19, Article 3 of the Clean Air Act of 1999 aims to address the air pollution coming from stationary sources such as factories, industrial facilities, and power plants. These stationary sources emit large quantities of harmful pollutants including sulfur dioxide (SO₂), nitrogen oxides (NO_x), particulate matter (PM), and volatile organic compounds (VOCs). This provision mandates the emission standards on mass rate of emission for all stationary sources of air pollution which should be based on internationally-accepted standards. The standards, whichever is applicable, shall be the limit on the acceptable level of pollutants emitted from a stationary source for the protection of the public's health and welfare. With respect to any trade, industry, process, and fuel-burning equipment or industrial plant emitting air pollutants, the concentration at the point of emission shall not exceed the limits provided in the statute.⁸

In addition, Incineration which is defined as the burning of municipal, bio-medical, and hazardous wastes, that emit poisonous and toxic fumes, is declared prohibited in Section 20 of the same act.⁹

⁶ Philippine Clean Air Act of 1999, Republic Act No. 8749, <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

⁷ *Id.*

⁸ *Id.*

⁹ *Id.*

d. Regulation of Pollution from Motor Vehicles

One of the major goals of the act is to mitigate air pollution from vehicular sources. Vehicles such as cars, trucks, jeepneys, and buses are emitting harmful substances such as nitrogen oxides, carbon monoxide, and particulate matter. This provision addresses air pollution concerns by providing a comprehensive framework of regulations and standards designed to reduce emissions from motor vehicles.¹⁰

Section 21, Article 4 of the Clean Air Act of 1999 provides that “for Pollution from Motor Vehicles, the DOTC shall implement the emission standards for motor vehicles. To further improve the emission standards, the DENR shall review, revise, and publish the standards every two (2) years, or as the need arises. It shall consider the maximum limits for all major pollutants to ensure substantial improvement in air quality for the health, safety and welfare of the general public. The standard for exhaust emission limit for gaseous pollutants shall be applicable for light-duty vehicles, light commercial vehicles, and heavy-duty vehicles. This section of the act also highlighted that the fuel evaporative emission for spark-ignition engines shall not exceed 2.0 grams of hydrocarbons per test.”¹¹

Furthermore, Article 4, Section 22 of this Clean Air Act prescribes for the Regulation of All Motor Vehicles and Engines. It strengthens the registration requirements of any imported new or locally-assembled new motor vehicle which shall not be registered unless it complies with the emission standards, as evidenced by a Certificate of Conformity (COC) issued by the Department.¹²

International Context

a. Comparative Analysis of the Clean Air Act Amendment of 1990 by the United States and the Clean Air Act of 1999 of the Philippines

The Clean Air Act of Amendment of the United States 1990 and the Clean Air Act of 1999 of the Philippines are environmental laws that aim to protect the welfare of its people by ensuring air quality. However, they differ in terms of scope, implementation, and key provisions.

Scope

The U.S. The Clean Air Act originally passed in 1963 and significantly amended in 1970, 1977, and 1990, addresses a wide range of air quality issues across

¹⁰ Philippine Clean Air Act of 1999, Republic Act No. 8749, <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

¹¹ *Id.*

¹² *Id.*

the country. It aims to protect and enhance the nation's air resources to promote public health and welfare and the productive capacity of its population.¹³

On the other hand, The Clean Air Act of the Philippines of 1999 was primarily enacted to solve the country's urgent air pollution problems, especially in major cities like Metro Manila. Its primary objective is to protect and advance the right of the people to a balanced and healthful ecology by addressing air pollution issues.

Implementation

The regulating body in The Clean Air Act of the United States is the Environmental Protection Agency (EPA) which focuses on the Extensive coverage of stationary sources, mobile sources, and hazardous air pollutants. Included in the provisions are the setting of National Ambient Air Quality Standards (NAAQS), New Source Performance Standards (NSPS), and vehicle emissions standards. The act emphasizes transparency and public access to information by making the information obtained by the Board available to the public, except information that would be harmful to a person's competitive position. In terms of public participation, the act includes mechanisms for public engagement and citizen lawsuits to enforce compliance.¹⁴

In the Philippines, the Department of Environment and Natural Resources (DENR) is the regulation body of The Clean Air Act of 1999. The law includes both stationary sources (factories, power plants) and mobile sources (motor vehicles) which focus on setting air quality standards, promoting cleaner technologies, and encouraging alternative energy sources. Public participation is encouraged by placing a strong emphasis on community involvement and transparency, with provisions for public access to air quality data and participation in environmental assessments.¹⁵

Focus Areas

The Philippine Clean Air Act of 1999 focuses on highlighting the connection between sustainable development and air quality objectives. It also aims to lessen reliance on fossil fuels by supporting alternative energy sources. And it has a significant focus on community and public involvement in air quality management.¹⁶

¹³ Evolution of the Clean Air Act, [from https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act](https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act)

¹⁴ United States H.R.3030 - Clean Air Act Amendments of 1990.

¹⁵ Philippine Clean Air Act of 1999, Republic Act No. 8749, <https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

¹⁶ *Id.*

In comparison to the Clean Air Act of 1990 by the United States, it focuses on a comprehensive approach covering a wide range of pollutants and sources. Also, it uses a combination of regulatory measures and market-based approaches to achieve air quality goals and emphasizes technological innovation and continuous improvement in pollution control measures.¹⁷

In a nutshell, both the Clean Air Act of the Philippines and the Clean Air Act of the United States aim to safeguard public health and ensure quality air across the nation. The Philippine law is primarily concerned with incorporating sustainable development and public engagement while the U.S. law is broader in scope, with a longer history of amendments and a more detailed regulatory framework that includes market-based mechanisms.

Alignment to the United Nations Sustainable Development Goals (UNSDG)

The 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, provides a shared goal of achieving peace and prosperity for the people and planet. At its heart are the seventeen (17) Sustainable Development Goals (SDGs), which are an urgent call for action by all countries - developed and developing - in a global partnership. One of its main goals is to attack climate change and preserve the environment.¹⁸

The R.A. No. 8749 “Clean Air Act of 1999” aligns with several goals of the United Nations Sustainable Development Goals (UNSDGs), reflecting its comprehensive approach to improving air quality, protecting public health, and promoting sustainable development.

The Clean Air Act of 1999 has a significant positive impact on several goals. It is directly contributing to SDG “Good Health and Well-Being” by decreasing the incidence of respiratory and cardiovascular diseases associated with poor air quality. Furthermore, by regulating the emission standards and promoting cleaner technologies the health risks can be reduced and it can help in improving the well-being of the people. SDG 7 which is “Affordable and Clean Energy” can also be achieved by promoting the use of cleaner fuels and the development of renewable energy sources.¹⁹

¹⁷ United States H.R.3030 - Clean Air Act Amendments of 1990, <https://www.congress.gov/bill/101st-congress/house-bill/3030#:~:text=Prohibits%20any%20lead%20content%20and,when%20using%201990%20certification%20fuel>

¹⁸ United Nations Sustainable Development, <https://sdgs.un.org/goals>

¹⁹ Sustainable Development Goals and air pollution, <https://www.cleanairfund.org/news-item/sustainable-development-goals/>

In addition, the act is indirectly contributing to SDG 11, “Sustainable Cities and Communities” by regulating the emissions from industrial activities and vehicles. It can also help in achieving SDG 12 or “Responsible Consumption and Production” because the act mandates the use of pollution control technologies and cleaner production processes, aligning with the goals of minimizing waste and promoting sustainable consumption and production. By reducing emissions and encouraging responsible environmental practices, the act supports the broader agenda of sustainable management of resources and waste. Lastly, SDG 13 “Climate Action” affected the reduction of emissions of greenhouse gasses and other pollutants. Hence, mitigating the risks of climate change.²⁰

The Clean Air Act of 1999 is undeniably aligned with the United Nations Development Goals as it addresses air pollution, promotes public health, encourages the use of clean energy, supports sustainable urban development, fosters responsible consumption and production, combats climate change, and protects terrestrial ecosystems. With the proper implementation program and stringent application of its regulations, cleaner air can be achieved.

Stakeholders involvement and initiatives

Acknowledging the importance of the Clean Air Act of 1999 and the rigid implementation of the programs of the Department of Environment and Natural Resources, the stakeholders are becoming more involved and conducting initiatives to help address the issues on air pollution.

The National Air Quality Status Report 2016-2018 of the Department of Environment and Natural Resources (DENR) showed that the regional agencies are becoming more involved in air quality management, as reflected in the completion of emissions inventories and regional monitoring results.²¹

There are also initiatives from other government agencies such as the Department of Energy (DoE) and the Department on Trade and Industries (DTI) on the standards for fuels and additives; the Department of Transportation (DoTr) on their programs for Euro 4 standards for fuels and engines; and Department of Health on the implementing guidelines of Inter-agency Committee of Environmental Health.²²

For the health aspect, the Lung Center of the Philippines and the World Health Organization (WHO) initiated a project on the Air Pollution Monitoring

²⁰ Sustainable Development Goals and air pollution, <https://www.cleanairfund.org/news-item/sustainable-development-goals/>

²¹ National Air Quality Status Report 2016-2018, <https://air.emb.gov.ph/wp-content/uploads/2021/07/NAQSR-2016-2018-FV.pdf>

²² *Id.*

Project which provides assistance on the PM_{2.5} breakpoints. In the academe, the University of the Philippines Diliman, De La Salle State University, and local and international academies have contributed to aerosol characterization research. For non-governmental organizations, there are several projects which assist in measurement and action plans.²³

Limitations and Challenges

The Clean Air Act of 1999 has significant actions in addressing air pollution in the country, however, it cannot be denied that there are several limitations and challenges that affect the effective implementation of the act.

The National Air Quality Status Report 2016-2018 of the Department of Environment and Natural Resources (DENR) shows that despite the consolidated efforts to improve air quality status, there are still remaining challenges and needed actions in air quality management.

Some of the notable challenges based on the DNRs report are the following:

- Lack of Air Quality Index (AQI) Breakpoints for PM 2.5
- Lack of siting criteria for air quality monitoring stations
- Lack of Data Handling Guidelines for Ambient Air Quality Monitoring Data
- Dependence expensive reference/equivalent AQM equipment
- Lack of an On-Line Real-Time Data Acquisition System in monitoring Large Industrial Facilities with Continuous Emission/ Opacity Monitoring Systems (CEMS/COMS)
- Lack of Online System for the Issuance of Permit to Operate (PTO) for Air Pollution Source Installation (APSI)
- Decreasing the uncertainties in emissions inventory
- Unable to capture the 75% data capture rate
- Assessment of air quality and its health impact at the regional level
- Weak inter-agency participation on air quality and health actions
- Management of mobile emissions sources ²⁴

These limitations suggest that the actual and effective implementation of the Clean Air Act of 1999 after how many years is still a challenge. These challenges are mostly caused by weak monitoring systems; these are gaps in the air quality monitoring network, including inadequate coverage and outdated equipment, which hinder accurate and comprehensive assessment of air quality across the country. There are also coordination issues among the government agencies and the LGUS leading to fragmented efforts and overlapping responsibilities. The economic and technological barriers are also the contributing factors. The high cost for compliance triggers the non-compliance of business from different industries, particularly small and medium enterprises (SMEs) due to lack of financial resources to invest in the

²³ National Air Quality Status Report 2016-2018, <https://air.emb.gov.ph/wp-content/uploads/2021/07/NAQSR-2016-2018-FV.pdf>

²⁴ *Id.*

required upgrades. The technological limitations of the Department of Environment and Natural Resources is also one of the barriers because the access to and adoption of advanced pollution control technologies are limited, particularly in rural and economically disadvantaged areas.

Conclusion

The Philippine Clean Air Act of 1999, Republic Act No. 8749 remains to be relevant for more than two decades after its enactment. Its comprehensive framework which aims to improve air quality in the country is essential in the protection of public health. Its alignment with the United Nations Sustainable Development Goals (UNSDGs) provides more importance that this act must be upheld to further improve the air quality in the Philippines. The Clean Air Act of 1999, even if it is timely and relevant, still faces a lot of limitations and challenges. Even though the DENR provided strategic actions and timelines to combat these predicaments, it cannot be denied that these occurrences happened ever since the effectivity of the Philippine Clean Air Act of 1999. These challenges will not be addressed unless the government strengthens the enforcement and compliance of the law, increases in the resources for regulatory bodies, and investments to advance technology in such a way as to ensure comprehensive and accurate data collection.

In the end, it is the government and its people that is responsible for taking good care of the environment and ensuring clean air quality. The Republic Act No. 8749 or the Philippine Clean Air Act of 1999 can be effectively and efficiently implemented when the government agencies and stakeholders collaborate to achieve the law's goal of cleaner air and a more environmentally friendly atmosphere for the common good and public welfare.

Bibliography

Statutes

An Act Providing for A Comprehensive Air Pollution Control Policy and For Other Purposes, Republic Act No. 8749,
<https://emb.gov.ph/wp-content/uploads/2015/09/RA-8749.pdf>

United States H.R.3030 - Clean Air Act Amendments of 1990,
<https://www.congress.gov/bill/101st-congress/house-bill/3030#:~:text=Prohibits%20any%20lead%20content%20and,when%20using%201990%20certification%20fuel.>

Reports

National Air Quality Status Report 2016-2018, Department of Environment and Natural Resources,
<https://air.emb.gov.ph/wp-content/uploads/2021/07/NAQSR-2016-2018-FV.pdf>

Online Sources:

Clean Air Act after 23 years: Time to revisit the law,
<https://mb.com.ph/2022/11/12/clean-air-act-after-23-years-time-to-revisit-the-law>

Evolution of the Clean Air Act <https://www.epa.gov/clean-air-act-overview/evolution-clean-air-act>

United Nations Sustainable Development,
<https://sdgs.un.org/goals>

Sustainable and Developmental Goals,
<https://www.cleanairfund.org/news-item/sustainable-development-goals/>

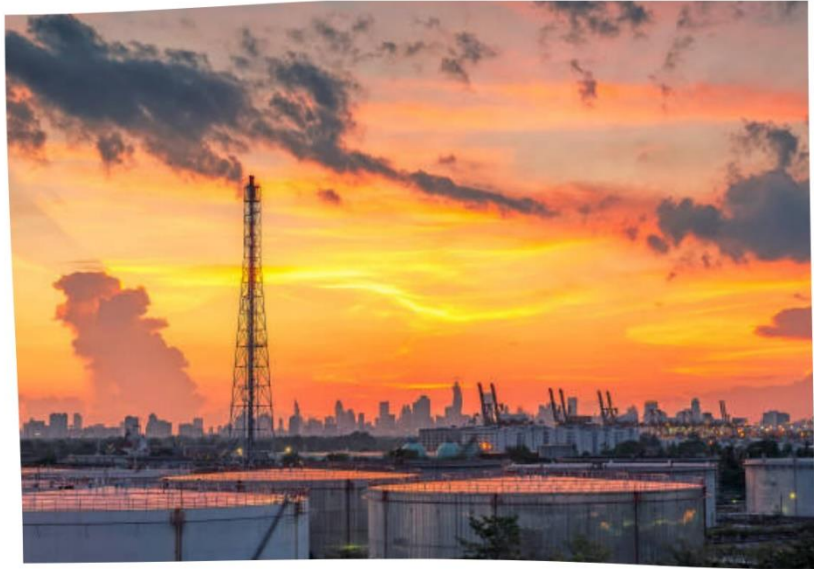


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NAVIGATING THE NEXUS: WIND POWER DEVELOPMENT OR NABAOY RIVER, AND THE ASPIRATIONS FOR NATIONAL LAND USE ACT

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Introduction

The right to water is recognized as a fundamental human right by the United Nations.¹ It entitles everyone to sufficient, safe, acceptable, physically accessible, and affordable water for personal and domestic use. Similarly, the right to sanitation ensures physical and affordable access to safe, hygienic facilities that respect dignity. This recognition emphasizes that access to clean drinking water and sanitation is essential for sustaining every person's life and well-being such a right is consistent with the principles enshrined in our constitution to which the constitution provides that:

“Section 16. The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.”²

This provision is expanded more so in the celebrated case of *Oposa v. Factoran*,³ the supreme court has recognized this right, although not belonging under our Bill of Rights, as no lesser rights for “Such a right belongs to a different category of rights altogether for it concerns nothing less than self-preservation and self-perpetuation — aptly and fittingly stressed by the petitioners — the advancement of which may even be said to predate all governments and constitutions. Such right as the supreme court has so advised need not be written as they exist from the inception of humankind. *“These rights to a balanced and healthful ecology carries with it the correlative duty to refrain from impairing the environment”*”

The NLUA's promise of sustainable development and harmonious land management extends beyond theory—it directly intersects with real-world challenges. One such challenge involves the contentious relationship between a wind power company and the ecologically vital Nabaoy River in Aklan. As we delve into this case, we'll explore how the NLUA's principles can guide us toward a balanced resolution that considers both economic progress and environmental preservation.

¹ Human Rights to Water and Sanitation (n.d) <https://www.unwater.org/water-facts/human-rights-water-and-sanitation>

² CONST. (1987), art. II §16 (Phil.).

³ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.).

The Nabaoy Watershed

The Nabaoy Watershed, with an approximate land area of 2,469 hectares, is situated the Municipality of Malay and covers Barangays of Cubay Sur, Motag, Napaan, Pawa, Nabaoy, Malay, and a portion of Tag-osip, Buruanga.⁴ The Nabaoy watershed is the main source of potable water in the community, neighboring barangays, and municipalities as well as the source of water to the famous Island of Boracay. This small watershed is being drained by its river system and tributaries with a common outlet that discharges downstream of Nabaoy River flowing its water out into the Sibuyan Sea.

Based on recent studies,⁵ point to Nabaoy River as a reliable source of food and livelihood not only for the local inhabitants that regularly depend on them, but also for the Municipality of Malay, who thanks to the “unique ecosystem” that allows better conservation of groundwater, this unique characteristic allows the limestone liquefiers to hold on water for at least 16 years from the top before it could be downloaded to its river tributaries, allowing the Nabaoy River maintain a steady stream of water especially during dry season where water is most essential, leading researchers like Raymond Sugang, a senior researcher at the Philippine Nuclear Research Institute (PNRI) of the Department of Science and Technology (DOST), in a Facebook post,⁶ has since warned against any attempts to alter, disrupt or disturb the cracks in this formation, such by simple drillings that create more fracture would result in more water being released, permanently altering the unique ecology in the area.

The Nabaoynon Perspective

Hundreds of Nabaoynon's have called on the Malay local government unit (LGU) to stop the turbine expansion of the Nabas Wind Power Project, a renewable energy project of PetroWind Energy Incorporated (PWEI).

⁴ Señeris, G. T. Nabaoy River Watershed potential impact to flooding using Geographic Information System remote sensing. *AIMS Environmental Science*, 9(3) (2022), <https://www.aimspress.com/article/doi/10.3934/environsci.2022024>

⁵ Maliao, R. J., Cahilig, R. C., Cahilig, R. R., & Jaspe, B. T. Climate Change Awareness and Indigenous Knowledge Systems and Practices (Iksp) of Riverine Fishers in the Nabaoy River, Malay, Aklan, Philippines: Linking Local Social Capital to Socio-Ecological Resilience Amidst the Changing Environment and Climate. Malay, Aklan, Philippines: Linking Local Social Capital to Socio-Ecological Resilience Amidst the Changing Environment and Climate (February 16, 2022), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4034376

⁶ Raymond Jacinto Sugang, “How the Nabaoy River Maintains a Steady Baseflow Year-Round Despite Summer Conditions, Despite Its Rock Composition's Limited Water Storage Capacity” Facebook (February 17, 2024), <https://www.facebook.com/photo?fbId=7120348081376858&set=a.132826973462372>

Villagers and natives of the Nabaoy River have staged multiple protest⁷ condemning the supposed project because of fears that its construction near the said river would nevertheless take a toll on their environment, the protesters has since called on to their local government official to denounce the project noting that completion of such project would inevitably lead to irreparable damage to the Nabaoy River and would not only impact the healthy ecology in the area but the livelihood and socio-economic life of the villagers that rely on them.

The fears of the locals rather ignited by the horror of what has already occurred in the neighboring river of Napaan when during the 1st phase of the Nabas wind project which was launched June 2015, and by July of 2023, furnished by local reports had already shown siltation in the area, that when it rained hard on the August month of that same year, shorelines of several barangays in Malay especially in Argao turned muddy, the clear waters of which has been only comparable to that of Nabaoy, now turned brown from the heavy siltation that caused by the wind farm construction on a neighboring mountain,⁸ the age old practice of catching *patuyaw* (native shrimp) the most decimated, as similar species of freshwater prawns, used to thrive abundantly in Napaan, significantly declining, because of their prone sensitivity to dirty water.⁹ Local leaders in Nabaoy proposed that if the Malay government intends to proceed with the project, it should be relocated away from the rivers and mountains of the Northwest Panay Peninsula Natural Park, the last low-lying forest in Panay.

The Petrowind Perspective

In an article published on their website, PetroEnergy.Co, the company has assured all of its stakeholders that the Nabas 2 wind power project (Phase 2) in Nabas-Malay, Aklan of PetroWind Energy Inc. (PWEI) is fully compliant with all national and local environmental rules, the Department of Environment and Natural Resources - Environmental Management Bureau Region 6 (DENR-EMB 6),¹⁰ it promised to generate clean and more environment-friendly energy that will contribute to the fight against climate change, the company has also promised to continue with its robust implementation of impact mitigation measures, measures recommended by local and international bioengineering experts, such measures, it said were applied in the first phase of the project, which was completed in October 2023.

⁷ Harmé, J. N. Villagers protest planned wind farm expansion project in Aklan. (February 3, 2024). RAPPLER, <https://www.rappler.com/philippines/visayas/villagers-protest-planned-wind-farm-expansion-project-aklan-january-31-2024/>

⁸ Harmé, J. N. Online and offline, this Aklan village fights a wind farm to save their river. (April 30, 2024). RAPPLER, <https://www.rappler.com/newsbreak/in-depth/online-offline-nabaoy-community-struggles-renewable-energy-project/>

⁹ Arnaldo, M. S. F. Threatening the environmental gains of Boracay | Ma. Stella F. Arnaldo. Business Mirror. (March 2, 2024), <https://businessmirror.com.ph/2024/02/23/threatening-the-environmental-gains-of-boracay/>

¹⁰ PetroWind is fully compliant -- DENR-EMB (n.d.), <https://petroenergy.com.ph/#!/newsitem/51>

The company also boasts the compliance of the project by the recent inspections done by environmental agencies confirmed by the project's compliance with Environmental Compliance Certificate (ECC), Special Agreement on Protected Area (SAPA), and Forestland Use Agreement conditions. The compliance, it said, was affirmed in a hearing by the town council of Malay on January 31, 2024.

Petrowind also stressed the project's importance in stabilizing Panay's power supply through sustainable energy sources. It urged stakeholders to rely on accurate information and avoid speculation,¹¹ it also warned that the "Propaganda based on purely ideological grounds without hard scientific and technical basis can result in less investments in Aklan, loss of additional tax revenues for the LGUs, and loss of employment and business opportunities for host communities."

After getting a nod anew from the Sangguniang Bayan of Malay, when the council re-endorsed the project last February 22, 2024, the Petrowind Energy Inc., (PWEI), the company has since confirmed that *gabions*¹² and silt barriers have since been installed, at the same time numerous Barangay Officials, specifically the ones in Argao and Cogon have expressed support for the Phase 2, provided that all mitigating measures will be put in place. Meanwhile, officials of Barangay Cogon have downplayed all possible threats, rationalizing that all development activities have certain effects on the environment, but such effects are temporary.¹³

The Contention and the Present situation

After receiving a citation from the Department of Natural Resources, commending its Nabas-1 Wind farm site, the company has been granted a greenlight on the 14-megawatt second phase of the Nabas wind farm project of Yuchengco-led PetroWind Energy Inc. (PWEI) which will add an additional six (6) wind turbines to the existing eighteen (18) wind turbines of its 36MW Nabas 1, which has started shoring up power supply for the energy-strained Panay island after securing warranted clearances from relevant authorities and entities in the market.¹⁴ At the time of this writing, the Petrowind Phase 2 project has already finished 1.5kms of road connecting

¹¹ Job growth and environmental responsibility: Yuchengcos defend PetroWind's expansion in Aklan (n.d.), <https://bilyonaryo.com/2024/02/21/job-growth-and-environmental-responsibility-yuchengcos-defend-petrowinds-expansion-in-aklan/power/>

¹² a cage, cylinder or box filled with rocks, concrete, or sometimes sand and soil for use in civil engineering, road building, military applications and landscaping.

¹³ PetroWind's Phase 2 Re-endorsed by Malay SB Following; Support from DENR and Barangay Officials (April 16, 2024), <https://petroenergy.com.ph/#!/newsitem/57>

¹⁴ Yuchengco-owned wind farm starts shoring up power supply for Panay. (n.d.). Manila Bulletin, <https://mb.com.ph/2024/4/5/yuchengco-owned-wind-farm-starts-shoring-up-power-supply-for-panay>

the first three (3) turbines and another 800 meters to be finished that will connect the three (3) other turbines, which the renewable energy company has claimed to be done without any effects at all to the water source and belying the claims of detractors as mere misinformation.¹⁵

Despite the claims made by the renewable energy corporation, the Villagers remain undeterred and unconvinced about the benefits of the project, this even if Petrowind has promised program to give back to the community, the villagers in the area continue to reiterate that they were not part of the project's impact assessment, which means that Petrowind may not be held responsible for the damage done to the river if ever there be any consequences,¹⁶ a fact that the Local Government Unit has otherwise advised, stating that Nabaoy River is the direct impact site, because of the close proximity of the project site and the fact that it is the river which will be hit by mud coming from the mountain where the turbines will be affected.

National Land Use Act

As the Philippines grapples with a significant dilemma in effectively managing its land resources due to the absence of comprehensive national legislation governing land use planning. This lack of a unified framework results in conflicting land utilization, environmental decline, and suboptimal allocation of resources. The proposed National Land Use Act (NaLUA) stands as a pivotal remedy, offering the promise of sustainable development and balanced land management.¹⁷ Although the welfare of the Philippines' natural resources and national parks are secured under the now expanded National Integrated Protected Areas System Act (E-NIPAS Act), first passed in 1992, subsequent amendment to the law has added concessions allowing the exploration and establishment of renewable energy projects in natural parks, if the Protected Area Management Board and the secretary of the Philippines' Department of Environment and Natural Resources (DENR) both approve it.¹⁸ Looking into the focuses of each system, with the Expanded National Integrated Protected Areas System Act (ENIPAS) alone, it fails to sustain the balancing economic development with environmental stewardship, when it should have been that the two acts whole have been played a complementary role, where ENIPAS specifically safeguards

¹⁵ Nabas Wind Farm. Facebook (February 19, 2024). <https://www.facebook.com/NabasWindFarm/posts/pfbid.0BmwkMpRiP3dcjWgySwKcmEpJFeJQACVQifA86nVQnFVFbEoAgykqxWqSJqG5ivqml>

¹⁶ Harme, J.N. *supra* note 7, at 3.

¹⁷ PIDS study urges passage of National Land Use Act to address land use conflicts. (February 26, 2024), <https://www.pid.s.gov.ph/details/news/press-releases/pid.s-study-urges-passage-of-national-land-use-act-to-address-land-use-conflicts#:~:text=This%20lack%20of%20a%20unified,development%20and%20harmonious%20land%20management>

¹⁸ Albay, Rick Lars. Tensions heighten as Philippine clean energy projects encroach on conservation areas (May 31, 2024). <https://www.wind-watch.org/news/2024/06/01/tensions-heighten-as-philippine-clean-energy-projects-encroach-on-conservation-areas/>

protected areas¹⁹ the NLUA addresses broader land management challenges, emphasizing sustainable development and harmonious land use nationwide.

For one the National Land Use Act as proposed²⁰ provides a structured approach to land use planning, ensuring that development aligns with environmental protection. By designating land for specific purposes—such as agriculture, industry, or conservation—it minimizes ecological harm. Watersheds, forests, and biodiversity receive due consideration, contributing to disaster resilience. With climate change impacts looming, the NLUA’s systematic approach helps communities adapt and build resilient infrastructure. The Efficient land allocation under the NLUA supports economic development. It encourages investment, infrastructure projects, and urbanization while preventing haphazard growth. By optimizing land utilization, the NLUA fosters sustainable economic value addition. Moreover, it addresses cross-cutting issues that span various sectors, bridging conflicting interests. Whether in rural or urban contexts, planned land use ensures safe settlements, housing, and livelihood opportunities. With the issue at hand, it can only be imagined that the National Land Use Act (NLUA) holds significant potential to address the ongoing dispute concerning the Nabaoy River in Aklan.

To be specific, among the ways it may contribute to resolving this is the Law’s innate ability to establish a comprehensive national land use policy, guiding zoning and allocation of land resources. By delineating specific zones for various purposes (e.g., residential, agricultural, industrial), it can prevent conflicting land uses near the Nabaoy River. Zoning can designate buffer zones along the riverbanks, protecting them from encroachment, pollution, and unsustainable development. This approach ensures that land near the river remains ecologically sensitive and resilient. Secondly, the law emphasizes sustainable land management. It encourages practices that balance economic growth with environmental protection. In the case of the Nabaoy River, the Law could promote eco-friendly land use practices, such as reforestation, wetland conservation, and responsible tourism.

It would discourage activities that harm the river ecosystem, such as unregulated construction or mining. Lastly, as the Law envisions participatory land use planning, such steps may respond strongly to the Local communities, including those affected by the Nabaoy River dispute, can actively engage in decision-making. As the rule imagines stronger involvement of stakeholders, the law itself may foster dialogue, negotiation, and consensus-building. It provides a legal framework for resolving conflicts, ensuring that the river’s interests align with community needs and environmental well-being.

¹⁹ Dela Cruz, Beverly Mae N., National Integrated Protected Areas System (NIPAS) Act and E (expanded) - NIPAS Act of the Philippines (May 20, 2019), <https://www.slid.eshare.net/slid.eshow/national-integrated-protected-areas-system-nipas-act-and-e-expanded-nipas-act-of-the-philippines/146668636>

²⁰ Preservation and protection of our lands is our duty as people, and we are failing miserably. (n.d.), <https://www.greenprint.ph/national-land-use-act>

Conclusion

Striking a delicate balance between development and environmental preservation remains a critical challenge in today's world. As societies progress, the demand for economic growth, infrastructure, and urbanization intensifies. Simultaneously, concerns about ecological degradation, climate change, and the depletion of natural resources escalate. This dilemma prompts us to reflect on the interplay between human progress and the well-being of our planet. In the landmark case of *Oposa v. Factoran*,²¹ the Philippine Supreme Court recognized the doctrine of intergenerational responsibility regarding the environment. The court emphasized that the right to a balanced and healthful ecology is not merely aspirational but also enforceable, even without specific enabling legislation. This ruling underscore the need to harmonize development with environmental stewardship, ensuring a sustainable future for generations to come.

The National Land Use Act (NLUA) offers a holistic approach to land management, potentially mitigating the Nabaoy River conflict by harmonizing development goals with ecological preservation. Its implementation would require collaboration among government agencies, local authorities, and concerned citizens to safeguard this vital water resource.

²¹ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.).

Bibliography

Constitution

1987 Constitution of the Philippines, art. II, §16 (Phil.).

Jurisprudence

Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.).

Online Sources

Albay, Rick Lars. Tensions heighten as Philippine clean energy projects encroach on conservation areas (May 31, 2024),

<https://www.wind-watch.org/news/2024/06/01/tensions-heighten-as-philippine-clean-energy-projects-encroach-on-conservation-areas/>

Arnaldo, M. S. F. Threatening the environmental gains of Boracay | Ma. Stella F. Arnaldo. Business Mirror. (March 2, 2024),

<https://businessmirror.com.ph/2024/02/23/threatening-the-environmental-gains-of-boracay/>

Dela Cruz, Beverly Mae N., National Integrated Protected Areas System (NIPAS) Act and E (expanded) - NIPAS Act of the Philippines (May 20, 2019),

<https://www.slideshare.net/slideshow/national-integrated-protected-areas-system-nipas-act-and-e-expanded-nipas-act-of-the-philippines/146668636>

Harme, J. N., Online and offline, this Aklan village fights a wind farm to save their river. (April 30, 2024), RAPPLER,

<https://www.rappler.com/newsbreak/in-depth/online-offline-nabaoy-community-struggles-renewable-energy-project/>

Harme, J. N. Villagers protest planned wind farm expansion project in Aklan. (February 3, 2024), RAPPLER,

<https://www.rappler.com/philippines/visayas/villagers-protest-planned-wind-farm-expansion-project-aklan-january-31-2024/>

Human Rights to Water and Sanitation (n.d.),

<https://www.unwater.org/water-facts/human-rights-water-and-sanitation>

Job growth and environmental responsibility: Yuchengcos defend PetroWind's expansion in Aklan (n.d.),

<https://bilyonaryo.com/2024/02/21/job-growth-and-environmental-responsibility-yuchengcos-defend-petrowinds-expansion-in-aklan/power/>

-
- Maliao, R. J., Cahilig, R. C., Cahilig, R. R., & Jaspe, B. T.. Climate Change Awareness and Indigenous Knowledge Systems and Practices (Iksp) of Riverine Fishers in the Nabaoy River, Malay, Aklan, Philippines: Linking Local Social Capital to Socio-Ecological Resilience Amidst the Changing Environment and Climate. Malay, Aklan, Philippines: Linking Local Social Capital to Socio-Ecological Resilience Amidst the Changing Environment and Climate. (February 16, 2022), https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4034376
- Nabas Wind Farm. Facebook. (February 19, 2024), <https://www.facebook.com/NabasWindFarm/posts/pfbid0BmwkMpRiP3dcjWgySwKcmEpJFeJQACVQifA86nVQnFVFbEoAgykqxWqSJqG5ivqml>
- PetroWind is fully compliant -- DENR-EMB (n.d.), <https://petroenergy.com.ph/#!/newsitem/51>
- PetroWind's Phase 2 Re-endorsed by Malay SB Following; Support from DENR and Barangay Officials (April 16, 2024), <https://petroenergy.com.ph/#!/newsitem/57>
- PIDS study urges passage of National Land Use Act to address land use conflicts. (February 26, 2024), <https://www.pids.gov.ph/details/news/press-releases/pids-study-urges-passage-of-national-land-use-act-to-address-land-use-conflicts#:~:text=This%20lack%20of%20a%20unified,development%20and%20harmonious%20land%20management>
- Preservation and protection of our lands is our duty as people, and we are failing miserably. (n.d.), <https://www.greenprint.ph/national-land-use-act>
- Raymond Jacinto Sugang, "How the Nabaoy River Maintains a Steady Baseflow Year-Round Despite Summer Conditions, Despite Its Rock Composition's Limited Water Storage Capacity" Facebook (February 17, 2024), <https://www.facebook.com/photo?fbid=7120348081376858&set=a.132826973462372>
- Señeris, G. T. Nabaoy River Watershed potential impact to flooding using Geographic Information System remote sensing. AIMS Environmental Science, 9(3), (2022), <https://www.aimspress.com/article/doi/10.3934/environsci.2022024>

Yuchengco-owned wind farm starts shoring up power supply for Panay. (n.d.). Manila Bulletin,
<https://mb.com.ph/2024/4/5/yuchengco-owned-wind-farm-starts-shoring-up-power-supply-for-panay>

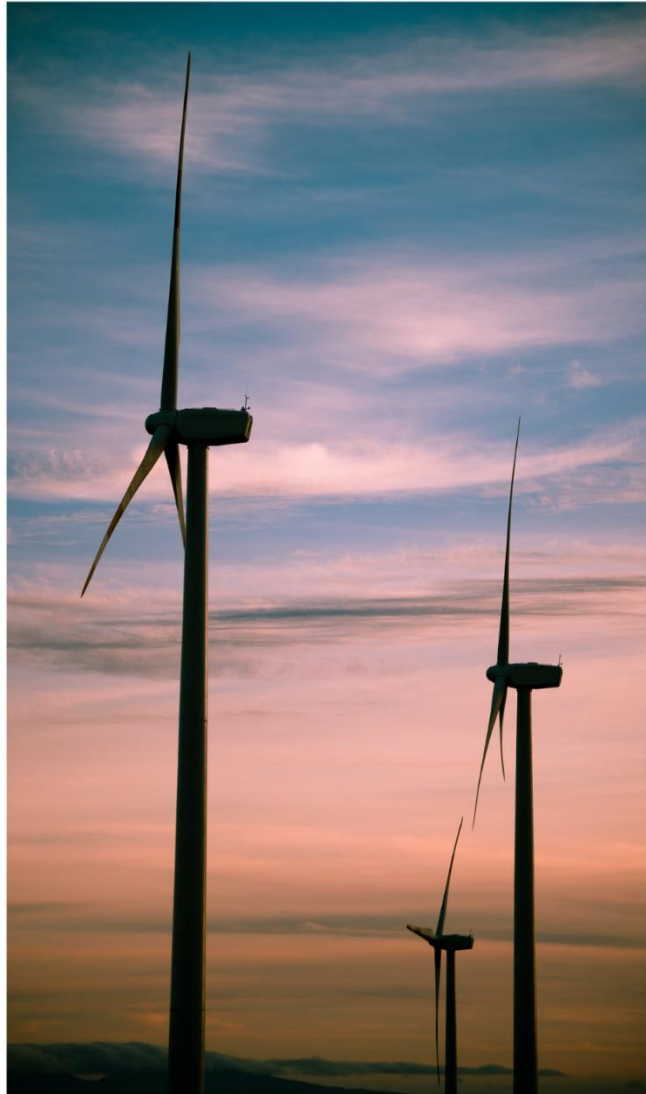


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PARIS AGREEMENT ON CLIMATE CHANGE: RISING SEA LEVELS THREATENING PHILIPPINE CITIES

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“Even if we stop all greenhouse emissions today – which is not exactly realistic – sea levels will continue to rise. That’s not to say that what we do doesn’t matter. In fact, it matters a great deal.”¹

Climate change is a significant issue in the Philippines that is already causing significant economic, social, and environmental impacts. The country is particularly vulnerable to the impacts of climate change due to its location and its high population density and one of the environmental issues is the sea level rise because of climate change.²

The Philippines aims to combat climate change and adapt to its effects. Thus, Philippines was a signatory to the Paris Agreement, a landmark treaty in the multilateral climate change process which is a binding agreement for all nations together to combat climate change and adapt to its effects.³

Moreover, the 1987 Philippine Constitution upholds preservation and protection of our patrimony in its Preamble, and the right to a balanced and healthful ecology enshrined in Article II, Section 16 which provides that *“The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.”⁴*

Moreover, the Supreme Court in the case of *Oposa v. Factoran* stated that *“the right to a balanced and healthful ecology need not even be written in the Constitution for it is assumed, like other civil and political rights guaranteed in the Bill of Rights, to exist from the inception of mankind and it is an issue of transcendental importance with intergenerational implications.”⁵*

Thus, the government must take steps to mitigate greenhouse gas emissions and adapt to the changing climate, but much more needs to be done to reduce the

¹ Sebesvari, Rita, Sea level rise is inevitable – but what we do today can still prevent catastrophe for coastal regions. (September 26, 2019) <https://theconversation.com/sea-level-rise-is-inevitable-but-what-we-do-today-can-still-prevent-catastrophe-for-coastal-regions-124129>

² Philippine Atmospheric Geophysical and Astronomical Services Administration (PAG ASA) (n.d.), <https://www.pagasa.dost.gov.ph/information/climate-change-in-the-philippines>

³ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.), <https://unfccc.int/process-and-meetings/the-paris-agreement>

⁴ CONST. (1987), art. II, §16 (Phil.).

⁵ *Oposa v. Factoran*, G.R. No. 101083, (July 30, 1993) (Phil.).

impacts of climate change on the country and its people. The Philippines must continue to prioritize climate action to build a more sustainable and resilient future.

Climate Change and its Impact in the Philippines

Today, the worst impacts are mostly felt in places where sea level rise has combined with sinking land, also known as subsidence. Subsidence is a natural process, but the weight of high-rise buildings and the emptying of underground water reservoirs tends to speed it up. That's why sinking coastal megacities may suffer a great deal from just 20-40cm of sea level rise.⁶

Climate change is having serious impacts in the Philippines such as increased frequency and severity of natural disasters, sea level rise, extreme rainfall, resource shortages, and environmental degradation.⁷ All of these impacts together have greatly affected the Philippines' agriculture, water, infrastructure, human health, and coastal ecosystems and they are projected to continue having devastating damages to the economy and society of the Philippines.⁸

According to the UN Office for the Coordination of Humanitarian Affairs (OCHA), the Philippines is one of the most disaster-prone countries in the world.⁹ The archipelago is situated along the Pacific Ocean's typhoon belt, leaving the country vulnerable to around 20 typhoons each year, a quarter of which are destructive.¹⁰ The December 2021 typhoon known colloquially as Typhoon Odette caused around a billion dollars (₱51.8 billion) in infrastructure and agricultural damages and displaced about 630,000 people. The United Nations estimated that Typhoon Odette affected the livelihoods of 13 million people, destroying their homes and leaving them without adequate food or water supplies.¹¹ More tragically, the

⁶ Sebesvari, Rita Sea level rise is inevitable – but what we do today can still prevent catastrophe for coastal regions (September 26, 2019), <https://theconversation.com/sea-level-rise-is-inevitable-but-what-we-do-today-can-still-prevent-catastrophe-for-coastal-regions-124129>

⁷ United States Agency for International Development (USAID) CLIMATE CHANGE RISK PROFILE PHILIPPINES (n.d.), https://www.climatelinks.org/sites/default/files/asset/document/2017_Climate%20Change%20Risk%20Profile_Philippines.pdf

⁸ *Id.*

⁹ UNOCHA "About OCHA in The Philippines" (March 2019) <https://www.unocha.org/philippines>

¹⁰ Asian Disaster Reduction Center Information on Disaster Risk Reduction of the Member Countries (n.d.) <https://www.adrc.asia/nationinformation.php?NationCode=608&Lang=en>

¹¹ ABS-CBN News "PAGASA redefines 'super typhoon', revises wind signals". (March 23, 2022) <https://news.abs-cbn.com/news/03/23/22/pagasa-redefines-super-typhoon-tweaks-wind-signals>

physical and economic repercussions of Typhoon Odette led to the death of over 400 people as of December 2021.¹²

Compounding these issues, the impacts of climate change, such as accelerated sea level rise, exacerbate the state's high susceptibility to natural disasters, like flooding and landslides.¹³ Aside from geography, climate change impacts regions with a history of colonization more intensely than regions without a history of colonization.¹⁴ Colonized regions experience the repercussions of climate change most jarringly "because of their high dependence on natural resources, their geographical and climatic conditions and their limited capacity to effectively adapt to a changing climate."¹⁵ Since low-income countries have a history of colonialism and resource exploitation, their environment lacks the diversity necessary to prevail against natural disasters.¹⁶ A lack of biodiversity reduces the resilience of a specific region, leaving them more susceptible to natural disasters and the effects of climate change. With its history of Spanish colonization, the Philippines is not environmentally nor economically equipped to overcome issues it is currently dealing with, such as natural disasters and climate change. This inability to recover exacerbates the problem, creating a cycle of environmental and economic devastation in the country.¹⁷

The Philippines said it was revising up its target to cut greenhouse gas emissions to a 75% reduction by 2030 under its commitment to the Paris Agreement on Climate Change, up from a target of 70% set four years ago.¹⁸

Finance Secretary Carlos Dominguez III, who also chairs the country's Climate Change Commission, described the new target as "ambitious", though some environmental activists called on the government to do more.¹⁹

¹² Typhoon Rai. Wikipedia https://en.wikipedia.org/w/index.php?title=Typhoon_Rai&oldid=1087095376

¹³ Perez, Rosa T.; et al., "Climate Change Impacts and Responses in the Philippines Coastal Sector" (August 1999), *Climate Research*, 12 (2/3): 97–107, Bibcode:1999ClRes..12...97P, doi:10.3354/cr012097, JSTOR 24866004

¹⁴ Smith, Paul. "Climate Change, Weak States and the "War on Terrorism" in South and Southeast Asia" (2007). *Contemporary Southeast Asia*. 29 (2): 264–285. doi:10.1355/CS29-2C. ISSN 0129-797X. JSTOR 25798831.

¹⁵ *Id.*

¹⁶ Das Gupta, Monica. "Population, Poverty, and Climate Change". (2014). *The World Bank Research Observer*. 29 (1): 83–108. doi:10.1093/wbro/lkt009. hdl:10986/22565. ISSN 0257-3032. JSTOR 24582389.

¹⁷ *Id.*

¹⁸ Reuters Philippines raises carbon emission reduction target to 75% by 2030 | Philippines raises carbon emission reduction target to 75% by 2030 | Reuters (2021) <https://www.reuters.com/business/environment/philippines-raises-carbon-emission-reduction-target-75-by-2030-2021-04-16/>

¹⁹ *Id.*

Breaking down the target, 72.29% is conditional on the support of climate finance, technologies and capacity development provided by developed countries, as prescribed by the Paris Agreement.²⁰

The Paris Agreement, which came into force in 2016, aims to transform the world's fossil-fuel-driven economy within decades and slow the pace of a global temperature increase to well below 2 degrees Celsius.

The Philippines ratified the agreement in 2017, allowing Manila access to the Green Climate Fund, which aims to channel billions of dollars to help poor nations tackle global warming.²¹

The remaining 2.71% of the target should be implemented mainly through domestic resources, the Department of Finance said in a statement.²²

Under a "business-as-usual" scenario, the government estimates the Southeast Asian country's overall emissions during the 2020-2030 period will be 3,340.3 metric tons of carbon dioxide equivalent (MTCO2e).²³

Greenpeace, however, called on the government to do more and for the Department of Energy to speed up the country's energy transition.

"Following the administration's pronouncements to fast-track renewable energy and for a moratorium on new coal proposals, the energy department should be doing much more to put their words into action," the environmental organization said in a statement.²⁴

Sea Level Rise

Globally, an estimated 150 million people live in communities that will be submerged in water by 2050. One of the problems of environmental issues is the sea level rise. The rise in sea level is an increase in the level of the world's oceans because of global warming. Burning fossil fuels is one of the causes of global warming because it releases carbon dioxide and other heat-trapping gases into the atmosphere. The oceans then absorb most of this heat. Sea levels are rising because of climate change. This rise is likely to accelerate over the coming century and continue for centuries. The impacts of sea level rise include permanent flooding (inundation) of low-lying

²⁰ Reuters Philippines raises carbon emission reduction target to 75% by 2030 | Philippines raises carbon emission reduction target to 75% by 2030 | Reuters (2021) <https://www.reuters.com/business/environment/philippines-raises-carbon-emission-reduction-target-75-by-2030-2021-04-16/>

²¹ *Id.*

²² *Id.*

²³ *Id.*

²⁴ *Id.*

areas, and increased frequency, extent and depth of tidal inundation. Sea level rise will also cause most sandy beaches to recede (where beaches will move further inland) and erode.²⁵

But the sea level rise we see today is the consequence of global warming that started from emissions released decades ago. Large bodies of water like the oceans have a great heat capacity – they warm up slowly and keep that temperature for a long time. This means that changes in sea level lag warming of the atmosphere.²⁶ It was forecasted that over 410 million people could be at risk from rising sea levels by 2100. The climate crisis has already caused global sea levels to rise by over 10cm between 1993 and 2024. In its Global Risks Report 2024, the World Economic Forum identified sea level rise from collapsing ice sheets as a key threat over the coming decade. Homes, livelihoods and ultimately lives are under threat from rising sea levels. By 2100, up to 410 million people could be at risk from coastal flooding as the climate crisis causes sea levels to rise even higher.²⁷

Impacts of Sea Level Rise in the Philippines

The Philippines is among the countries most affected by rising sea levels²⁸ and is not spared as it is also experiencing sea-level rise, which is threatening more than 40% of the population that live in coastal areas. The rising sea levels are causing increased flooding, erosion, and salinization of freshwater resources. These impacts have significant economic, social, and environmental consequences, including the displacement of people and destruction of coastal infrastructure. To address the impacts of climate change, the Philippine government has taken steps to mitigate greenhouse gas emissions and adapt to the changing climate. The country has committed to reducing its greenhouse gas emissions by 75% by 2030, which will require significant changes in the energy and transportation sectors.²⁹

The children are the most vulnerable. As of 2023, records from three decades show sea levels rising by 7 to 10 centimeters per decade along Philippine coasts, which is three times faster than the world average.³⁰

²⁵ World Economic Forum, Sea level rise: Everything you need to know (July 11, 2024)., <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

²⁶ *Id.*

²⁷ *Id.*

²⁸ Garcia, Ma Angelica, "LOOK: How rising sea levels could affect PHL cities, towns by 2050" (November 4, 2019), <https://www.gmanetwork.com/news/scitech/science/714173/how-rising-sea-levels-could-affect-phl-cities-towns-by-2050/story/>

²⁹ *Id.*

³⁰ Ines, Jezreel "In troubled waters: Rising sea levels threaten sinking town's survival". (July 4, 2023), <https://www.rappler.com/environment/in-troubled-waters-rising-sea-levels-threaten-sinking-town-survival/>

Philippine coastlines, one of the longest in the world, stretch more than 36,000 kilometers. Coastal areas and their communities are threatened by rising sea levels and gradual submersion. These endanger the livelihood of fishing communities as well as the country's food security.³¹

The report of the United Nations-backed Intergovernmental Panel on Climate Change released in September showed that the world's oceans and cryosphere—glaciers and polar regions—are getting warmer, more acidic and less productive.

In only three decades, areas in the Philippines now home to around 6.8 million people will likely be inundated as the sea levels continue to rise.³² By the end of the century, land currently occupied by 8.6 million Filipinos could be lower than the height of average annual coastal flood. The figures are among the findings of the study produced by New Jersey-based science organization Climate Central.

The research, published in the journal *Nature Communications*, found that 250 million people across the globe currently live on land below current annual flood levels and 110 million live below the high tide line at present.³³

Using a more accurate way of calculating land elevation called CoastalDEM, the study of authors Scott Kulp and Benjamin Strauss suggests there are 5.4 million Filipinos already occupying land below annual flood levels. Annual flood level is used to denote the water level at the shoreline that local coastal floods exceed on average once per year.³⁴

Conservative Estimates. The new projections for the Philippines—an archipelago in the Pacific Ocean—are based on a relatively conservative scenario of expected changes. This assumes humans will moderately reduce warming emissions roughly in line with the 2015 Paris climate agreement's two-degree Celsius target and there will be a fairly stable Antarctica. The numbers also used by the authors were based on a 2010 population data—100 million.³⁵

³¹ Ines, Jezreel "In troubled waters: Rising sea levels threaten sinking town's survival". (July 4, 2023), <https://www.rappler.com/environment/in-troubled-waters-rising-sea-levels-threaten-sinking-town-survival/>

³² Philstar, In 30 years, rising seas will threaten Philippine cities, towns home to 6.8 million (November 6, 2019) <https://www.philstar.com/headlines/2019/11/06/1966508/30-years-rising-seas-will-threaten-philippine-cities-towns-home-68m#:~:text=Sea%20levels%20are%20projected%20to%20rise%20nearly%20half%20a%20meter>

³³ Kulp, Scott A, et al., New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding (October 29, 2019), <https://www.nature.com/articles/s41467-019-12808-z>

³⁴ *Id.*

³⁵ *Id.*

But assuming the high greenhouse gas-emissions continue unabated along with Antarctic instability, seven million Filipinos live on land that could be threatened by inundation by mid-century. By 2100, the number will rise to 13 million.³⁶

Seven million Filipino below high tide line by 2050. The new data showed that 4.3 million people in the Philippines live in places that are below the high tide line. Tideline is used to indicate the long-term average of the highest daily local tide level. *“This suggests that this land is protected today, possibly by coastal defenses. It is also possible that our data underestimate elevation in some places,”* Peter Girard, director of communications at Climate Central.³⁷

Even under a highly optimistic scenario, 5.2 million Filipinos will occupy land that will be underwater at high tide by 2050. The number will increase to 6.9 million by the end of the century.³⁸

If emissions continue to rise, land currently home to 5.4 million Filipinos will fall below the level of high tide by mid-century. By 2100, the number will double to 11 million.³⁹

Sea Level Rise affecting Iloilo City

The sea level in the Philippines is rising three times faster than the global average, putting many of its coastal villages in peril, according to a climate scientist of the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA).⁴⁰

*“Based on a report done by Pagasa, we saw that the sea-level rise of the Philippines is rising three times faster compared to the global average.” “So, this can cause inundation of low-lying areas, especially many of our countrymen who live on the beach.”*⁴¹

The World Bank said rising sea levels “create not only stress on the physical coastline, but also on coastal ecosystems.”⁴²

³⁶ Kulp, Scott A, et al., New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding (October 29, 2019), <https://www.nature.com/articles/s41467-019-12808-z>

³⁷ *Id.*

³⁸ *Id.*

³⁹ *Id.*

⁴⁰ Corales, Nestor PH Sea Level Rising 3 times faster than World Average (September 23, 2022), <https://newsinfo.inquirer.net/1669093/ph-sea-level-rising-3-times-faster-than-world-average-says-expert>

⁴¹ *Id.*

⁴² *Id.*

The urbanized center of Iloilo City suffers from the same diminishing rates, particularly in the easter coastal districts of La Paz to Villa Arevalo. Here, the highest rates of subsidence were found within Mandurriao and Molo Districts and La Paz District. Such subsidence has also caused worse flooding and tidal disruption in the city proper.⁴³ Iloilo City is not spared as sea levels are projected to rise nearly half a meter by 2100 if Earth warms 2 degrees Celsius above pre-industrial levels and 84 centimeter in a 3C to 4C world.⁴⁴

Sea Level Measurement

Sea level is the measurement of the sea's surface height. Between the 1800s and early 1990s, tide gauges attached to structures such as piers measured global sea level, as research organization the Smithsonian Institution explains. Now satellites carry out this task by bouncing radar signals off the ocean's surface.⁴⁵

Because local weather conditions and other factors can affect sea level, measurements are taken globally and then averaged out.

In 2021, scientists discovered a sea-level "fingerprint" from the Greenland ice sheet, confirming fears about the extent to which ice is melting. These fingerprints are "detectable patterns of sea level variability around the world resulting from changes in water storage on Earth's continents and in the mass of ice sheets", according to NASA.⁴⁶

The Greenland ice sheet is now losing around 30 million tons of ice an hour, reports the Guardian, and a recent study predicts its total collapse could happen by 2025.⁴⁷

Extent of the Sea Level Rise

With the ice sheet at "*a tipping point of irreversible melting*", scientists currently expect an unavoidable sea level rise of 1-2 meters.⁴⁸

⁴³Esquire Mag Philippine Cities Are Sinking Faster Than We Realize. What Now? (August 27, 2024), <https://www.esquiremag.ph/politics/news/land-subsidence-climate-change-study-philippine-cities-a1057-20240827?s=5s8nj6gf8k6qrh22s92ef389ln>

⁴⁴ *Id.*

⁴⁵ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁴⁶ *Id.*

⁴⁷ *Id.*

⁴⁸ *Id.*

Global sea levels have already risen by over 10cm between 1993 and 2024, according to NASA, which says sea levels have been rising at unprecedented rates over the past 2,500 years.⁴⁹

The US Space Agency and other US government agencies warned in 2022 that levels along the country's coastlines could rise by another 25-30cm by 2050.⁵⁰

The global sea level has risen by about 21cm since records began in 1880. While measuring in centimeters or even millimeters might seem small, these rises can have big consequences. This is particularly true where storm surges sweep further inland than they would have previously.⁵¹

Causes of Sea Level Rise

Two main factors cause sea level rise. These are melting ice from glaciers, and seawater expanding because of rises in global temperatures, explains the US National Oceanic and Atmospheric Administration.⁵²

Greenland and Antarctica are losing around 270 billion and 150 billion tons of ice a year, respectively.⁵³ There are also likely negative feedback loops that could speed up glacier ice melt. For example, the Thwaites Glacier in Antarctica is disintegrating more quickly than anticipated. It's nicknamed the 'doomsday glacier' because sea levels could rise more than three meters without it and its supporting ice shelves.⁵⁴

Heat stored in the ocean is responsible for between a third and half of global sea level rise, NASA says. The past decade has been the ocean's warmest since at least 1800, and ocean temperatures reached a new high in 2023.⁵⁵ Because local weather conditions and other factors can affect sea level, measurements are taken globally and then averaged out.⁵⁶

⁴⁹ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁵⁰ *Id.*

⁵¹ *Id.*

⁵² *Id.*

⁵³ *Id.*

⁵⁴ *Id.*

⁵⁵ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁵⁶ *Id.*

Philippines as a signatory to the Paris Agreement on Climate Change

The Paris Agreement is a legally binding international treaty on climate change. It was adopted by 196 Parties at the UN Climate Change Conference (COP21) in Paris, France, on 12 December 2015. It entered into force on November 4, 2016.⁵⁷

Its overarching goal is to hold “*the increase in the global average temperature to well below 2°C above pre-industrial levels*” and pursue efforts “to limit the temperature increase to 1.5°C above pre-industrial levels.”⁵⁸

However, in recent years, world leaders have stressed the need to limit global warming to 1.5°C by the end of this century. That’s because the UN’s Intergovernmental Panel on Climate Change indicates that crossing the 1.5°C threshold risks unleashing far more severe climate change impacts, including more frequent and severe droughts, heat waves and rainfall. To limit global warming to 1.5°C, greenhouse gas emissions must peak before 2025 at the latest and decline 43% by 2030.⁵⁹

The Philippines has already taken steps to pursue national actions to address climate change through its intended nationally determined contribution (INDC).⁶⁰

Despite financial limitations, Philippines has already demonstrated its commitment to meeting its pledge with the implementation of the national climate action plan and the integration of climate change in national and local development plans and budgets. It will continue to cascade climate change mitigation and adaptation actions to sub-national levels and to invest in climate – resilient local economies, consistent with post-2015 international frameworks such as the 2030 agenda on sustainable development, Sendai framework for disaster risk reduction, and most especially, the Paris agreement on climate change.⁶¹

To secure a sustainable future, the Philippines is pursuing value-adding strategies to ensure the provision of ecosystem services and green growth to address pollution and environmental degradation. To this end, a greenhouse gas inventory management and reporting system is being developed to create a transparent, accurate and comparable baseline of emissions.⁶²

⁵⁷ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.), <https://unfccc.int/process-and-meetings/the-paris-agreement>

⁵⁸ *Id.*

⁵⁹ World Economic Forum Sea level rise: Everything you need to know. (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>

⁶⁰ Department of Environment and Natural Resources (DENR) (n.d.), <https://ncr.denr.gov.ph/news-events/ph-signs-the-paris-climate-change-agreement/>

⁶¹ *Id.*

⁶² *Id.*

It has also submitted recently its articles of accession to the Doha amendment to the Kyoto protocol.

As a highly vulnerable country, Philippines has committed early to contributing its fair share to the global action consistent with the climate vulnerable forum's Manila-Paris declaration which calls for a mitigation goal of below 1.5 degrees Celsius and full decarbonization and 100% renewable energy production by 2050.⁶³

Too much is at stake. Developed countries therefore need to do more in terms of dramatically raising their NDC ambition compatible with the 1.5-degrees Celsius threshold, as well as raising funding contribution to the climate action plans of vulnerable countries, which require adequate, predictable and sustainable financing. The Philippines also urges developed and developing countries alike to take urgent pre-2020 action.⁶⁴

Climate Change Commission (CCC)

Climate Change Commission⁶⁵ is the lead policy-making body of the government tasked to coordinate, monitor and evaluate government programs and ensure mainstreaming of climate change in national, local, and sectoral development plans towards a climate-resilient and climate-smart Philippines.⁶⁶

Vision

The Commission envisions a climate-resilient and climate-smart Philippines with highly adaptive communities.⁶⁷

Mission

To take the lead in the development and mainstreaming of evidence-based climate adaptation and mitigation policies through optimum coordination among key stakeholders towards achieving a climate-resilient and climate-smart Philippines with healthy, safe, prosperous, and self-reliant communities.

The Climate Change Commission (CCC) has vowed to exert its utmost efforts to finalize expeditiously the Philippines' commitment to the Paris agreement to reduce carbon emissions and adapt to the impacts of climate change.⁶⁸

⁶³ Department of Environment and Natural Resources (DENR) (n.d.), <https://ncr.denr.gov.ph/news-events/ph-signs-the-paris-climate-change-agreement/>

⁶⁴ *Id.*

⁶⁵ Climate Change Commission (n.d.), <https://climate.gov.ph/about-us>

⁶⁶ *Id.*

⁶⁷ *Id.*

⁶⁸ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/cc-c-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

*"The country's first Nationally Determined Contribution (NDC) under the Paris Agreement shall convey the shared vision and collective strategies of our sectors for the immediate pursuit of low carbon development and a sustainable future for the Filipino nation," the CCC said.*⁶⁹

*"It shall also communicate to the developed world our readiness for transformative green investments and finance, consistent with the President's call for climate justice," it noted.*⁷⁰

The CCC, pursuant to its mandate under the law, has spearheaded the development and finalization of the NDC through a whole-of-government-and-society approach.⁷¹

*"We shall ensure that the NDC reflects information that is not only evidence-based, but also verifiable and accurate," it said.*⁷²

To ensure the technical integrity of the NDC, the Commission noted that the national government, with support from development partners, has devoted expertise, time, and resources to strengthen the capacities of the lead sectoral agencies to undertake the NDC analytics.⁷³

Moreover, the CCC said government agencies have required more time to develop and enhance the sectoral policies and measures, and to determine the country's economy-wide target for emission reduction and avoidance.⁷⁴

However, some civil society organizations had earlier expressed their discontent on the country's unambitious proposed commitment to reduce greenhouse gas (GHG) emissions to address climate change. They were citing the CCC's Dec. 23, 2020 presentation of the country's three-page draft NDC text.⁷⁵

In the latest version, the Philippines aims to reduce its GHG emissions by 30 percent by 2040 from its business-as-usual scenario (2020-2040), with at least two percent of these reductions to be done unconditionally. The groups highlighted that the reduction of 30 percent of GHG by 2040 is much lower than what the country had

⁶⁹ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

⁷⁰ *Id.*

⁷¹ *Id.*

⁷² *Id.*

⁷³ *Id.*

⁷⁴ *Id.*

⁷⁵ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

committed to reduce in its Intended Nationally Determined Contribution or INDC in 2015. The Philippines then had committed to reduce by 70 percent its carbon emissions by 2030 to achieve a below two degrees Celsius increase in global temperatures.⁷⁶

Climate Finance

Climate finance refers to local, national or transnational financing—drawn from public, private and alternative sources of financing—that seeks to support mitigation and adaptation actions that will address climate change. The Convention, the Kyoto Protocol and the Paris Agreement call for financial assistance from Parties with more financial resources to those that are less endowed and more vulnerable. This recognizes that the contribution of countries to climate change and their capacity to prevent it and cope with its consequences vary enormously. Climate finance is needed for mitigation, because large-scale investments are required to significantly reduce emissions. Climate finance is equally important for adaptation, as significant financial resources are needed to adapt to the adverse effects and reduce the impacts of a changing climate.⁷⁷

In accordance with the principle of “common but differentiated responsibility and respective capabilities” set out in the Convention, developed country Parties are to provide financial resources to assist developing country Parties in implementing the objectives of the UNFCCC. The Paris Agreement reaffirms the obligations of developed countries, while for the first time also encouraging voluntary contributions by other Parties. Developed country Parties should also continue to take the lead in mobilizing climate finance from a wide variety of sources, instruments and channels, noting the significant role of public funds, through a variety of actions, including supporting country-driven strategies, and taking into account the needs and priorities of developing country Parties. Such mobilization of climate finance should represent a progression beyond previous efforts.⁷⁸

It is important for all governments and stakeholders to understand and assess the financial needs of developing countries, as well as to understand how these financial resources can be mobilized. Provision of resources should also aim to achieve a balance between adaptation and mitigation.⁷⁹

Overall, efforts under the Paris Agreement are guided by its aim of making finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development. Assessing progress in provision and mobilization of support is also part of the global stock take under the Agreement. The Paris Agreement

⁷⁶ De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>

⁷⁷ United Nations Framework Convention on Climate Change (UNFCCC) (n.d.) <https://unfccc.int/topics/introduction-to-climate-finance>

⁷⁸ *Id.*

⁷⁹ *Id.*

also places emphasis on the transparency and enhanced predictability of financial support.⁸⁰

Climate Financial Mechanism & Sources of Fund

To facilitate the provision of climate finance, the Convention established a financial mechanism to provide financial resources to developing country Parties. The financial mechanism also serves the Kyoto Protocol and the Paris Agreement.⁸¹

The Convention states that the operation of the financial mechanism can be entrusted to one or more existing international entities. The Global Environment Facility (GEF) has served as an operating entity of the financial mechanism since the Convention's entry into force in 1994. At COP 16, in 2010, Parties established the Green Climate Fund (GCF) and in 2011 also designated it as an operating entity of the financial mechanism. The financial mechanism is accountable to the COP, which decides on its policies, program priorities and eligibility criteria for funding.⁸²

In addition to providing guidance to the GEF and the GCF, Parties have established two special funds—the Special Climate Change Fund (SCCF) and the Least Developed Countries Fund (LDCF), both managed by the GEF—and the Adaptation Fund (AF) established under the Kyoto Protocol in 2001.⁸³

At the Paris Climate Change Conference in 2015, the Parties agreed that the operating entities of the financial mechanism – GCF and GEF – as well as the SCCF and the LDCF shall serve the Paris Agreement. Regarding the Adaptation Fund serving the Paris Agreement negotiations are underway in the Ad hoc Working Group on the Paris Agreement (APA).⁸⁴

Nationally Determined Contribution (NDC), the NDC Implementation Plan (NDCIP) 2020-2030⁸⁵

To further realize the Philippines' set ambition in the first Nationally Determined Contribution (NDC), the NDC Implementation Plan (NDCIP) 2020-2030 outlines key actions, phasing, costs, and implementation arrangements for delivering

⁸⁰ United Nations Framework Convention on Climate Change (UNFCCC) (n.d.) <https://unfccc.int/topics/introduction-to-climate-finance>

⁸¹ *Id.*

⁸² *Id.*

⁸³ *Id.*

⁸⁴ *Id.*

⁸⁵ Implementation Plan for the Republic of the Philippines Nationally Determined Contribution (NDC) (2020-2030), [https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines' %20NDC%20Implementation%20Plan%20-%2020072024.pdf](https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines'%20NDC%20Implementation%20Plan%20-%2020072024.pdf)

the Philippines' NDC under the identified policies and measures in the transport, energy, waste, industry, and agriculture sectors.⁸⁶

In 2021, the Philippines submitted its Nationally Determined Contribution (NDC) to the United Nations Framework Convention on Climate Change (UNFCCC). Through the NDC, the Philippines commits to a projected greenhouse gas (GHG) emissions reduction and avoidance of 75%, for 2020 to 2030, of which 2.71% is unconditional and 72.29% is conditional. This NDC Implementation Plan (NDCIP) sets out a roadmap and actions for implementing the Philippines' NDC.⁸⁷

It outlines policies and measures (PAMs) to reduce or avoid emissions in five sectors: agriculture, waste, industry, transport, and energy. NDC PAMs include actions to be implemented in each sector, with implementation overseen by four sector departments: the Department of Agriculture (DA), Department of Energy (DOE), Department of Environment and Natural Resources (DENR), and Department of Transportation (DOTr). The PAMs were developed through a bottomup approach, building actions and needs at the sector level, to drive necessary action at the national level.⁸⁸

Regular updates to the plan will align with modifications to the NDC, ensuring continued relevance and efficacy.⁸⁹

Nationally Determined Contribution Policies and Measures

The initial phase of mitigation actions, encapsulated as NDC PAMs, are articulated in this plan. These initiatives set the Philippines on a pathway to low-carbon growth, delivering, once fully implemented, an aggregate reduction of approximately 990 million metric tons of carbon dioxide equivalent (mmtCO₂e) against the baseline. The delivery of the PAMs thus ensures that the unconditional NDC target of 2.71% is met and considerably exceeded, while also contributing to the partial fulfillment of the conditional target.⁹⁰

As additional support becomes available, the PAMs will be further enhanced through periodic reviews, enabling reductions toward achieving the NDC's conditional target. This plan will then be updated to reflect changes to and strengthen current and future PAMs.⁹¹

⁸⁶ Implementation Plan for the Republic of the Philippines Nationally Determined Contribution (NDC) (2020-2030), [https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines' %20NDC%20Implementation%20Plan%20-%20072024.pdf](https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines'%20NDC%20Implementation%20Plan%20-%20072024.pdf)

⁸⁷ *Id.*

⁸⁸ *Id.*

⁸⁹ *Id.*

⁹⁰ *Id.*

⁹¹ *Id.*

Future projections

Future projections for the current trajectory of climate change predict that global warming is likely to exceed 3 °C, potentially 4 °C, by 2060.⁹² Specifically in the Philippines, average temperatures are "virtually certain" to see an increase of 1.8 to 2.2 °C.⁹³ This temperature increase will stratify the local climate and cause the wet and dry seasons to be wetter and drier, respectively.⁹⁴ Most areas in the Philippines will see reduced rainfall from March to May, while Luzon and Visayas will see increased heavy rainfall.⁹⁵ There will also be an increase in: the number of days that exceed 35 °C; that have less than 2.5 mm of rainfall; and that have more than 300mm of rainfall.⁹⁶ Additionally, climate change will continue to increase the intensity of typhoons and tropical storms. Sea levels around the Philippines are projected to rise 0.48 to 0.65 meters by 2100, which exceeds the global average for rates of sea level rise. Combined with sea level rise, this stratification into more extreme seasons and climates increases the frequency and severity of storm surge, floods, landslides, and droughts. These exacerbate risks to agriculture, energy, water, infrastructure, human health, and coastal ecosystems.

Climate Change: The “greatest threat affecting our nations and people”

Former President Rodrigo Duterte called on industrialized nations to pay smaller countries like the Philippines reparations for the damage caused by climate change.⁹⁷

“We produce the least but we suffer the most. We are really the one paying for it the most. The pollution produced by other countries, we are the ones who catch it,” he said.⁹⁸

Addressing world leaders at the United Nations General Assembly in New York, President Marcos amplified his predecessor’s point, calling climate change the “greatest threat affecting our nations and peoples.”

⁹² World Bank. Getting a Grip on Climate Change in the Philippines (June 25, 2013). <https://www.worldbank.org/en/country/philippines/publication/getting-a-grip-on-climate-change-in-the-philippines>

⁹³ *Id.*

⁹⁴ United States Agency for International Development (USAID) CLIMATE CHANGE RISK PROFILE PHILIPPINES. (n.d.), https://www.climatelinks.org/sites/default/files/asset/document/2017_Climate%20Change%20Risk%20Profile_Philippines.pdf

⁹⁵ *Id.*

⁹⁶ *Id.*

⁹⁷ Galvez, Daphne, Rich countries must pay for climate change damage – Duterte. (April 19, 2022) <https://newsinfo.inquirer.net/1584618/rich-countries-must-pay-for-climate-change-damage-duterte#ixzz8mBA6HJCM>

⁹⁸ *Id.*

*“The Philippines, for example, is a net carbon sink. We absorb more carbon dioxide than we emit. And yet, we are the fourth most vulnerable country to climate change,” he said.*⁹⁹

*“There is no other problem so global in nature that it requires a united effort, one led by the United Nations. The effects of climate change are uneven and reflect a historical injustice,” Marcos said.*¹⁰⁰

In response to calls from the leaders of developing nations, including the Philippines, United Nations Secretary General Antonio Guterres said that *“it was time for “meaningful action” on the issue of compensation for damage wrought by the climate crisis.*”¹⁰¹

Recommendations

The government has been implementing measures to increase the resilience of the population to the impacts of climate change. These measures include the construction of flood control infrastructure, disaster risk reduction policies, and programs to promote sustainable agriculture but more should be done.

Implementation of the Paris Agreement requires economic and social transformation, based on the best available science. The Paris Agreement works on a five-year cycle of increasingly ambitious climate action -- or, ratcheting up -- carried out by countries. Since 2020, countries have been submitting their national climate action plans, known as nationally determined contributions (NDCs). Each successive NDC is meant to reflect an increasingly higher degree of ambition compared to the previous version.¹⁰²

Recognizing that accelerated action is required to limit global warming to 1.5°C, the COP27 cover decision requests Parties to revisit and strengthen the 2030 targets in their NDCs to align with the Paris Agreement temperature goal by the end of 2023, considering different national circumstances.¹⁰³

In their NDCs, countries communicate actions they will take to reduce their greenhouse gas emissions to reach the goals of the Paris Agreement. Countries also

⁹⁹ Galvez, Daphne, Rich countries must pay for climate change damage – Duterte. (April 19, 2022) <https://newsinfo.inquirer.net/1584618/rich-countries-must-pay-for-climate-change-damage-duterte#ixzz8mBA6HJCM>

¹⁰⁰ *Id.*

¹⁰¹ *Id.*

¹⁰² *Id.*

¹⁰³ *Id.*

communicate in their NDCs actions they will take to build resilience to adapt to the impacts of climate change.¹⁰⁴

Long-Term Strategies: To better frame the efforts towards the long-term goal, the Paris Agreement invites countries to formulate and submit long-term low greenhouse gas emission development strategies (LT-LEDS).¹⁰⁵

LT-LEDS provide the long-term horizon to the NDCs. Unlike NDCs, they are not mandatory. Nevertheless, they place the NDCs into the context of countries' long-term planning and development priorities, providing a vision and direction for future development.¹⁰⁶

Paris Agreement support. The Paris Agreement provides a framework for financial, technical and capacity building support to those countries that need it,¹⁰⁷ to wit:

Finance. The Paris Agreement reaffirms that developed countries should take the lead in providing financial assistance to countries that are less endowed and more vulnerable, while for the first time also encouraging voluntary contributions by other Parties. Climate finance is needed for mitigation, because large-scale investments are required to significantly reduce emissions. Climate finance is equally important for adaptation, as significant financial resources are needed to adapt to the adverse effects and reduce the impacts of a changing climate.¹⁰⁸

Technology. The Paris Agreement speaks of the vision of fully realizing technology development and transfer for both improving resilience to climate change and reducing GHG emissions. It establishes a technology framework to provide overarching guidance to the well-functioning Technology Mechanism. The mechanism is accelerating technology development and transfer through its policy and implementation arms.¹⁰⁹

Capacity-Building. Not all developing countries have sufficient capacities to deal with many of the challenges brought by climate change. As a result, the Paris Agreement places great emphasis on climate-related capacity-building for developing countries and requests all developed countries to enhance support for capacity-building actions in developing countries.¹¹⁰

¹⁰⁴ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.) <https://unfccc.int/process-and-meetings/the-paris-agreement>

¹⁰⁵ *Id.*

¹⁰⁶ *Id.*

¹⁰⁷ *Id.*

¹⁰⁸ *Id.*

¹⁰⁹ *Id.*

¹¹⁰ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.) <https://unfccc.int/process-and-meetings/the-paris-agreement>

¹¹⁰ *Id.*

Tracking Progress. With the Paris Agreement, countries established an enhanced transparency framework (ETF). Under ETF, starting in 2024, countries will report transparently on actions taken and progress in climate change mitigation, adaptation measures and support provided or received. It also provides for international procedures for the review of the submitted reports.¹¹¹ The information gathered through the ETF will feed into the Global stock take which will assess the collective progress towards the long-term climate goals.¹¹² This will lead to recommendations for countries to set more ambitious plans in the next round.¹¹³

Current Achievements. Although climate change action needs to be massively increased to achieve the goals of the Paris Agreement, the years since its entry into force have already sparked low-carbon solutions and new markets. More and more countries, regions, cities and companies are establishing carbon neutrality targets. Zero-carbon solutions are becoming competitive across economic sectors representing 25% of emissions. This trend is most noticeable in the power and transport sectors and has created many new business opportunities for early movers.¹¹⁴ By 2030, zero-carbon solutions could be competitive in sectors representing over 70% of global emissions.¹¹⁵

Lastly, no matter how seemingly small your contributions are, in the grand scheme of things, they make a difference that could create a significant effect towards achieving your goal. When it comes to environmental issues, small efforts create ripples, which could eventually form waves and finally shake the core of every individual to stir a tide of lifestyle change for the betterment of Mother Earth. Anchored on the concept of “what goes around comes around” and the law of cause and effect.

Quoting a Miss Earth who was asked “*Who or what do you think is the biggest enemy of Mother Earth and why?*” To which, she replied, “*I believe that the real problem in this world is not climate change. The real problem is us because of our ignorance and apathy. What we have to do is to start changing our ways, to start recalibrating our minds and redirecting our steps because together, as a global community, our micro-efforts will have a macro-effect to help save our home, our planet.*”¹¹⁶ In our individual capacities, let us be advocates of the use of reusable, sustainable energy to reduce emissions in the atmosphere and, ultimately, lessen the effects of global warming.

¹¹¹ United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.) <https://unfccc.int/process-and-meetings/the-paris-agreement>

¹¹² *Id.*

¹¹³ *Id.*

¹¹⁴ *Id.*

¹¹⁵ *Id.*

¹¹⁶ Libero- Cruz, Grace Metro Style: Miss Earth 2017 Karen Ibasco: “Our Micro-Efforts Will Have a Macro-Effect” (November 9, 2017) <https://metro.style/people/celebrities/miss-earth-2017-karen-ibasco-our-micro-efforts-wil/7340>

To emphasize, “*Our micro-efforts will surely have a macro-effect to help save our home, our planet.*”

Bibliography

Constitution

1987 Constitution of the Philippines, art. II §16 (Phil.).

Conventions

United Nations Framework Convention on Climate Change (UNFCCC) The Paris Agreement. (n.d.),
<https://unfccc.int/process-and-meetings/the-paris-agreement>

Jurisprudence

Oposa v. Factoran, G.R. No. 101083, (July 30, 1993) (Phil.).

Journals

Das Gupta, Monica. "Population, Poverty, and Climate Change". (2014). The World Bank Research Observer. 29 (1): 83–108. doi:10.1093/wbro/lkt009. hdl:10986/22565. ISSN 0257-3032. JSTOR 24582389.

Perez, Rosa T.; et al., "Climate Change Impacts and Responses in the Philippines Coastal Sector" (August 1999), Climate Research, 12 (2/3): 97–107, Bibcode:1999ClRes..12...97P, doi:10.3354/cr012097, JSTOR 24866004

Smith, Paul. "Climate Change, Weak States and the "War on Terrorism" in South and Southeast Asia" (2007). Contemporary Southeast Asia. 29 (2): 264–285. doi:10.1355/CS29-2C. ISSN 0129-797X. JSTOR 25798831.

Online Sources

ABS-CBN News "PAGASA redefines 'super typhoon', revises wind signals". (March 23, 2022),
<https://news.abs-cbn.com/news/03/23/22/pagasa-redefines-super-typhoon-tweaks-wind-signals>

Asian Disaster Reduction Center. Information on Disaster Risk Reduction of the Member Countries (n.d.),
<https://www.adrc.asia/nationinformation.php?NationCode=608&Lang=en>

Climate Change Commission (n.d.),
<https://climate.gov.ph/about-us>

Corales, Nestor PH Sea Level Rising 3 times faster than World Average. (September 23, 2022),
<https://newsinfo.inquirer.net/1669093/ph-sea-level-rising-3-times-faster-than-world-average-says-expert>

-
- De Vera-Ruiz, Ellalyn, CCC vows to fast-track PH commitment to Paris climate agreement. (January 18, 2021), <https://mb.com.ph/2021/01/18/ccc-vows-to-fast-track-ph-commitment-to-paris-climate-agreement/>
- Department of Environment and Natural Resources (DENR) (n.d.), <https://ncr.denr.gov.ph/news-events/ph-signs-the-paris-climate-change-agreement/>
- Esquire Mag Philippine Cities Are Sinking Faster Than We Realize. What Now? (August 27, 2024), <https://www.esquiremag.ph/politics/news/land-subsidence-climate-change-study-philippine-cities-a1057-20240827?s=5s8nj6gfk6qrh22s92ef389ln>
- Galvez, Daphne, Rich countries must pay for climate change damage – Duterte. (April 19, 2022), <https://newsinfo.inquirer.net/1584618/rich-countries-must-pay-for-climate-change-damage-duterte#ixzz8mBA6HJCM>
- Garcia, Ma Angelica, "LOOK: How rising sea levels could affect PHL cities, towns by 2050" (November 4, 2019), <https://www.gmanetwork.com/news/scitech/science/714173/how-rising-sea-levels-could-affect-phl-cities-towns-by-2050/story/>
- Implementation Plan for the Republic of the Philippines Nationally Determined Contribution (NDC) (2020-2030), <https://climate.gov.ph/public/ckfinder/userfiles/files/Knowledge/The%20Philippines%20NDC%20Implementation%20Plan%20-%20072024.pdf>
- Ines, Jezreel "In troubled waters: Rising sea levels threaten sinking town's survival". (July 4, 2023), <https://www.rappler.com/environment/in-troubled-waters-rising-sea-levels-threaten-sinking-town-survival/>
- Kulp, Scott A, et al., New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding. (October 29, 2019), <https://www.nature.com/articles/s41467-019-12808-z>
- Libero- Cruz, Grace Metro Style: Miss Earth 2017 Karen Ibasco: “Our Micro-Efforts Will Have a Macro-Effect” (November 9, 2017), <https://metro.style/people/celebrities/miss-earth-2017-karen-ibasco-our-micro-efforts-wil/7340>
- Philippine Atmospheric Geophysical and Astronomical Services Administration (PAGASA) (n.d.), <https://www.pagasa.dost.gov.ph/information/climate-change-in-the-philippines>

Philstar, In 30 years, rising seas will threaten Philippine cities, towns home to 6.8 million. (November 6, 2019), <https://www.philstar.com/headlines/2019/11/06/1966508/30-years-rising-seas-will-threaten-philippine-cities-towns-home-68m#:~:text=Sea%20levels%20are%20projected%20to%20rise%20nearly%20half%20a%20meter>

Reuters Philippines raises carbon emission reduction target to 75% by 2030 | Reuters (2021), <https://www.reuters.com/business/environment/philippines-raises-carbon-emission-reduction-target-75-by-2030-2021-04-16/>

Sebesvari, Rita, Sea level rise is inevitable – but what we do today can still prevent catastrophe for coastal regions. (September 26, 2019), <https://theconversation.com/sea-level-rise-is-inevitable-but-what-we-do-today-can-still-prevent-catastrophe-for-coastal-regions-124129>

Typhoon Rai. Wikipedia, https://en.wikipedia.org/w/index.php?title=Typhoon_Rai&oldid=1087095376

United States Agency for International Development (USAID) CLIMATE CHANGE RISK PROFILE PHILIPPINES. (n.d.), https://www.climatelinks.org/sites/default/files/asset/document/2017_Climate%20Change%20Risk%20Profile_Philippines.pdf

UNOCHA "About OCHA in The Philippines" (March 2019), <https://www.unocha.org/philippines>

World Bank. Getting a Grip on Climate Change in the Philippines (June 25, 2013), <https://www.worldbank.org/en/country/philippines/publication/getting-a-grip-on-climate-change-in-the-philippines>

World Economic Forum, Sea level rise: Everything you need to know (July 11, 2024), <https://www.weforum.org/agenda/2024/07/rising-sea-levels-global-threat/>



Image Source: Canva for Education

THE E-WASTE LAW DISCONNECT: CHALLENGES IN TACKLING ELECTRONIC WASTE IN THE PHILIPPINES

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Connection, work productivity, effective learning, convenience, and, most prominently, entertainment—a short list of the diverse uses of our gadgets, seamlessly integrating into nearly every aspect of modern life. Gadgets have become indispensable tools from enabling instant communication across the globe to providing access to information and enhancing personal productivity. They support learning, offer fitness insights, simplify home management, and bring endless entertainment options to our fingertips. These benefits, however, also bring challenges like the environmental impact of electronic waste.

But how do we truly cut off our old gadgets? Have we ever considered the impact on how we dispose of them?

In my honest opinion, I do not give it much thought. I often toss them aside, thinking they are just pieces of junk with no further use, without regard to the fact that I have contributed to the growing e-waste crisis. I know that I am not the only one guilty of such acts. These small contributions are aggregated and form part of the global crisis that we are currently facing and addressing the conservation of our environment for the present and future generations.

According to the UN's fourth Global E-waste Monitor (GEM), electronic waste (e-waste) generation is surging, with a record of 62 million tons produced in 2022 which contributes to an increase of 82% since 2010. This figure is projected to rise to 82 million tons by 2030. Alarming, only 22.3% of e-waste was properly collected and recycled in 2022, leaving about \$62 billion worth of recoverable resources unaccounted for and heightening pollution risks globally.¹

The Philippines is facing a growing e-waste crisis, with e-waste reaching five hundred thirty-seven million (537,000,000) kilograms in 2022. This surge raises serious concerns about environmental and health risks. According to the latest UN Global E-waste Monitor, the country is among the top producers of e-waste in Southeast Asia.²

Before diving further, let us first define what e-waste is. Electronic waste, or e-waste, refers to all items of Electrical and Electronic Equipment (EEE) and its parts

¹ United Nations Institute for Training and Research; The Global E-waste Monitor 2024, <https://ewastemonitor.info/the-global-e-waste-monitor-2024/>

² Geneva Internet Platform, March 27, 2024, D Philippines faces escalating e-waste crisis | Digital Watch Observatory, <https://dig.watch/updates/philippines-faces-escalating-e-waste-crisis>

that have been discarded by its owner as waste without the intent of re-use. E-waste is also referred to as Waste Electrical and Electronic Equipment (WEEE), electronic waste, or e-scrap in different regions and under different circumstances in the world. It includes a wide range of products that consist of almost any household or business item with circuitry or electrical components with power or battery supply.³

Environmental Protection in the Philippines

The 1987 Philippine Constitution specifically embodied the protection of the environment in its provisions. Article 2, Sections 15 and 16 clearly state that the Constitution protects the right of the people to a balanced and healthful ecology.⁴ In his speech at the Judges Forum on Environmental Protection, former Justice Puno encouraged his fellow judges in disputes about environmental issues to equip themselves with a commanding armor of the emerging substantive body of Environmental Law, especially considering the depth and breadth of contemporary environmental issues. He also nudged them to be creative and to shed their fears even when they find themselves in some unfenced spaces of our environmental law.⁵

Former Justice Puno even acknowledged that the Philippine environmental laws are somehow inadequate. Times might have changed; however, it seems like we still lack the statute addressing electronic waste problems.

In 2000, the Philippine legislature enacted R.A. No. 9003 also known as the "Ecological Solid Waste Management Act of 2000," it aims to provide a systematic and comprehensive approach to waste management in the Philippines. The law establishes the framework for managing solid waste, emphasizing environmental protection and sustainability. It has established a national solid waste framework regarding the proper ways to collect and segregate solid waste, and provide public awareness and education. The law also gives incentives to encourage the participation of the general public and imposes penalties to its violators.⁶

While R.A. No. 9003 promotes general waste management practices such as reduction, reuse, and recycling, it falls short in actively engaging manufacturers and producers in the e-waste management process. A comprehensive approach is needed to hold these stakeholders accountable and ensure they contribute to mitigating the impacts of e-waste on the environment.

³ The Global E-waste Statistic Partnership, <https://globalewaste.org/what-is-e-waste/#:~:text=Electronic%20waste%2C%20or%20e-waste%2C%20refers%20to%20all%20items,owner%20as%20waste%20without%20the%20intent%20of%20re-use>

⁴ CONST. (1987), art. II, §15 and §16 (Phil.).

⁵ Philippine Environmental Law Practice and the Role of the Courts, Aug. 14, 2003, <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/2/4800>

⁶ Ecological Solid Waste Management Act of 2000, Rep. Act No. 9003 (Phil.).

E-waste should be addressed differently from other solid waste because it imposes a different level of threat both to the health and environment as evidenced by related studies. Generally solid waste is any unwanted or unusable material that is solid in its state of matter.⁷ Which is generally a broader category of waste, while electronic wastes are items that contain many different toxic substances⁸ that are discarded from electronic devices. Unfortunately, the Ecological Solid Waste Act of 2000 does not provide enough solutions to address the crisis being presented, a detailed and well-defined statute should be passed to address the issue.

Effects of E-waste

Various research shows the bad effects of e-waste, both on the environment and human body as it causes air pollution from burning e-waste, water pollution due to toxic chemicals, and soil contamination that affects agriculture. These pollutants can harm ecosystems and enter the food chain, posing risks to human health.⁹ E-waste is non-biodegradable and toxic, leading to accumulation in the environment, including soil, air, water, and living organisms. Practices like open-air burning and acid baths to extract valuable materials release harmful substances into the environment.¹⁰

One of the most common effects of E-waste on the environment is air pollution. E-waste can pollute the air when improperly shredded, melted, or burned. These wrong practices release dust particles or toxins, such as dioxins, into the environment and cause air pollution.¹¹ When the improper disposal of e-waste occurs in regular landfills or in places where it is dumped illegally, both heavy metals and flame retardants can leach directly from the e-waste into the soil, causing contamination of underlying groundwater or contamination of crops that may be planted nearby or in the area in the future.¹² After soil contamination, toxins can eventually make their way into nearby water. Heavy metals from improperly disposed e-waste, such as mercury, lithium, lead, and barium, can leach through the earth, even reaching the groundwater. Consequently, when these heavy metals reach groundwater, they eventually make their way into ponds, streams, rivers, and lakes.¹³ Its improper disposal affects both land and sea animals. It is also worth considering the effects that electronic waste has on climate change. Every device ever produced has a carbon footprint and is contributing to human-made global warming. The production of a ton

⁷ The Daily Eco, Jan. 8, 2003- What is Solid Waste, <https://www.thedailyeco.com/what-is-solid-waste-definition-and-examples-41.html>

⁸ Electronic Waste, Oct. 1, 2004, <https://www.who.int/news-room/fact-sheets/detail/electronic-waste-%28e-waste%29>

⁹ Wasted: The Effects of E-Waste on the Environment, Cleanlites Recycling, Jun. 22, 2019, <https://cleanlites.com/wasted-effects-of-ewaste>

¹⁰ GENEVA Network, The Growing Environmental Risks of E-Waste, (2024) <https://www.genevaenvironmentnetwork.org/resources/updates/the-growing-environmental-risks-of-e-waste/>

¹¹ Environment Go; 10 Environmental Impacts of Electronic Waste, Jan. 9, 2023, <https://environmentgo.com/environmental-impacts-of-electronic-waste/>

¹² *Id.*

¹³ *Id.*

of laptops emits up to ten metric tons of carbon dioxide. When the carbon dioxide released over a device's lifetime is considered, it predominantly occurs during production and usually before consumers buy a product.¹⁴ Agricultural land use is significantly impacted by soil and water pollution. Heavy metals and flame-retardant chemicals can stick around in crops. Livestock, wildlife, and human populations are at increased risk of consuming crops contaminated with these pollutants.¹⁵

E-waste presents serious environmental challenges, from polluting air, soil, and water to contributing to climate change. Improper disposal releases harmful chemicals and heavy metals, contaminating ecosystems and entering the food chain, which can have adverse effects on human health, agriculture, wildlife, generally, the environment.

The Proposed Bill

On July 22, 2018, Senator Francis Tolentino introduced Senate Bill No. 751¹⁶ which is the "E-Waste Management Act," and it aims to regulate e-waste disposal in the Philippines. It emphasizes the need for manufacturers to adopt recovery plans for electronic equipment and mandates convenient, accessible recycling options. Manufacturers must label products for tracking, set up collection points, and report yearly on waste management. The bill includes penalties for non-compliance, outlines roles for local governments, and allocates fines imposed to an "E-Waste Recycling Fund" to support its initiatives. It seeks to align with the existing ecological laws for a comprehensive approach to e-waste management.

The bill promotes a policy of the State to protect the environment from the hazards of the improper disposal of obsolete electronic equipment, encourage the responsible manufacturing and sale thereof, and assist consumers in the recycling and disposal of their electronic waste.¹⁷ As of August 30, 2022, this bill is still pending.¹⁸

In my opinion, the delay in passing Senate Bill No. 751 into law is caused by the doubt and possible effects of it on the economy. Particularly establishing the necessary infrastructure for recycling will be costly and enforcing manufacturers' responsibility may exponentially increase its costs which may soon lead to the lack of interest of potential and current investors, from both local and international.

¹⁴ Environment Go; 10 Environmental Impacts of Electronic Waste, Jan. 9, 2023, <https://environmentgo.com/environmental-impacts-of-electronic-waste/>

¹⁵ *Id.*

¹⁶ Senate of the Philippines, 19th Congress; Senate Bill No. 751, <https://legacy.senate.gov.ph/lisdata/3844734904!.pdf>

¹⁷ *Id.*

¹⁸ *Id.*

Other Philippine Statutes Addressing the Issue

R.A. No. 6969 or the Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990 is a law that aims the regulating, importation, manufacture, handling, and disposal of hazardous substances and waste. It empowers the Department of Environment and Natural Resources (DENR) to oversee toxic chemicals and manage hazardous waste disposal. Furthermore, it also seeks to protect human health and the environment by controlling exposure to toxic chemicals and ensuring proper waste treatment and disposal methods.¹⁹

R.A. No. 9512, known as the *National Environmental Awareness and Education Act of 2008*, aims to increase environmental awareness among Filipinos through education. The law mandates integrating environmental education into school curricula at all levels, including both public and private institutions. It covers topics such as natural resource conservation, waste management, pollution prevention, and climate change. It encourages environmental programs and partnerships among schools, government agencies, and the private sector to foster a culture of environmental protection and sustainable development in the Philippines.²⁰

Both laws are too broad and general in their context. They fail to provide a concrete solution addressing the specific issue of e-waste crisis. The e-waste crisis needs targeted attention as it poses threats to the environment and health. Unlike general solid waste, e-waste contains toxic chemicals and heavy metals, such as mercury, lead, and cadmium, that can leach into soil and water, harming ecosystems and human health. Addressing e-waste separately allows for distinct policies, recycling programs, and handling protocols that would ensure the protection of both public health and the environment more effectively.

State Responsibility of the Philippines

To address the growing threat being imposed by hazardous waste, including electronic waste, the Philippines signed multiple treaties to address the e-waste and one of them is the Basel Convention. Basel Convention, established in 1989 is a global treaty aimed at controlling the transboundary movement and disposal of hazardous wastes, including electronic waste. It requires countries to manage hazardous waste in environmentally sound ways and prohibits the export of such waste to countries lacking the capacity for safe disposal. The convention emphasizes minimizing hazardous waste generation and promotes reducing cross-border waste transfers. It fosters international cooperation and capacity-building to protect human health and the environment from hazardous waste impacts.²¹

¹⁹ Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990 (Phil.), <https://www.officialgazette.gov.ph/1990/10/26/republic-act-no-6969/>

²⁰ National Environmental Awareness and Education Act of 2008 Rep. Act No. 9512 (Phil.), <https://www.officialgazette.gov.ph/2008/12/12/republic-act-no-9512/>

²¹ Basel Convention 1989 (n.d.), <https://www.basel.int/TheConvention/Overview/tabid/1271/Default.aspx>

The Philippines also has the responsibility to conserve the environment for the present and future generations. Although not yet codified, the principle of intergenerational responsibility to preserve the environment is increasingly recognized as part of customary international law. It is supported by the Stockholm Declaration²² which was later affirmed by the Rio Declaration²³ and enshrined in the UN Framework Convention on Climate Change.²⁴ It reflects a widespread understanding that present generations must act as stewards of the environment for the benefit of future generations, an obligation that many nations integrate into their domestic laws and policies.

As part of its commitment to international environmental standards, the Philippines holds a responsibility to manage and mitigate the impact of e-waste. This involves ensuring environmentally sound disposal of hazardous materials and actively reducing harmful emissions and waste. The nation also upholds an intergenerational duty to protect natural resources, emphasizing sustainable practices that protect the environment for future generations, aligned with both international expectations and national constitutional mandates.

Conclusion

Man has been constantly discovering, inventing, creating, and advancing a lot of things. In the present time, man's capability to transform his surroundings, if used wisely, can be beneficial to people in terms of the benefits of development and the opportunity to enhance the quality of life. Wrongly or heedlessly applied, the same power can do incalculable harm to human beings and the environment. We see around us growing evidence in many regions of the earth of the actual dangerous levels of pollution in living water, air, earth, and beings; major and undesirable disturbances to the ecological balance of the biosphere; destruction and depletion of irreplaceable resources; and gross deficiencies, harmful to the physical, mental and social health of man, in the man-made environment, and working environment.²⁵

The Philippines faces a significant challenge in tackling the growing e-waste crisis. Existing laws fall short of addressing the specific environmental and health threats posed by electronic waste, highlighting the urgent need for more targeted regulations in this time of globalization.

²² Basel Convention 1989 (n.d.), <https://www.basel.int/TheConvention/Overview/tabid/1271/Default.aspx>

²³ Rio Declaration, Report of the United Nations Conference on Environment and Development June 1992 (n.d.), https://www.un.org/en/development/desa/population/migration/generalassembly/docs/globalcompact/A_CONF.151_26_Vol.I_Declaration.pdf

²⁴ United Nations Framework Convention on Climate Change, 1992, <https://unfccc.int/resource/docs/convkp/conveng.pdf>

²⁵ Stockholm Conference, United Nations Conference on the Human Environment, June 1972, <https://documents.un.org/doc/undoc/gen/nl7/300/05/pdf/nl730005.pdf>

A separate law for e-waste, distinct from general solid waste, is indispensable for the reason that electronic waste poses unique challenges and risks not addressed by traditional waste laws. E-waste contains toxic substances like mercury, lead, and cadmium, which can have severe environmental and health damage when improperly disposed of. E-waste requires specialized handling, recycling, and disposal processes to prevent contamination of soil, water, and air. A dedicated law would establish clear guidelines and regulations, focusing on safe e-waste management, producer responsibility, and recycling infrastructure tailored to address the specific hazards and recycling needs of electronic waste. This targeted approach ensures a safer, more effective response to the growing e-waste problem. The enactment of such would also promote accountability among manufacturers, requiring them to take responsibility for e-waste through mechanisms like recycling and safe disposal. Additionally, it would enable a structured, consistent approach to e-waste management nationwide, aligning with international commitments and protecting future generations from suffering long-term ecological harm.

Good intentions alone are insufficient for protecting the environment and addressing e-waste. These efforts must be clearly defined and enforced through well-defined legislation that is binding to everyone.

Bibliography

Constitution

1987 Constitution of the Philippines, art. II, §15 and §16 (Phil.).

International Law

United Nations Framework Convention on Climate Change
<https://unfccc.int/resource/docs/convkp/conveng.pdf>

United Nations Conference on the Human Environment, 5-16 June 1972, Stockholm,
<https://www.un.org/en/conferences/environment/stockholm1972>

Rio Declaration, Report of the United Nations Conference on Environment and
Development June 1992 (n.d.)
https://www.un.org/en/development/desa/population/migration/generalassembly/docs/globalcompact/A_CONF.151_26_Vol.I_Declaration.pdf

Legislation

Ecological Solid Waste Management Act of 2000, Rep. Act No. 9003 (Phil.).

National Environmental Awareness and Education Act of 2008, Rep. Act No. 9512
(Phil.).

Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990, Rep. Act
No. 6969 (Phil.).

Online Sources

Basel Convention on the Control of Transboundary Movements of Hazardous Wastes
and their Disposal,
<https://www.basel.int/TheConvention/Overview/tabid/1271/Default.aspx>

Environment Go; 10 Environmental Impacts of Electronic Waste, Jan. 9, 2023,
<https://environmentgo.com/environmental-impacts-of-electronic-waste/>

Electronic Waste, Oct. 1, 2004,
<https://www.who.int/news-room/fact-sheets/detail/electronic-waste-%28e-waste%29>

Geneva Internet Platform, March 27, 2024, D Philippines faces escalating e-waste
crisis | Digital Watch Observatory, <https://dig.watch/updates/philippines-faces-escalating-e-waste-crisis>

GENEVA Network, The Growing Environmental Risks of E-Waste (2024),
<https://www.genevaenvironmentnetwork.org/resources/updates/the-growing-environmental-risks-of-e-waste/>

Philippine Environmental Law Practice and the Role of the Courts, Aug. 14, 2003,
[https://philja.judiciary.gov.ph/files/learning_materials/Phil Envi Law Pun o.pdf](https://philja.judiciary.gov.ph/files/learning_materials/Phil_Env_Law_Pun_o.pdf)

Senate of the Philippines, 19th Congress; Senate Bill No. 751
<https://legacy.senate.gov.ph/lisdata/3844734904!.pdf>

The Daily Eco, Jan. 8, 2003- What is Solid Waste,
<https://www.thedailyeco.com/what-is-solid-waste-definition-and-examples-41.html>

United Nations Institute for Training and Research; The Global E-waste Monitor 2024,
<https://ewastemonitor.info/the-global-e-waste-monitor-2024/>

Wasted: The Effects of E-Waste on the Environment,
<https://cleanlites.com/wasted-effects-of-ewaste>



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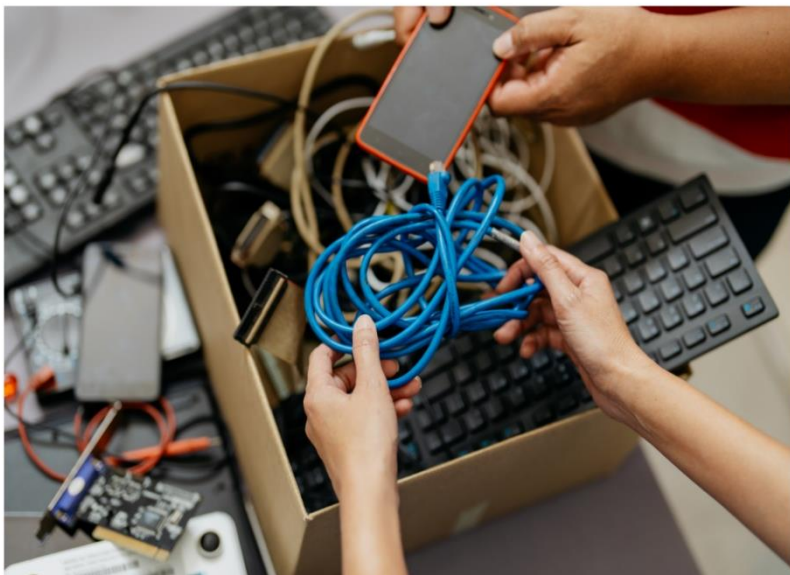


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IS BIOTECHNOLOGY A THREAT TO THE ENVIRONMENT?

VANESSA G. SILLA, JD

The commercial release of genetically modified organisms, specifically, Golden Rice and Bt (*Bacillus thuringiensis*) eggplant, is no more, at least for now.

On April 18, 2023, the Supreme Court issued a *Writ of Kalikasan* in the case of *Magsasaka at Siyentipiko Para sa Pag-Unlad ng Agrikultura, et al. v. Secretary of the Department of Agriculture, et. al.*¹

In a briefer released on April 19, 2023, the Court required the respondents Secretary of the Department of Agriculture, the Secretary of the Department of Environment and Natural Resources, the Secretary of the Department of Health, the Director of the Bureau of Plant Industry of the Department of Agriculture, the Philippine Rice Research Institute, and University of the Philippines – Los Baños (UPLB) to file a verified return within 10 days from service.²

It all started when

The Department of Agriculture (DA) issued Department of Agriculture Administrative Order No. 08, series of 2002 (DAO-08-2002) providing rules and regulations for the importation and release into the environment of plants and plant products derived from the use of modern biotechnology.

DAO-08-2002 covers the importation or release into the environment of: (1) any plant which has been altered or produced through the use of modern biotechnology if the donor organism, host organism, or vector or vector agent belongs to the genera or taxa classified by the Bureau of Plant Industry (BPI) as meeting the definition of plant pest or is a medium for the introduction of noxious weeds; or (2) any plant or plant product altered through the use of modern biotechnology which may pose significant risks to human health and the environment based on available scientific and technical information.

Farmers, Non-Governmental Organizations (NGOs), organic traders and other sectors of society expressed their opposition, noting the threats Genetically

¹ *Magsasaka at Siyentipiko Para sa Pag-Unlad ng Agrikultura, et al. v. Secretary of the Department of Agriculture, et. al.* G.R. 263595, (April 18, 2023) (Phil). The Supreme Court Public Information Office will upload a copy of the ruling once it receives a copy of the same from the *En Banc* Office of the Clerk of Court.

² SC Issues Writ of Kalikasan on Genetically Modified Rice and Eggplant Products (April 19, 2023), <https://sc.judiciary.gov.ph/sc-issues-writ-of-kalikasan-on-genetically-modified-rice-and-eggplant-products/> (Portal of the Philippine Judiciary).

Modified Organisms (GMOs) pose to human beings and the environment, as well as the possible contamination that could affect the Philippine organic industry.³

Unconstitutional

In *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. vs. Greenpeace Southeast Asia (Phils.), et al.*,⁴ the Supreme Court declared DAO-08-2002⁵ as null and void. It ruled that “the failure of DAO 08-2002 to accommodate the NBF means that the Department of Agriculture lacks mechanisms to mandate applicants to comply with international biosafety protocols.”

Precautionary Principle

Notably, the Supreme Court discussed the Precautionary Principle.

It stated that “three conditions must be present: (a) settings in which the risks of harm are uncertain; (b) settings in which harm might be irreversible and what is lost is irreplaceable; and (c) settings in which the harm that might result would be serious.”

The precautionary principle bridges the gap in cases where scientific certainty in factual findings cannot be achieved. By applying the precautionary principle, the court may construe a set of facts as warranting either judicial action or inaction, with the goal of preserving and protecting the environment. This may be further evinced from the second paragraph where bias is created in favor of the constitutional right of the people to a balanced and healthful ecology. In effect, the precautionary principle shifts the burden of evidence of harm away from those likely to suffer harm and onto those desiring to change the status quo. An application of the precautionary principle to the rules on evidence will enable courts to tackle future environmental problems before ironclad scientific consensus emerges.

³ BPI's conflict of interest in GMO products, (February 13, 2012) <https://opinion.inquirer.net/22993/bpi%E2%80%99s-conflict-of-interest-in-gmo-products> (Philippine Daily Inquirer, a daily newspaper in the Philippines).

⁴ *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. v. Greenpeace Southeast Asia (Phils.), et al.*, G.R. No. 209271, (December 08, 2015) (Phil).

⁵ Department of Agriculture Administrative Order No. 8 Series of 2002 (April 3, 2002) <http://spsissuances.da.gov.ph/attachments/article/298/DA%20Administrative%20Order%20No.%208%20S2002.pdf> (Sanitary and Phytosanitary Regulations and related Issuances, Department of Agriculture).

As applied

Assessing the evidence on record, as well as the current state of GMO research worldwide, the Court finds all the three conditions present.

Eggplants (*talong*) are a staple vegetable in the country and grown by small-scale farmers, majority of whom are poor and marginalized. While the goal of increasing crop yields to raise farm incomes is laudable, independent scientific studies revealed uncertainties due to unfulfilled economic benefits from *Bt* crops and plants, adverse effects on the environment associated with use of GE technology in agriculture, and serious health hazards from consumption of GM foods. For a biodiversity-rich country like the Philippines, the natural and unforeseen consequences of contamination and genetic pollution would be disastrous and irreversible.

Alongside the aforesaid uncertainties, the non-implementation of the NBF in the crucial stages of risk assessment and public consultation, including the determination of the applicability of the EIS requirements to GMO field testing, are compelling reasons for the application of the precautionary principle. There exists a preponderance of evidence that the release of GMOs into the environment *threatens* to damage our ecosystems and not just the field trial sites, and eventually the health of our people once the *Bt* eggplants are consumed as food. Adopting the precautionary approach, the Court rules that the principles of the NBF need to be operationalized first by the coordinated actions of the concerned departments and agencies before allowing the release into the environment of genetically modified eggplant. The more prudent course is to immediately enjoin the *Bt talong* field trials and approval for its propagation or commercialization until the said government offices shall have performed their respective mandates to implement the NBF.

The Court has found the experience of India in the *Bt brinjal* field trials - for which an indefinite moratorium was recommended by a Supreme Court-appointed committee till the government fixes regulatory and safety aspects - as relevant because majority of Filipino farmers are also small-scale farmers. Further, the precautionary approach entailed inputs from all stakeholders, including the marginalized farmers, not just the scientific community. This proceeds from the realization that acceptance of uncertainty is not only a scientific issue, but is related to public policy and involves an ethical dimension. For scientific research alone will not resolve all the problems, but participation of different stakeholders from scientists to industry, NGOs, farmers and the public will provide a needed variety of perspective foci, and knowledge.

Turn of Events

In *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. vs. Greenpeace Southeast Asia (Phils.), et al.*,⁶ the Supreme Court overturned its Decision dated December 8, 2015 on the ground of mootness. It stated that “it was improper for the Court to resolve the merits of the case which had become moot in view of the absence of any valid exceptions to the rule on mootness, and to thereupon rule on the objections against the validity and consequently nullify DAO 08-2002 under the premises of the precautionary principle.”

And now I wonder...

What are the consequences of this jurisprudential trend?

Thoughts

I can think of two things.

First, scientists and their “inventions” are set to suffer from government regulations — at least until the government provides a clear guideline on the introduction of biotechnological innovations. This is bad because we should aim to grow with science, not resist them.

Second, the sweeping application of the precautionary principle might send a wrong message to the public. I am afraid that many will be discouraged to venture in an ambitious and probably life-changing undertaking for fear of admonishment. Do we really have to nip these activities in the bud?

My two cents

There is a need to carefully assess the possible impacts of biotechnology on the environment. It is undeniable that GMOs are still seen by the majority in a bad light. But I think that the public must give more weight to the opinions of the expert. I discourage the adoption of conservative thinking; we must adjust to the swiftly moving facts of our changing society. Thus, this subject must be evaluated with an open mind. ♦

⁶ *Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. v. Greenpeace Southeast Asia (Phils.), et al.*, G.R. No. 209271, (December 08, 2015) (Phil.).

Bibliography

Jurisprudence

Int'l. Service for the Acquisition of Agri-Biotech Applications, Inc. v. Greenpeace Southeast Asia (Phils.), et al., G.R. No. 209271, (December 08, 2015) (Phil.).

Magsasaka at Siyentipiko Para sa Pag-Unlad ng Agrikultura, et al. v. Secretary of the Department of Agriculture, et. al. G.R. 263595, (April 18, 2023) (Phil). The Supreme Court Public Information Office will upload a copy of the ruling once it receives a copy of the same from the *En Banc* Office of the Clerk of Court.

Online Sources

BPI's conflict of interest in GMO products, (February 13, 2012), <https://opinion.inquirer.net/22993/bpi%E2%80%99s-conflict-of-interest-in-gmo-products> (Philippine Daily Inquirer, a daily newspaper in the Philippines).

Department of Agriculture Administrative Order No. 8 Series of 2002 (April 3, 2002) <http://spsissuances.da.gov.ph/attachments/article/298/DA%20Administrative%20Order%20No.%208%20S2002.pdf> (Sanitary and Phytosanitary Regulations and related Issuances, Department of Agriculture).

SC Issues Writ of Kalikasan on Genetically Modified Rice and Eggplant Products (April 19, 2023) <https://sc.judiciary.gov.ph/sc-issues-writ-of-kalikasan-on-genetically-modified-rice-and-eggplant-products/> (Portal of the Philippine Judiciary).



Image Source: Canva for Education

BRAVING THE STORM: STRENGTHENING THE PHILIPPINES’ DISASTER PREPAREDNESS AND RESILIENCE

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The highest human right is the right to life. One has the expectation to have a healthful and sustainable environment, to live without any fear of destruction, and to have the capacity to resist any disastrous effects of inevitable natural calamities. This is supported by Section 16 of Article II of the 1987 Philippine Constitution which provides that *the State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.*¹

The Philippines is one of the world’s most disaster-prone countries. Located along the boundary of major tectonic plates and at the center of a typhoon belt, its islands are regularly affected by floods, typhoons, landslides, and earthquakes. The Philippine government has developed strong coping mechanisms over their long history of experience with disasters. Yet, significant gaps remain in natural disaster management capacities across different regions of the Philippines and surprisingly little data are available referencing local levels of disaster resilience and preparedness.

In a news report, The World Risk Index, a part of the annual risk report which calculates the disaster risk for 193 countries, showed that the Philippines has the highest disaster risk, with an index score of 46.82. The Philippines’ current ranking in the World Risk Index was a huge jump from its previous records. In 2018, the country ranked third on the list of countries with the highest disaster risk, with an index score of 26.70. In 2019, the country dropped to ninth after registering an index value of 20.69. In 2020, as the COVID-19 pandemic hit countries across the globe, the Philippines was still among the top 10 countries with the highest disaster risk while remaining in ninth place with the same index score. Last year, the Philippines ranked eighth with an index score of 21.39.²

Exposure

The report also found that the Philippines was among the top 10 countries with the highest exposure to disasters. The country ranked fourth with 39.99 exposure, trailing China, Japan, and Mexico. “Thus, exposure consists of the aspects of

¹ CONST. (1987), art II, §16 (Phil.).

² Baclig, C. E. *PH most disaster-prone country in the world—study* | *Inquirer News*. INQUIRER.net., (October 7, 2022), <https://newsinfo.inquirer.net/1676227/ph-most-disaster-prone-country-in-the-world-study>

hazardousness, which include the frequency and intensity of earthquakes, tsunamis, coastal and river floods, cyclones, droughts, and sea-level rise in an area (hazard zone) and populations (hazard object),” the report added.³

Natural disasters are natural hazards that result in great injury and damage. Natural hazards (such as extreme weather, wildfires, or earthquakes) may be inevitable, but disasters – with loss of life and high economic costs – are not.⁴ This is also referred to as natural hazards that are extreme sudden events caused by environmental factors such as storms, floods, droughts, and fires-now occurring with increasing severity, scope, and impact.

How Disaster Affects the Philippine Environment

For decades, the Philippine environment has been under assault from a fast-growing population and practices that have degraded the country’s air, land, and water. Despite laws requiring trees to be replanted, forests were clear cut and denuded areas abandoned, damaging watersheds. Consequently, this created and increased the desire to introduce sound environmental practices and various policies. The landmark statute that has been passed, which is the *Philippine Disaster Reduction and Management Act*,⁵ is however, underfunded and poorly implemented. These are successfully achieved when started in a community.

According to the Asian Disaster Reduction Center (ADRC), the Philippines is visited by an average of 20 typhoons each year, and most are destructive.⁶ This is supported by records of the Nationwide Operational Assessments & Hazards at the University of the Philippines (NOAH) Center that on average, disasters happen once a year and at most twice per year. But from 2017-2018, there were six in the span of one year.⁷ The impacts of typhoons can range from extensive property damage and economic loss to loss of human life. Record reveals that Typhoon Yolanda, which

³ Baclig, C. E., *PH most disaster-prone country in the world—study* | *Inquirer News*. INQUIRER.net., (October 7, 2022), <https://newsinfo.inquirer.net/1676227/ph-most-disaster-prone-country-in-the-world-study>

⁴ Scheunemann, C., *What is disaster risk reduction and why is it important?* Sphera., (June 24, 2024), <https://sphera.com/blog/what-is-disaster-risk-reduction-and-why-is-it-important/>

⁵ Philippine Disaster Reduction and Management Act (May 27, 2010) (Phil.), <https://www.officialgazette.gov.ph/2010/05/27/republic-act-no-10121/>.

⁶ Asian Disaster Reduction Center (ADRC). (2017), <https://www.adrc.asia/nationinformation.php?NationCode=608&Lang=en>

⁷ Novio, E., *Climate Change and Disasters in the Philippines* | Heinrich Böll Foundation | Southeast Asia Regional Office. Heinrich Böll Foundation | Southeast Asia Regional Office (January 21, 2022), <https://th.boell.org/en/2022/01/21/climate-disasters-philippines>

occurred in November 2013, was the most devastating typhoon in the Philippines in terms of the total value of damages. Damages caused by the typhoon reached about 95.48 billion Philippine pesos. With the international name Haiyan, this was also the costliest typhoon that has ever occurred in the country since 1990.⁸ This has been a great result of all the sudden and unexpected occurrence of disaster, which could have been prevented when everybody was prepared and equipped.

Disaster Legislation of the Philippines: How Effective is it?

With all these disastrous effects of calamities in the country, everybody seemed to be resilient and emotionally prone to every event. Everyone acts as if typhoons have already become part of our lives. But I guess, the word *resiliency* is now overused and should not be taken for granted. Life matters and that should be given the highest priority. Learning from its previous experiences in dealing with disasters, the Philippines has gradually shifted its approach from reactive to proactive risk management with a significant focus on preparedness and resilience building.

Known as the *writ of kalikasan*, or a ‘writ of nature’, this legal remedy that is unique to Philippine law provides for the protection of one’s right to “a balanced and healthy ecology in accordance with the rhythm and harmony of nature”, and is accorded as a constitutional right to the people of the Philippines. With this, the Philippine government has already implemented various legislation in combating and preventing harmful effects of disasters, particularly the program in *Disaster Risk and Reduction Management*. The bottom line is, how effective it is and what are the loopholes?

The Philippine Disaster Reduction and Management Act (R.A. No. 10121) or an act strengthening the Philippine disaster risk reduction and management system, providing for the national disaster risk reduction and management framework and institutionalizing the national disaster risk reduction and management plan, appropriating funds therefor and for other purposes,⁹ is an act of 2010 that was approved by previous president Gloria Macapagal-Arroyo.

⁸ Statista, Most disastrous typhoons Philippines 2011-2022, by value of damages, (January 4, 2024), <https://www.statista.com/statistics/1278568/philippines-most-devastating-typhoon-by-value-of-damages/#:~:text=Most%20disastrous%20typhoons%20Philippines%202011%2D2022%2C%20by%20value%20of%20damages&text=Typhoon%20Yolanda%2C%20w high%20occurred%20in,about%2095.48%20billion%20Philippine%20pesos>

⁹ Philippine Disaster Reduction and Management Act, Rep. Act No. 10121, (May 27, 2010) (Phil.), <https://www.officialgazette.gov.ph/2010/05/27/republic-act-no-10121/>

Its purpose primarily focused on implementing an efficient response regarding risks and disasters. It aims to give justice to the people's constitutional rights to life and property by developing disaster risk and management plans. The core values to which it adheres are good governance such as transparency and accountability within the context of environmental protection. It also provides assistance to those individuals and families affected by the disasters with the objective of normalizing and resuming social and economic activities.

Organization at the local government level is also discussed, which is the composition of the Local Disaster Risk Reduction and Management Council (LDRRMC) from the local executives to the representatives. Along with this are the offices that should be established at every province, city, municipality, and the barangay. The law also includes the mandate of integrating of Disaster Risk Reduction Education into the school curricula and Sangguniang Kabataan, which encourages community specifically the youth in participating into disaster, risk and management activities. This is where individual participation should come in, in order to successfully achieve this business. The Act shifted the policy environment and the way the country deals with disasters from mere response to preparedness. It provides a comprehensive, all-hazard, multi-sectoral, inter-agency, and community-based approach to disaster risk management through the formulation of the National Disaster Risk Management Framework. This is being formulated, developed, and implemented as the master plan that will provide the strategies, organization, tasks of concerned agencies and local government units, and other guidelines in dealing with disasters or emergencies. Through this plan, coherent, integrated, efficient, and responsive disaster risk management at all levels will hopefully be achieved.

It also recognizes local risk patterns and trends and decentralization of resources and responsibilities and thus the full participation of the Local Government Units (LGUs) and communities in governance. The approach tends to be 'response-oriented' or 'reactive.' This is evidenced by the widespread emphasis on post-disaster relief and short-term preparedness, such as forecasting and evacuation, rather than on mitigation and post-disaster support for economic recovery. Moreover, the Act mandates the establishment of a Disaster Risk Reduction and Management Office (DRRMO) in every province, city and municipality, and a Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in every barangay. This provides for the calamity fund to be used in support of disaster risk reduction or mitigation, prevention, and preparedness activities for the potential occurrence of disasters and not just for response, relief, and rehabilitation efforts.

Reflected in the United Nations Office for Disaster Risk Reduction (UNDRR), the Disaster Risk Reduction (DRR) report provides a summary of the DRR

progress the Philippines has achieved under all of the priorities.¹⁰ Recently in their 2023 report, it shows that at a global level, there have been significant advances in developing national DRR strategies. The number reported increased from 55 in 2015 to 125 in 2022 (127%). Training and capacity-building opportunities to monitor implementation contributed to this progress. At a regional level, major progress has also been witnessed. Regions adopted regional strategies at regional platforms, which played important roles in improving risk understanding and awareness, and accelerating DRR implementation. At a local level, the number almost doubled, from 51 in 2015 to 91 in 2022.¹¹

Challenges and Lapses

Disaster preparedness plans reduce future damages but may lack testing to assess their effectiveness in operation. The local governments were found highly vulnerable to tropical cyclones and floods while vulnerable to earthquakes, droughts, and landslides. They were partially prepared regardless of profile, but the coastal, middle-earning, most populated, having the least number of villages, and middle-sized had higher levels of preparedness. Those highly vulnerable to earthquakes and forest fires were prepared, yet only partially prepared to floods, storm surges, droughts, tropical cyclones, tornadoes, tsunamis and landslides. The diverse attitude of stakeholders, insufficient manpower, and poor database management were the major problems encountered in executing countermeasures. Appointing full-time disaster managers, developing a disaster information management system, and massive information drive, organizing village-based volunteers, integrating disaster management into formal education, and mandatory trainings for officials, preparing for a possible major volcanic eruption, and crafting a comprehensive plan against emerging emergencies may lead to a 360° preparedness.

Community Engagement Should be Enhanced

A person's safety during calamities is mainly dependent on their community's disaster preparedness. No matter how ready they are for all possible catastrophic scenarios, every bit of preparation can go down the drain if their community cannot respond effectively to such events. Community participation has been recognized as the additional element in disaster management necessary to reverse the worldwide

¹⁰ Disaster risk reduction in the Philippines: Status Report 2019, UNDRR.org., <https://www.undrr.org/publication/disaster-risk-reduction-philippines-status-report-2019>

¹¹ STATUS REPORT ON TARGET E 2023, UNDRR.org., <https://www.undrr.org/publication/status-report-target-e-2023>

trend of increasing frequency and loss from disasters, build a culture of safety and disaster resilient communities, and ensure sustainable development for all.

The above progress is encouraging and brings promises. But there are still opportunities for improvement. There is a pressing need for strengthening and empowering community-based disaster management (CBDM) as showcased by good practices in community-based preparedness.

Why disaster risk reduction is everyone's business? It involves life- your life. One must engage in community preparedness initiatives, such as neighborhood emergency response teams or local disaster preparedness programs. Collaborating with neighbors and local authorities allows one owner to share resources, information, and support during emergencies.

A study shows that community engagement is very effective. The early delivery of a warning that affects the alertness of the residents had been confirmed effective by most of the respondents. In a 2020 study on *the Effectiveness of Community-Based Disaster Management in Cabanatuan City*, results showed that the prevention and minimizing the impact of natural disasters made by the barangay were properly prepared and necessary actions were taken properly. In terms of conducting a short-term recovery period there is a sufficient supply of relief goods that helps the residents to recover faster and the rescue team conducts their job without further delay. Thus, effective training is needed to improve the performance of the rescue team and incentives might also be used in improving it.¹²

Also, in the 2018 study on *the Level of Effectiveness of Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in the Selected Municipalities in the 3rd Congressional District of Quezon*, the result of the study evidently confirmed that the BDRRMC are effective in performing their duties/functions and responsibilities in terms of setting direction, development, implementation and coordination, but based on the response of the stakeholders the BDRRMCs appeared to be mostly moderately effective specifically in terms of implementation and coordination. Furthermore, the BDRRMC still encountered different problems and challenges, which include insufficient training and capability building, road access, communication, awareness, politics, and the unavailability of Disaster Risk Reduction and Management supplies and equipment.¹³

¹² Santos, K. E., *Effectiveness of Community-Based Disaster Management in Cabanatuan City*. Research Gate., (2020), https://www.researchgate.net/publication/338856470_Effectiveness_of_Community-Based_Disaster_Management_in_Cabanatuan_City

¹³ Glindo, M. S. A., *Level of effectiveness of Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in the selected municipalities in the 3rd*

Utilizing the findings of these studies, one could propose a program enhancement for the Barangay Risk Reduction and Management Committee which was deemed very effective. Strengthening local institutional capacity for DRR should be the highest priority. Understanding the different capacities and constraints of local governments is critical. Local governments and authorities are often knowledgeable about the local needs and gaps, and their support in the operations, especially in the remote regions, is important to guarantee the actualization of intended aspirations for future resilience and sustainability not only in recovery, but also in preparedness and mitigation. There is a need to develop a harmonized and comprehensive planning process with a clear understanding of the linkages at the local level.

Disaster risk reduction is everyone's business. Every individual is a material particle in pushing through this significant objective towards achieving a healthy environment. Participating in the various advocacies of environmental law will create in every individual this altruistic feeling, which would be very beneficial to its country. Recognizing the constitutional guarantee of protecting and advancing the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature, would lead us all in a most prepared community. Implementing environmental legislations will only remain dormant and ineffective unless all of these loopholes and gaps are fully implemented with proper focus on gray areas. National government must start from the localities; community engagement should be enhanced.

Bibliography

Constitution

1987 Constitution of the Philippines, art II, §16 (Phil.).

Statutes

Philippine Disaster Reduction and Management Act, Rep. Act 10121, (May 27, 2010) (Phil.),
<https://www.officialgazette.gov.ph/2010/05/27/republic-act-no-10121/>

Online Sources

Asian Disaster Reduction Center (ADRC) (2017),
<https://www.adrc.asia/nationinformation.php?NationCode=608&Lang=en>

Baclig, C. E. *PH most disaster-prone country in the world—study* | Inquirer News. INQUIRER.net., (October 7, 2022),
<https://newsinfo.inquirer.net/1676227/ph-most-disaster-prone-country-in-the-world-study>

Disaster risk reduction in the Philippines: Status Report 2019, UNDRR.org.,
<https://www.undrr.org/publication/disaster-risk-reduction-philippines-status-report-2019>

Glindo, M. S. A., *Level of effectiveness of Barangay Disaster Risk Reduction and Management Committee (BDRRMC) in the selected municipalities in the 3rd Congressional District of Quezon: Basis for a proposed program enhancement.*, (November 18, 2018),
<https://ojs.aaresearchindex.com/index.php/AAJMRA/article/view/3398>

Novio, E., *Climate Change and Disasters in the Philippines* | Heinrich Böll Foundation | Southeast Asia Regional Office. Heinrich Böll Foundation | Southeast Asia Regional Office (January 21, 2022),
<https://th.boell.org/en/2022/01/21/climate-disasters-philippines>

Santos, K. E. *Effectiveness of Community-Based Disaster Management in Cabanatuan City.* Research Gate., (2020),
https://www.researchgate.net/publication/338856470_Effectiveness_of_Community-Based_Disaster_Management_in_Cabanatuan_City

Scheunemann, C., *What is disaster risk reduction and why is it important?* Sphera, (June 24, 2024), <https://sphera.com/blog/what-is-disaster-risk-reduction-and-why-is-it-important/>

Statista. Most disastrous typhoons Philippines 2011-2022, by value of damages, (January 4, 2024), <https://www.statista.com/statistics/1278568/philippines-most-devastating-typhoon-by-value-of-damages/#:~:text=Most%20disastrous%20typhoons%20Philippines%202011%2D2022%2C%20by%20value%20of%20damages&text=Typhoon%20Yolanda%2C%20which%20occurred%20in,about%2095.48%20billion%20Philippine%20pesos>

STATUS REPORT ON TARGET E 2023. UNDRR.org, <https://www.undrr.org/publication/status-report-target-e-2023>



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QUARRYING AS AN ENVIRONMENTAL ETHIC CRISIS

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History

Stone extraction for construction materials is the primary use of quarries. The use of quarries dates back thousands of years. The Great Pyramids were constructed by the inhabitants of Kemet, ancient Egypt, using enormous hand-cut limestone and granite stones from surrounding quarries.¹ These blocks weigh many tons each. In marble, granite, and limestone quarries in ancient Rome, the incredibly hard task of cutting stones was frequently assigned to enslaved and imprisoned individuals.

The techniques for removing stone and other commodities from quarries have evolved since the first mines were established in the Egyptian region of Aswan.² Hammers, picks, and chisels made of stone or metals like iron and bronze were used to mine the first quarries. It took a lot more work to quarry material for use in building materials. Stones had to be hand-hauled or transported out of quarries.³ Moreover, pulley systems with ropes and movable wooden sleds and tracks might be used to lift stones.

Definition

A quarry is a place where rocks, sand, or minerals are extracted from the surface of the Earth.⁴ A quarry is a type of mine called an open-pit mine because it is open to Earth's surface. Another type of mine, a subsurface mine, consists of underground tunnels or shafts. Additionally, asphalt and concrete are made with aggregate.⁵ Because of this, concrete and asphalt plants are frequently constructed next to quarries. When building roads, aggregate is combined with an oily material called asphalt. The ancient Romans created concrete, which is a combination of aggregate and sticky stone cement. The extensive network of roads and aqueducts constructed by the Romans, much of which is still in use today, was made possible by concrete and aggregate.

¹ Rutledge, et. al., National Geographic, (October 19, 2023), <https://education.nationalgeographic.org/resource/quarry/>

² *Id.*

³ *Id.*

⁴ *Id.*

⁵ *Id.*

The Constitution

Article II, Section 16 of the Philippine Constitution provides that:

“The State shall protect and advance the right of the people to a balanced and healthful ecology in accord with the rhythm and harmony of nature.”⁶

Policy of the State

Section 2 of Presidential Decree No. 1151 provides that:

“it shall be the responsibility of the Government, in cooperation with concerned private organizations and entities, to use all practicable means, consistent with other essential considerations of national policy, in promoting the general welfare to the end that the Nation may (a) recognize, discharge and fulfill the responsibilities of each generation as trustee and guardian of the environment for succeeding generations, (b) assure the people of a safe, decent, healthful, productive and aesthetic environment, (c) encourage the widest exploitation of the environment without degrading it, or endangering human life, health and safety or creating conditions adverse to agriculture, commerce, and industry, (d) preserve important historic and cultural aspects of the Philippine heritage, (e) attain a rational and orderly balance between population and resource use, and (f) improve the utilization of renewable and non-renewable resources.”⁷

The same Decree likewise provides that:

“It shall be the duty and responsibility of each individual to contribute to the preservation and enhancement of the Philippine environment.”⁸

While the Philippines needs to strengthen its law enforcement capacity, it will also benefit from building on its rich well-crafted environmental policies with a new model of environmental governance focused on well-being and guided by principles of sustainability, ecosystem health, precaution, interconnectedness, and inclusiveness.

An Environmental Concern

For an extended period, environmental issues have been escalating and widespread worldwide, particularly in the Philippines. The Philippines is renowned for both its breathtaking scenery and its wealth of natural resources, which include stunning mountains, unspoiled islands, riverbanks, and other features that offer vital

⁶ CONST. (1987) art. II, §16 (Phil.).

⁷ Philippine Environmental Policy, Pres. Dec. 1151 (June 06, 1977) (Phil),

⁸ *Id.*

ecosystem services to the local population.⁹ It is recognized for having a distinct and abundant biodiversity, where uncommon flora and animals can be found around the entire country. But because of its many non-eco-friendly activities that worsen the environment and cause typhoons, landslides, and other natural disasters, the Philippines is also one of the many countries that regularly experience these natural disasters.¹⁰ It is also affected by climate change. Quarrying is one of the most destructive man-made activities commonly carried out in the Philippines. The process of quarrying involves taking stones or building materials out of the ground and along riverbanks.¹¹

There are different kinds of quarrying: river and mountain quarrying.¹² They both have a significant detrimental impact on the environment. Issues surrounding quarrying have been covered by the media and are still ongoing. Indeed, reports of illicit quarrying operations have surfaced nationwide, operating on Philippine beaches, riverbanks, and mountains in particular regions and rural areas. These operations have resulted in coastal erosion, flooding in coastal communities, and even accidents and fatalities.

Mountain quarrying entails removing highland soil and plants, destroying the value of the removed region. In contrast, river quarrying is the most common type of quarrying in the Philippines, and it extracts sand and gravel. River quarrying converts riverbeds into big and deep holes, lowering the groundwater table and causing drinking water wells atop the river's embankments to dry up. In coastal areas, saltwater water may seep into freshwater bodies. Aside from that, quarrying reduces soil cohesiveness, widens riverbanks, and lowers their height. It also causes uneven riverbed deepening, which can lead to riverbank degradation and environmental damage.¹³

There is a greater potential for pollution, soil erosion, floods, and siltation with these types of quarry operations. In addition, the transportation of these quarrying

⁹ De Burca, J. (March 21, 2024), <https://constructive-voices.com/philippines-biodiversity-and-the-built-environment-2/>

¹⁰ World Bank Group., Towards a Comprehensive Disaster Risk Management System for the Philippines (March 07, 2023), <https://www.worldbank.org/en/country/philippines/brief/towards-a-comprehensive-disaster-risk-management-system-for-the-philippines>

¹¹ The Rise of Environmental Sustainability in the Philippines (April 24, 2023), <https://standard-insights.com/blog/rise-of-environmental-sustainability-in-the-philippines/>

¹² Reyes, M.A., Bad Eggs (September 24, 2023), <https://www.pids.gov.ph/details/news/in-the-news/bad-eggs>

¹³ Das et.al., Riverbank Erosion (2014), <https://www.preventionweb.net/understanding-disaster-risk/terminology/hips/gh0028>

activities and the noise pollution in the quarry locations produce dust.¹⁴ Furthermore, throughout the aggregate extraction and processing phases, engineering activities directly contribute to a portion of the environmental disturbances caused by quarrying. Even though there are some short-term effects of quarrying that are simple to predict and monitor, it is still preferable to remain vigilant and avoid complacency.

Currently, it is undeniable that quarrying plays a vital role in society by creating necessary structures like roads, schools, bridges, and buildings, thus aiding in the overall development of the country. It also creates employment opportunities and plays a significant role in the growth of the contemporary economy.

Nevertheless, it is also noted that mining operations pose a danger to the daily lives of people. The World Health Organization states that substandard environmental conditions are a significant factor in poor health and quality of life, impairing sustainable development, and emphasizing the vital role of the environment in human health. Everyday life may be affected by some of the adverse consequences of quarrying.¹⁵

Air Pollution from Quarrying

Dust produced by quarries is a significant contributor to air pollution, with the impact varying based on factors such as local microclimate conditions, dust concentration in the air, particle size, and chemical composition. For instance, limestone quarries generate alkaline and reactive dust, while coal mines produce acidic dust.¹⁶

Air pollution is not only bothersome as it can affect surfaces and health, especially for those with respiratory issues, but also harms plants by blocking and damaging internal structures, causing abrasion on leaves and cuticles, and potentially affecting long-term survival with chemical effects.¹⁷

¹⁴ Worku, H., Environmental and socioeconomic impacts of cobblestone quarries in Addis Ababa and implication for resource use efficiency, environmental quality, and sustainability of land after-use (March 02, 2018), <https://onlinelibrary.wiley.com/doi/10.1002/tqem.21524>

¹⁵ Health and the Environment (n.d.) World Health Organization, <https://www.who.int/westernpacific/about/how-we-work/programmes/health-and-the-environment>

¹⁶ The Impact of Quarrying: Cause and Effect (January 7, 2024), <https://organiconaturalpaint.co.uk/blogs/the-impact-of-quarrying-cause-and-effect/>

¹⁷ *Id.*

Apart from dust pollution, quarrying also generates carbon emissions from heavy machinery, material transportation, and energy-intensive stone processing. These actions emit greenhouse gasses, which add to the problem of climate change.

Noise Pollution from quarrying

Regrettably, mining includes multiple activities that produce substantial levels of noise. The process begins with preliminary tasks like building roads or railway access, setting up a compound, and constructing mineral processing plants. Afterward comes the stage of revealing the mineral to be taken out, typically accomplished by clearing away the surface soil and other layers with equipment like scrapers hydraulic excavators, and dump trucks. The extraction of the mineral will create significant noise, especially when blasting techniques are utilized. Afterward, the utilization of motorized equipment to move the materials and the potential installation of factories to crush and sort the minerals, both add to the noise pollution in the surroundings.¹⁸

Damage to Biodiversity from quarrying

One major downside of quarrying on the environment is the harm it causes to biodiversity. Biodiversity mainly involves a variety of living organisms, such as fish, insects, invertebrates, reptiles, birds, mammals, plants, fungi, and microorganisms. Preserving biodiversity is crucial because all species are connected, even if not obvious or acknowledged, and our survival relies on the delicate equilibrium in nature.

Quarrying can devastate habitats and the wildlife they sustain. While excavations may not directly eliminate habitats, they can still cause harm by indirectly impacting the environment, leading to changes in groundwater or surface water levels that could result in habitats drying up or becoming flooded. Noise pollution can profoundly impact certain species and hinder their ability to reproduce successfully. However, by carefully planning and managing, it is feasible to reduce the impact on biodiversity. Additionally, quarries can offer a chance to establish new habitats or restore current ones.¹⁹

Additionally, mining in riverbeds mainly impacts the environment, the community, and the wildlife. The destruction of both public and private properties can

¹⁸ The Impact of Quarrying: Cause and Effect (January 7, 2024), <https://organicnaturalpaint.co.uk/blogs/the-impact-of-quarrying-cause-and-effect/>

¹⁹ *Id.*

harm the lives of residents living nearby, particularly in rural areas where people rely heavily on the resources of their surroundings. Additionally, it has the potential to result in fatalities and serious incidents by triggering deadly landslides. Quarrying activities can cause significant harm to the environment, such as the destruction of plants, erosion of soil, siltation of rivers, pollution from dust, and landslides. In 2018, it was reported that quarrying operations in seven regions in the Philippines were temporarily closed because an island in Cebu City close to a limestone quarry was affected by a landslide triggered by typhoon Mangkhut which killed approximately twenty-nine (29) people and some were trapped under the debris. This alone served as a warning that quarry operation is indeed deadly and harmful.²⁰

While not everyone may be aware of this environmental problem, quarrying is widely recognized as a highly hazardous activity that impacts the lives of numerous individuals. This activity leaves significant long-term economic, social, and environmental marks.²¹ The rise in quarrying activities is linked to various social challenges such as health and safety risks, displacement of communities, destruction of cultural sites, and utilization of villages for quarries.

How has the issue arisen?

Quarrying operations in the Philippines have become vital for economic growth and infrastructure development due to increasing demands from development, population growth, and construction needs. However, this activity has also raised concerns about its impact on the environment.²²

This problem is now a major concern and quite worrying as illegal quarrying continues in the Philippines despite the presence and enforcement of environmental laws, due to inadequate supervision by authorities, strict regulations on quarry rehabilitation, and the absence of post-quarrying programs to mitigate its negative

²⁰ Marquez B., Philippines: At least 29 dead in Cebu landslide. Aljazeera. (September 21, 2018), <https://www.aljazeera.com/news/2018/9/21/philippines-at-least-29-dead-in-cebu-landslid.e>

²¹ Lad R.J. and Samant J.S. Environmental And Social Impacts Of Stone Quarrying- A Case Study Of Kolhapur District (March 25, 2014), https://www.academia.edu/6548627/ENVIRONMENTAL_AND_SOCIAL_IMPACTS_OF_STONE_QUARRYING_A_CASE_STUDY_OF_KOLHAPUR_DISTRICT

²² Tamayo, P., Characterizing the Effects of Quarrying Industry in Northern Philippines: A Mixed-methods Study. (2023), https://www.researchgate.net/publication/371342275_Characterizing_the_Effects_of_Quarrying_Industry_in_Northern_Philippines_A_Mixed-methods_Study

effects on the environment.²³ Additionally, many developing countries have neglected to rehabilitate most quarrying sites after the cessation of quarrying activities. Should these issues and the mining activities continue without control, there is no question that this ecological issue will escalate into a more severe threat to the lives of Filipinos.

Political Factor

The government plays various roles in environmental issues. Their responsibility is to safeguard the environment and its resources, ensuring that environmental activities are sustainable and benefit everyone equally. They also work to protect the properties of those living near quarry sites, particularly in indigenous communities. Politics plays a role in shaping environmental laws, and there is a contentious debate about the potential political influence on quarrying activities that have a detrimental impact on the environment. Certain government departments neglected to monitor quarry operations and did not conduct proper reviews of quarry owners' permits, allowing illegal quarrying to persist and thrive. Companies must still obtain permits to legally extract minerals from the city when quarrying. Getting permits approved and signed by the government is necessary, but it often leads to complications.

The main task in mining operations is transporting materials using dump trucks.²⁴ This is a daily operation that drivers repeat through the same routes, making it a tedious and challenging task for them. Dump trucks utilized in quarries require a significant investment, making it crucial to prioritize road maintenance to boost transportation efficiency, decrease maintenance expenses, and extend the lifespan of tires and machinery overall. Road maintenance involves flattening gravel roads, but quarry road conditions change daily due to mining activities. As stone extraction operations continue, new roads are built and existing roads are altered, making current maps outdated.

²³ Department of Environment and Natural Resources (DENR) (July 2, 2021), <https://denr.gov.ph/news-events/denr-enforcement-operatives-swoop-down-on-illegal-quarrying/>

²⁴ Types of Mining and Their Respective Equipment (n.d.), <https://www.cashmanequipment.com/about/the-dirt-blog/common-types-of-mining-equipment-used>

Issues in the Government

In 2024, the Provincial Government of Camarines Norte investigated reports of illicit quarry activities in different locations throughout the province.²⁵ A contractor reportedly engaged in a river control project in Barangay Bakiad and Sta is said to have conducted unauthorized excavation along the shores. Even though authorities carried out a joint inspection, no individuals were arrested at the location. The Inter-Agency Task Force on Environmental Protection has shown dissatisfaction regarding the discovery.²⁶ As stated by the Department of Public Works and Highways (DPWH), it is strictly against the rules to conduct any extraction or quarrying activities within 10 meters of government buildings or one kilometer of a bridge.

In 2018, Iloilo mulls mountain quarrying amid a shortage of sand, and gravel. The use of mountain soil as an alternative to address the shortage of sand and gravel for the construction of various projects implemented by the DPWH in Iloilo is being explored. The Iloilo government considers mining in the mountains due to a lack of sand and gravel. Exploring the use of mountain soil as a substitute for sand and gravel in DPWH projects in Iloilo to combat shortages.²⁷

The Regional Director of DPWH-6 stated that there were approximately PHP11 billion worth of projects allocated for Iloilo province in 2018. The budget is higher than the PHP8 billion allocated for projects last year. He mentioned that the projects in the province for that year required 329,621 cubic meters of sand and 537,722 cubic meters of gravel, including roads, bridges, and schools. He also mentioned that he asked other provincial governors in the area to give priority to contractors when granting quarry permits.²⁸

In the same way, in 2014, DPWH Region 11 called on the Davao City government to prevent illegal quarrying near bridges. DPWH 11 regional director Mariano Alquiza said that the degradation of the bridge's foundation is largely due to illegal quarrying. Mayor Rodrigo R. Duterte of Davao City released Executive Order (EO) No. 40, also known as "An Act Establishing the Task Force Subasa Dakbayan (TF Suba)" to address environmental issues in line with current laws protecting the river. The command indicates that the extensive unauthorized quarry activities along

²⁵ GMA Network., Alleged illegal quarrying in Camarines Norte looked into (November 24, 2023), <https://www.gmanetwork.com/regionaltv/news/99638/alleged-illegal-quarrying-in-camarines-norte-looked-into/story/>

²⁶ *Id.*

²⁷ Panay News, Local Newspaper, Iloilo running out of sand, gravel (November 10, 2016), <https://www.panaynews.net/iloilo-running-out-of-sand-gravel/>

²⁸ Ferrer, C., Iloilo mulls mountain quarrying amid shortage of sand, gravel, Philippine News Agency (July 12, 2018), <https://www.pna.gov.ph/articles/1041295>

the primary river in the urban area have been observed and are expected to pose a threat not just to the people living nearby but also to the safeguarding and maintenance of the rivers.²⁹

Why is quarrying even needed?

Quarries, also referred to as open-cast mines,³⁰ are locations where raw mineral material is removed. This is crucial for numerous secondary products and industries, which are essential for achieving a clean, sustainable net-zero society. All things surrounding us are composed of either organic (biologically derived) or inorganic (mineral-derived) materials. The majority of products consist of a combination of both.

The demand for specific metallic minerals like lithium, selenium, and indium is increasing rapidly due to the production of EVs, wind turbines, and solar panels.³¹ Mined minerals are used to make high-performance glazing glass, insulating mineral wool, and other eco-friendly building materials, along with natural fertilizers that substitute petrochemical-derived ammonium nitrates.

Quarry Location

Shortening the distance from the quarry to the customer can decrease the traffic and emissions linked to transportation. Yet, when the site is close to a village or another populated area, inhabitants may be concerned about the potential increase in traffic, noise, and dust from a new quarry. If not controlled, wildlife can be greatly affected by silt runoff, reduced water table levels, and habitat loss.

On a worldwide scale, when minerals are imported from countries with lower costs, it often means transferring problems to countries with less regulation and enforcement of environmental, social, and ethical protections. In certain nations, this has affected not just animals, but also led to the relocation of individuals. Irregular and illegal mining.³²

²⁹ Edge Davao., DPWH asks city gov't to keep quarrying away from bridges (October 2, 2014), <https://edgedavao.net/the-big-news/2014/10/dpwh-asks-city-govt-to-keep-quarrying-away-from-bridges/>

³⁰ Atlantic Pumps, Quarrying & Mining – Good or Bad for the Environment? (January 11, 2024), <https://atlanticpumps.co.uk/blogs/news/quarrying-mining-good-or-bad-for-the-environment>

³¹ *Id.*

³² Atlantic Pumps, Quarrying & Mining – Good or Bad for the Environment? (January 11, 2024), <https://atlanticpumps.co.uk/blogs/news/quarrying-mining-good-or-bad-for-the-environment>

Government Approach

Recognizing the significance of our waterways, DPWH released Department Order No. 139, Series of 2014³³ to establish regulations for river dredging in flood-control projects and quarrying activities. To prevent harm to the environment caused by quarrying, project sites must undergo a thorough soil investigation, which involves conducting on-site bathymetric, topographic, and hydrographic surveys in line with the department's Design Guidelines Criteria and Standards.

A thorough investigation and analysis of the soil must be conducted to avoid alterations in the morphology of the stream. It is necessary to obtain the appropriate environmental permits, such as the Certificate of Non-Coverage (CNC) or Environmental Compliance Certificate (ECC).³⁴ An exhaustive plan and design report for dredging must also be presented to the DPWH offices as outlined by DENR's delegated authorities. Moreover, before the approval of projects, they must adhere to the current laws and regulations of the Water Code of the Philippines.³⁵ and the Philippine Mining Act of 1995,³⁶ among other requirements.

The DPWH Project Management Office (PMO), along with all regional and district offices, must follow instructions to protect waterways' morphology during flood-control projects, dredging, and quarrying operations nationwide.

Local officials are advised to follow DENR and DPWH recommendations before authorizing quarrying activities near rivers and water bodies.³⁷

Philippine Jurisprudence

In the case of the *Province of Pampanga v. Executive Secretary Romulo*,³⁸ Associate Justice Lazaro-Javier, in her concurring opinion, said that quarry operations

³³ DPWH Issues Guidelines for River Dredging Operations, (December 23, 2014), <https://www.dpwh.gov.ph/dpwh/news/1135>

³⁴ DPWH Issues Guidelines for River Dredging Operations, (December 23, 2014), <https://www.dpwh.gov.ph/dpwh/news/1135>

³⁵ A Decree Instituting a Water Code, Thereby Revising and Consolidating the Laws Governing the Ownership, Appropriation, Utilization, Exploitation, Development, Conservation and Protection of Water Resources, Pres. Dec. 1067, s. 1976

³⁶ An Act Instituting a New System of Mineral Resources Exploration, Development, Utilization, and Conservation, Rep. Act 7942, (March 3, 1995) (Phil.).

³⁷ DPWH Issues Guidelines for River Dredging Operations (December 23, 2014), <https://www.dpwh.gov.ph/dpwh/news/1135>

³⁸ *Province of Pampanga v. Executive Secretary Romulo*, G.R. No. 195987, (January 12, 2021) (Phil.).

do not involve massive undertakings. Quarry operations may likewise be done utilizing small-scale equipment and methods. Quarrying and small-scale mining are not mutually exclusive terms. Small-scale mining may include quarry operations, though quarry operations cannot extend to small-scale mining.³⁹ The description of quarrying as the extraction by light blasting and barring down with steel bars and crushing by manual labor and crushers is attuned with R.A. No. 7076's definition of small-scale mining, i.e., heavy reliance on manual labor using simple implements and methods, without using explosives or heavy mining equipment.

Also, the Supreme Court has issued a writ of kalikasan against the Department of Environment and Natural Resources (DENR), Mines and Geosciences Bureau (MGB), Ipilan Nickel Corporation (INC), and Celestial Nickel Mining and Exploration Corporation (Celestial Mining) ordering them to file a verified return of the writ within a non-extendible period of 10 days from receipt of the order.⁴⁰ On May 30, 2022, the DENR Community Environment and Natural Resources Office issued a cease-and-desist order⁴¹ directing INC to halt its port construction for the lack of a valid Miscellaneous Lease Agreement approved by the DENR MIMAROPA Regional Office. This was followed by a resolution by the Sangguniang Bayan of Brooke's Point requesting the MGB to investigate the mining areas claimed by INC and recall issued permits pending investigation. However, no action was taken by the MGB on the request. This prompted the Sangguniang Bayan to issue another resolution urging President Ferdinand R. Marcos, Jr., through the DENR Secretary, to investigate INC's operation despite the absence of the following crucial government permits: (1) clearance from the Protected Area Management Boards; and (2) Certificate of Precondition from NCIP, as well as the extension of the MPSA without the endorsement of the various local government units of Brooke's Point.

The Court also applied the precautionary principle, as laid down in the 2021 case of *PTK2 H2O Corp. v. Court of Appeals*.⁴² Under the principle, a project proponent is required to provide evidence to dispel concerns regarding the potentially harmful impact of a project on the environment, shifting the burden of evidence of harm away from those likely to suffer harm and onto those desiring to change the status quo.

³⁹ *League of Provinces of the Philippines v. Department of Environment and Natural Resources*, 709 Phil. 189, 229-230 (2013).

⁴⁰ *Supreme Court Issues Writ of Kalikasan Against DENR and Mining Operators in Mt. Mantalingahan, Palawan*, Supreme Court of the Philippines, (August 16, 2023), <https://sc.judiciary.gov.ph/supreme-court-issues-writ-of-kalikasan-against-denr-and-mining-operators-in-mt-mantalingahan-palawan/>

⁴¹ Pulta, B., SC issues writ of kalikasan v. mining ops in Palawan, (August 16, 2023), <https://www.pna.gov.ph/articles/1207906>

⁴² *PTK2 H2O Corporation v. Court of Appeals*, 915 Phil. Rep. 119 - 219 (November 16, 2021).

In the present case, the Court found that there is a possibility of serious and irreversible harm to the environment and the inhabitants of Brooke's Point located in the Mt. Mantalingahan Mountain Range. Thus, with the Court's issuance of a writ of kalikasan, the project proponents, INC and Celestial, are required to provide evidence to dispel concerns regarding the potentially harmful impact of a project on the environment.

Who Benefits from Quarrying?

The benefits of quarrying include an increase in the employment rate of the country. According to the Philippine Statistics Authority (PSA), in 2016, with a total of 130 mining and quarrying establishments, there were 28,992 who were employed in this sector. 11.1% of these people or 3,190 of them are employed in the quarrying of stone, sand, and clay. The city also benefits from quarrying. In Davao City, quarrying taxes contribute to the city's annual income by 10%. However, there are those establishments that quarry illegally, leaving the city with a loss of 50% of their potential income. Aside from that, erring government officials who are in great power and the business sectors can benefit from this situation because they can manipulate and control the resources and quarry operations.

Conclusion

It is a significant responsibility to plan and implement a new quarry or mine, as it involves balancing development and preservation needs. Decisions regarding planning and actions taken by quarry operators have the potential to either harm or improve local habitats, the environment, climate regulations, affordable living standards, and employment opportunities in the area.

Engaging in responsible mining and quarrying locally reduces the necessity of transporting important mineral resources worldwide, contributes to a net gain in biodiversity and ecosystems, and benefits the regional economy. Continuous research and innovation are still being conducted to discover novel methods to address current needs while also safeguarding the earth's resources for future generations.

River extraction presents potential harmful effects on the environment. Using heavy machinery like payloaders, backhoes, and dump trucks to push sand and gravel aggregates at the river banks and bed disrupts the soil's natural cohesion. Consequently, the riverbank is ruined and the river's mouth expands. In wet seasons, when there is a high volume of water flowing from upstream, flooding occurs. This leads to the accumulation of sand and gravel at the surface level of vegetation, causing the gradual erosion of topsoil and making it unsuitable for plants. This results in neighboring lands becoming infertile.

Recommendations

A more extensive and collaborative information dissemination effort, with the involvement of academia, Local Government Organizations (LGOs), and Non-Government Organizations (NGOs), is required to raise residents' awareness about the environmental impacts of quarrying and other activities that harm nature.

Laws should be created and enforced to require quarrying contractors to make compulsory contributions to fund environmental projects aimed at restoring the environment damaged by quarrying.

Bibliography

Constitution

1987 Constitution of the Philippines art. II, §16 (Phil.).

Statutes

A Decree Instituting a Water Code, Thereby Revising and Consolidating the Laws Governing the Ownership, Appropriation, Utilization, Exploitation, Development, Conservation and Protection of Water Resources, Pres. Dec. 1067, s. 1976.

An Act Instituting a New System of Mineral Resources Exploration, Development, Utilization, and Conservation, Rep. Act 7942 (March 3, 1995) (Phil.).

Philippine Environmental Policy, Pres. Dec. 1151 (June 06, 1977) (Phil.).

Jurisprudence

League of Provinces of the Philippines v. Department of Environment and Natural Resources, 709 Phil. 189, 229-230 (2013) (Phil.).

PTK2 H2O Corporation v. Court of Appeals, 915 Phil. Rep. 119 - 219 (November 16, 2021) (Phil.).

Province of Pampanga, v. Executive Secretary Romulo. G.R. No. 195987, (January 12, 2021) (Phil.).

Online Sources

Atlantic Pumps., Quarrying & Mining – Good or Bad for the Environment? (January 11, 2024), <https://atlanticpumps.co.uk/blogs/news/quarrying-mining-good-or-bad-for-the-environment>)

Das et.al., Riverbank Erosion (2014), <https://www.preventionweb.net/understanding-disaster-risk/terminology/hips/gh0028>

De Burca, J. (March 21, 2024), <https://constructive-voices.com/philippines-biodiversity-and-the-built-environment-2/>

DPWH Issues Guidelines for River Dredging Operations, (December 23, 2014),
<https://www.dpwh.gov.ph/dpwh/news/1135>

Edge Davao., DPWH asks city gov't to keep quarrying away from bridges (October 2, 2014),
<https://edgedavao.net/the-big-news/2014/10/dpwh-asks-city-govt-to-keep-quarrying-away-from-brId.ges/>

Ferrer, C. Iloilo mulls mountain quarrying amid shortage of sand, gravel (July 12, 2018). Philippine News Agency,
<https://www.pna.gov.ph/articles/1041295>

GMA Network., Alleged illegal quarrying in Camarines Norte looked into (November 24, 2023),
<https://www.gmanetwork.com/regionaltv/news/99638/alleged-illegal-quarrying-in-camarines-norte-looked-into/story/>

Health and the Environment (n.d.) World Health Organization,
<https://www.who.int/westernpacific/about/how-we-work/programmes/health-and-the-environment>

Lad R.J. and Samant J.S. Environmental And Social Impacts Of Stone Quarrying-A Case Study Of Kolhapur District (March 25, 2014),
https://www.academia.edu/6548627/ENVIRONMENTAL_AND_SOCIAL_IMPACTS_OF_STONE_QUARRYING_A_CASE_STUDY_OF_KOLHAPUR_DISTRICT

Marquez B., Philippines: At least 29 dead in Cebu landslide. Aljazeera. (September 21, 2018),
<https://www.aljazeera.com/news/2018/9/21/philippines-at-least-29-dead-in-cebu-landslide>

Panay News, Local Newspaper. Iloilo running out of sand, gravel (November 10, 2016),
<https://www.panaynews.net/iloilo-running-out-of-sand-gravel/>

Pulta, B., SC issues writ of kalikasan v. mining ops in Palawan, (August 16, 2023),
<https://www.pna.gov.ph/articles/1207906>

Reyes, M.A., Bad Eggs (September 24, 2023),
<https://www.pids.gov.ph/details/news/in-the-news/bad-eggs>

-
- Rutledge, et. al. National Geographic, (October 19, 2023),
<https://education.nationalgeographic.org/resource/quarry/>
- Tamayo, P., Characterizing the Effects of Quarrying Industry in Northern Philippines: A Mixed-methods Study. (2023),
https://www.researchgate.net/publication/371342275_Characterizing_the_Effects_of_Quarrying_Industry_in_Northern_Philippines_A_Mixed-methods_Study
- The Impact of Quarrying: Cause and Effect (January 7, 2024),
<https://organicnaturalpaint.co.uk/blogs/the-impact-of-quarrying-cause-and-effect/>
- The Rise of Environmental Sustainability in the Philippines. (April 24, 2023),
<https://standard-insights.com/blog/rise-of-environmental-sustainability-in-the-philippines/>
- Types of Mining and Their Respective Equipment (n.d.),
<https://www.cashmanequipment.com/about/the-dirt-blog/common-types-of-mining-equipment-used>
- Worku, H., Environmental and socioeconomic impacts of cobblestone quarries in Addis Ababa and implication for resource use efficiency, environmental quality, and sustainability of land after-use (March 02, 2018),
<https://onlinelibrary.wiley.com/doi/10.1002/tqem.21524>
- World Bank Group., Towards a Comprehensive Disaster Risk Management System for the Philippines (March 07, 2023),
<https://www.worldbank.org/en/country/philippines/brief/towards-a-comprehensive-disaster-risk-management-system-for-the-philippines>
- Supreme Court Issues Writ of Kalikasan Against DENR and Mining Operators in Mt. Mantalingahan, Palawan. Supreme Court of the Philippines, (August 16, 2023), <https://sc.judiciary.gov.ph/supreme-court-issues-writ-of-kalikasan-against-denr-and-mining-operators-in-mt-mantalingahan-palawan/>



Image Source: Canva for Education



Image Source: Canva for Education

NAVIGATING CHOPPY SEAS OF THE CONSTRUCTION INDUSTRY: DISSECTING THE CONTRACTOR-CLIENT RELATIONSHIP IN PRIVATE CONSTRUCTION CONTRACT DISPUTES

ATTY. MARY CHRISTINE S.C. FLORETE

Much has been told about the horrors of the proverbial handshake between two gentlemen who agree to bind themselves to perform a mutual obligation with one another without written proof. Decades-old friendship degenerate into internecine courtroom battles. Courtroom battles are nothing but a zero-sum game of documentation, and a contest of which side proves to be the skillful rapier of the rules of evidence.

With our legal systems overwhelmed, witnesses die, their recollections may wither or even evidence may disintegrate through the passage of time. Though one may prove the existence of a document through other means such as a recital in a public instrument or testimony of a witness,¹ this may not guarantee that the court may give this enough probative value so that the onus proving one's case may be enough to establish a cause of action. This may be true in construction contracts.

The lack of specificity of obligations of the respective parties reduced into writing is the cause why there are 88 construction industry-related arbitrations pending in the various institutions in 2021 alone involving megastructures.² In 2020, the global average value in dispute has risen to USD 54.26 million for every case.³ This a cogent reason why standards must be adopted and respective obligations must be set in stone to avoid being embroiled in a bitter legal dispute whether in court, or in an arbitral tribunal.

Just last year, I have attended a webinar sponsored by the Construction Industry Authority of the Philippines (“**CIAP**”), an attached agency of the Department of Trade and Industry which serves as the central authority in the Philippine construction industry policy, and program formulation, development and

¹ §5, Rule 130, 2019 Proposed Amendments to the Revised Rules on Evidence, <https://sc.judiciary.gov.ph/2019-amendments-to-the-1989-revised-rules-on-evidence/>

² Jus Mundi, Construction Arbitration Report 2021, available at <https://dailyjus.com/wp-content/uploads/2021/10/Construction-Arbitration-Report-20-Oct-2021.pdf>

³ Arcadis, 2021 Global Construction Disputes Report (2021), <https://www.arcadis.com/en/knowledge-hub/perspectives/global/global-construction-disputes-report>

implementation.⁴ The webinar was organized by the Philippine Institute of Construction Arbitrators and Mediators (“**PICAM**”). The PICAM is an organization composed of multidisciplinary professionals from law, architecture and engineering trained in Alternative Dispute Resolution. The beginning of the whole segment focused on construction contracts discussed by the late Atty. Roberto N. Dio (“**Atty. Dio**”) prior to his passing last 2023.

One of the segments discussed the purpose and advantages of adopting CIAP Document 102 in the whole scheme of the construction process. Otherwise known as the Uniform General Conditions of Contract for Private Construction, CIAP Document 102 was developed in May 1990 through the collaborative efforts of Here, Dio highlights the two purposes of the document, namely: (1) it establishes a fair contract that balances risk sharing between the client and contractor, and (2) encourages the balance of cost allocation of the project and the associated risks that come along with it.⁵ This will be discussed in detail further down below.

It can be recalled that even the Civil Code points out that contracts are obligatory in any form provided that all the essential requisites are present.⁶ This freedom however is only limited by the law’s requirement that a contract should manifest in a specific form such as the prerequisite that it must be written, be placed in a public instrument, or that for a real contract to be perfected, delivery must be shown.⁷ This disquisition will attempt to discuss the formation of the contract in all its possible permutations but emphasizes the superiority of the contract in its written form focusing primarily on the advantages of the CIAP Document 102. I will also discuss in passing certain issues in construction arbitration to provide more context into the succeeding discussions.

One must take into consideration that a construction is not a contract of sale according to Atty. Dio. Rather, it is a contract for a piece of work. This is a contract where the contractor binds himself to produce a project specifically in accordance with the particular specifications of his client. In other words, a contract for a piece of work contemplates a made-to-order contract of service.

Often, the real culprit to blame for why some contractors abandon projects or run away with their client’s money or clients get away with not paying their contractors is a lack of a legally enforceable document that contains all the necessary protections under the law.

⁴ Construction Industry Authority of the Philippines [CIAP], Mission/Vision, <http://construction.gov.ph/about-ciap/mission-vision/>

⁵ R.N. Dio, Know Your Contracts – Clark 2023, CIAP-PDCB Lecture (Mar. 30, 2023).

⁶ Civil Code, § 1356, Rep. Act 386, as amended (Phil.).

⁷ Civil Code, § 1357, 1358, 1316, Rep. Act 386, as amended (Phil.).

There are individuals who, for their ignorance, are simply ill-advised by friends and family who know nothing about the law. And the hapless project owner or contractor ends up doing themselves a greater harm than they had imagined due to underestimating the grave consequences of the law.

As construction contracts are contracts for a piece of work, the following principles govern the interpretation of these contracts.⁸

First: The contractor shall execute the work with the qualities agreed upon and has no defects that destroy or lessen its value or fitness for its ordinary or stipulated use in accordance with Article 1715 of the Civil Code, which provides:

Article 1715. The contract shall execute the work in such a manner that it has the qualities agreed upon and has no defects which destroy or lessen its value or fitness for its ordinary or stipulated use. Should the work be not of such quality, the employer may require that the contractor remove the defect or execute another work. If the contract fails or refuses to comply with this obligation, the employer may have the defect removed or another work executed, at the contractor's cost. (n)

Second: If the work is not of such quality, the employer may require the contract to remove the defect or execute another work pursuant to Article 1715 of the Civil Code as well. This may be harmonized with Article 1167 which states:

Article 1167. If a person obliged to do something fails to do it, the same shall be executed at his cost.

Third: If the contractor fails or refuses to comply, the employer may have the defect removed or another work executed at the contractor's cost also based on the same provisions discussed above.

Fourth: if the parties agreed that the work shall be accomplished to the satisfaction of the proprietor (project owner), in case of disagreement, the question shall be subject to expert judgment. One may refer to the first paragraph of Article 1730 which shows:

⁸ R.N. Dio, Know Your Contracts – Clark 2023, CIAP-PDCB Lecture (Mar. 30, 2023).

Article 1730. If it is agreed that the work shall be accomplished to the satisfaction of the proprietor, it is understood that in case of disagreement the question shall be subject to expert judgment.

xxx

The decision of the third-party neutral shall be final except in case of fraud or manifest error as provided in the second paragraph of Article 1730, as placed below:

xxx

If the work is subject to the approval of a third person, his decision shall be final, except in case of fraud or manifest error. (1598a)

The expert judgment is likened to expert determination, a mode of Alternative Dispute Resolution. In expert determination, parties mutually seek an expert's evaluation of emerging problems and possible solutions without using the forms of judicial process and without either side's defining an entrenched position.⁹ Expert determinations create binding decisions on technical rather than legal issues, and experts have the power to ask questions of the parties before rendering a decision. This mode of dispute resolution is common in intellectual property and construction disputes where estimates of valuers, scientists and engineers are critical.

Construction contracts come in three forms, namely: (1) oral or informal, (2) short form, and (3) long form. The most common contracts chosen by most consumers would be oral or informal. This is the easiest to form. The Civil Code provides that contracts are obligatory in whatever form they may have been entered into, provided all the essential requisites for their validity are present. However, when the law requires that a contract be in some form in order it may be valid or enforceable, or that a contract be provided in a certain way, that requirement is absolute and indispensable.¹⁰ Evidence of consent or acceptance may be express or implied.¹¹

⁹ Black's Law Dictionary.

¹⁰ Civil Code, § 1356, Rep. Act 386, as amended (Phil.).

¹¹ Civil Code, § 1320, Rep. Act 386, as amended (Phil.).

Based on requirements of the Civil Code, the oral form is the easiest to form. The, is the hardest to enforce. The Statute of Frauds found in Article 1403 of the Civil Code does not apply to construction contracts and to contracts that are performed beforehand.

Which is why it is essential to incorporate the bids, quotations, plans and specifications into the contract itself. It is also to be borne in mind that laws are deemed part of the contract pursuant to Article 1156 to 1422 of the Civil Code governing Obligations and Contracts. Articles 1713-1731 of the Civil Code also apply as construction contracts are Contracts for a piece of work. Other laws such as the National Building Code, Contractors' License Law and others are deemed incorporated into construction contracts.

Given the disputes that have arisen due to several issues in construction projects, the Construction Industry Authority of the Philippines (“**CIAP**”), an attached agency of the Department of Trade and Industry, promulgated the Uniform General Conditions of Contract for Private Construction or CIAP Document No. 102 on May 1990. The committee responsible for CIAP Document No. 102 is composed of the following: the Philippine Constructors Association, Inc., United Architects of the Philippines, Philippine Institute of Civil Engineers, Confederation of Filipino Consultants, Inc., and the Construction Industry Authority of the Philippines.

The mandate of such document is to recommend and encourage the adoption of equitable and realistic contract conditions for construction and to reflect the usages of the trade and generally-accepted construction industry standards.

On 30 March 2005, the CIAP promulgated CIAP Board Resolution No.1 that prescribes the application of CIAP Document 102. This document serves as the procedure, guidelines and criteria by contracting parties to a construction project. CIAP Document 102 applies particularly to adjudication and settlement of contract disputes in contract implementation where (1) there are no general conditions of contract, or (2) conditions of contract are incomplete, or (3) certain conditions are ambiguous.

Where there are no general conditions, incomplete conditions, and ambiguous provisions, then, CIAP Document No. 102 will fill in the gaps or ambiguities. Nineteen years later, on 18 February 2019, the CIAP created a Technical Working group to review CIAP Document No. 102.

CIAP revised its Document No. 102 for minor formatting, inclusion of gender-free language, and updates as of 30 September 2002. CIAP also added a new feature called Annotations with Case Synopses and Jurisprudence.

The purpose of CIAP Document No. 102 rests on two principles, namely: (1) Fair contract that balances risk sharing, and (2) balancing the cost allocation between the parties.

Fair contracts that balance risk sharing is one of the cornerstones of sound and harmonious business dealings. This allows for apportionment of risks and clearly defines the rights and obligations of the parties involved.¹² The clear determination of the individual rights and obligations of the parties is essential to the construction industry because significant investments are made by both project owners and contractors. Both also take on huge risks when the project fails. Inequitable contractual relationships in the industry have, more often than not, resulted in abandonment of projects, huge losses, and even bankruptcy, hence, posing a threat to the long-term viability of the construction sector.¹³

Second, the balancing of cost allocation is premised on the equitable sharing of risks between the project owner and contractor. This scheme allows both parties to achieve a return for its completed work, and to prevent costs from incurring a budget overrun.¹⁴

*Werr Corporation International v. Highlands Prime, Inc.*¹⁵

This case is about the development of a condominium complex named Tagaytay Midland Complex between developer Highlands Prime, Inc. (“**HPI**”) and contractor Werr Corporation International (“**Werr**”). The contract stipulates that Werr was to complete the project within two hundred eleven (211) days between 22 July 2005 until 19 February 2006.

HPI was bound to pay Werr a total contract price of Two hundred million seventy-one seven hundred ninety-seven million pesos (PhP271,797,900) upon the

¹² R.N. Dio, Know Your Contracts – Clark 2023, CIAP-PDCB Lecture (Mar. 30, 2023).

¹³ Foreword, Construction Industry Authority of the Philippines [CIAP], Revised CIAP Document 102 (2022 Ed.), [http://construction.gov.ph/wp-content/uploads/2023/02/Revised-CIAP-Document-102-2022-Ed.pdf].

¹⁴ Werr Corp. International v. Highlands Prime, Inc., G.R. No. 187543, (28 February 2017) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/62803>

¹⁵ Ibid.

completion of the project inclusive of labor, supply, transport of materials, and taxes. The payment of the total project cost was to be paid in three installments subject to deductions based on liquidated damages due to delays and costs of covering rectification of defects.

Werr commenced with construction upon Werr's payment of the first installment. Werr continued to pay its progress billings subject to deduction of retention money for every installment. However, the project was not completion on the stipulated date which was 19 February 2006. As a result, extensions were granted by HPI to complete the project. HPI, only agreed to an extension until 15 October 2006. Werr sought the assistance of HPI to pay for its obligations under a "Direct Payment Scheme" totalling P24,503,500.08 to which HPI only approved a portion of the total amount. The amount under the Direct Payment Scheme was charged to HPI's accumulated retention money. This amount corresponded to the 93.18% completion of the project.

Unfortunately, despite the extensions, Werr was not able to complete the project. Thus, HPI terminated the contract. Werr sent an extrajudicial demand for it to recover P36,078,652.90 from HPI. In response, the latter said that it owed Werr only P14,834,926.71. Werr did not reply to this declaration from HPI. Werr then commenced arbitration proceedings against HPI before the Construction Industry Arbitration Commission ("CIAC").

Eventually, the CIAC ruled that HPI owed Werr for the balance of the retention money, but at a reduced amount of P10,903,331.30 and for arbitration costs. Werr owed HPI for unrecouped retention and for unpaid cash advances. It denied HPI's prayer for counterclaim. HPI filed for a Petition for Review before the Court of Appeals assailing CIAC's factual findings. The Court of Appeals modified its decision to the extent of what was awarded by way of liquidated advances and arbitration costs.

The Supreme Court ruled that as the standard conditions for contract for private construction adopted is promulgated by the CIAP, CIAP Document No. 102 applies suppletorily to private construction contracts to remedy the conflict in inter internal documents of, or to fill in the omissions in, in the construction agreement.

The Court further ruled that the omission in the Agreement may be supplemented by the provisions of the Civil Code, industry practice and the CIAP Document No. 102. Hence, the industry practice that substantial compliance excuses the contractor from payment of liquidated damages applies to the Agreement.

*Diesel Construction Co., Inc. v. UPSI Property Holdings, Inc.*¹⁶

This dispute centers on the interior architecture works on the 14th and 16th floors of the UPSI Building 3 Meditel/Condotel Project (“**the Project**”). Diesel Construction Co., Inc., the contractor, undertook to complete the project by way of progress billing or payment by installments based on the amount of completion of the project. The contractor posted a performance bond in favor of the project owner UPSI Property Holdings, Inc.

The contract contained comprehensive provisions for contract works and project completion, extensions for completion, change orders, delays, and damages as a result of negative slippages. The parties originally agreed that the project would run for a period of 90 days. However, the parties agreed to a later commencement date which resulted to the end date being moved as well. At the very heart of the dispute was the agreement that the contractor’s obligation to pay liquidated damages in the event of a delay. The contractor likewise requested for an extension and for variation orders.

UPSI disallowed the extension and held that Diesel Construction Co., Inc. was in default. UPSI withheld Diesel’s 10% retention money and refused to pay the balance of the contract price. Diesel filed a complaint with the CIAC compelling UPSI to pay the balance, attorney’s fees and damages. The tribunal found for Diesel Construction Co., Inc. but awarded a lesser amount. The tribunal ordered UPSI to pay a reduced portion of the unpaid fees plus attorney’s fees. The counterclaims against Diesel Construction Co., Inc. were dismissed.

UPSI appealed the final award to the Court of Appeals which made alterations to the amount awarded. The Court of Appeals reduced the liquidated damages awarded in favor of Diesel Construction Co., Inc. The latter elevated the case to the Supreme Court based on the Court of Appeals factual rulings. Here, the Supreme Court ruled that the CIAC enjoys a presumption of regularity because the CIAC appointed tribunal is presumed to possess technical expertise in their line of authority.

¹⁶ Diesel Construction Co., Inc. v. UPSI Property Holdings, Inc, G.R. No. 154885, (24 March 2008) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/44845>

*New Kanlaon Construction, Inc. v. First Global BYO Corporation*¹⁷

This case on the other hand affirms the rulings in Diesel Construction Co., Inc. v. UPSI Property and Werr Corp. International v. Highlands Prime, Inc. The dispute stemmed from the construction of the Fort Palm Spring condominium project. The contractor was bound to accomplish the structural and architectural works for a set date.

Unfortunately, the contractor was not able to turn-over the project on time which lead to a dispute over payment and damages. The tribunal found that while the structural works were completed, the architectural works remained to be finished. As a result, the tribunal awarded both parties damages and made deductions on their respective claims.

When the dispute reached the Supreme Court, it ruled that factual findings made by quasi-judicial bodies such as the CIAC are accorded respect and finality when supported by substantial evidence. Thus, the Court of Appeals, when reviewing the awards rendered by the CIAC does not have unbridled discretion in its power of judicial review. The Court of Appeals must review the evidence on record to determine the correctness of its rulings.

*R.V. Santos Company, Inc. v. Belle Corporation*¹⁸

This involves the construction of the Tagaytay Woodlands Condominium project. In this case, Belle Corporation filed with the CIAC a request for a refund of overpayment to its contractor R.V. Santos Company, Inc. (“RVSCI”). To begin the project, Belle Corporation advanced 50% of the total construction cost to RVSCI. The latter delayed the project over and beyond its stipulated completion date. As a result of this, Belle Corporation request an audit over the project. The audit disclosed that Belle Corporation overpaid RVSCI. The latter unfortunately refused to refund the excess payments made to it by Belle Corporation. This lead to the project owner initiating arbitration proceedings against RVSCI.

RVSCI countered that the audit findings cannot be given weight as it was allegedly unilateral and is not sanctioned by their agreement. Additionally, the acquiescence of Belle Corporation to the final billing cannot be revoked as argued by

¹⁷ New Kanlaon Construction, Inc. v. First Global BYO Corporation, G.R. No. 231467-68, (16 October 2019) (Phil.), <https://legaldex.com/jurisprudence/new-kanlaon-construction-inc-v-first-global-byo-corp>

¹⁸ R.V. Santos Company, Inc. v. Belle Corporation, G.R. No. 159561-62, (3 October 2012) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/55286>

RVSCI and thus asks for damages. Belle Corporation, on the other hand, countered that the progress billing was subject for review and that the audit must be admissible. It insisted on the refund owed to them.

The tribunal finally ruled that the third-party audit was admissible and may be given weight. The provision of their contract provided no prohibition for any third-party audits conducted. Both the Supreme Court and the CIAC held that the progress billing was merely provisional subject to final review and acceptance of the project owner. And lastly, it was held that RVSCI was overpaid and was not therefore entitled to damages.

*Transcept Construction and Management Professionals, Inc. v. Aguilar*¹⁹

Here, Transcept and Teresita Aguilar (“**Aguilar**”) entered into an agreement to construct a vacation house. The dispute focused on billing, quality of work and additional works. Aguilar filed a notice of arbitration with the CIAC which awarded her for unaccomplished works, consultancy and additional works. Transcept appealed the final award to the Court of Appeals which made its own computation and reversed the CIAC Tribunal.

In the Supreme Court, it affirmed the ruling of the Court of Appeals with some modifications that the unfinished works be reduced to P30,076.72 instead of P198,916.02.

*William Golangco Construction Corporation v. Philippine Commercial International Bank*²⁰

William Golangco Construction Corporation (“**WGCC**”) and Philippine Commercial International Bank (“**PCIB**”) agreed to enter into a contract for the construction of PCIB Tower II on 20 October 1989. PCIB eventually accepted the completion of the work for PCIB Tower II through a letter dated 1 June 1992. WGCC submitted a one-year guarantee bond to answer for any defects within that one-year period.

The issues arose when the granite wash-out finish on the external walls began peeling off. PCIB requested WGCC to rectify these defects but the latter refused as it alleged that it was in the position to accomplish new finishing work. PCIB incurred

¹⁹ Transcept Construction and Management Professionals, Inc. v. Aguilar, G.R. No. 117556, (8 December 2010) (Phil.).

²⁰ William Golangco Construction Corporation v. Philippine Commercial International Bank, G.R. No. 142830, (24 March, 2006) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/40703>

expenses to repair these cosmetic defects. WGCC asserted that the defect took place beyond the one-year guarantee period. PCIB thereafter initiated arbitration proceedings before the CIAC. The tribunal eventually, held that WGCC was liable for such defect. The Court of Appeals upheld the tribunal's final award to which WGCC elevated to the Supreme Court. The Supreme Court held that the one-year guarantee period as specified in the contract is valid and enforceable. As the defect surfaced beyond the one-year period, WGCC cannot be held liable. The Court, pointed out the autonomous nature of contracts and the necessity of respecting the contractual stipulations made between the parties.

Final Note

The best way of protecting one's investment is to manage risks and maximizing returns is for the parties to reduce their agreements into writing and sign it. In the several cases mentioned, it bears stressing that the dispute arises when there are ambiguous agreements or unforeseen events parties failed to clearly agree on in writing. The CIAP Document 102 aims to address gaps in stakeholder engagements prior to commencing a construction project.

While the Civil Code will primarily govern the parties' relationships, the law does not take into account modern industrial practices pervading the construction space. In the age of monolithic cityscapes, developments and megastructures, the industry demands a more nuanced guideline to ensure that risks are balanced. With CIAP Documents 102 in place, we see more best practices adapted in contract formulation which allows more efficient and improved decision-making.

The adoption of these standards is based on successful approaches. The standards are a product of documented successful outcomes that are a result of consistent and uniform practices. While past performance does not guarantee future gains, the probability of adapting best practices results in consistency and high-quality results, and leads to less disputes. The evils of construction creep when the parties have a different understanding of the quality of work, variation of work, the determination when work is completed and is satisfactory to the project owner. These concepts must be quantitatively defined as qualitative notions of compliance with the obligations may turn out to be subjective, thus the proverbial demons of uncertainty are raised from the dead. When the project owner rises in arms in protest of the contractor's work, it would stop payment automatically which results in an acrimonious relationship. Here, we navigate further into the realms of lawsuits or in this case, construction arbitration disputes.

Eventually, parties are lost in the rabbit-hole of whodunnit and who must bear the cost of the injury. Specific quantifications, timelines, well-defined trigger events

and specific consequences of non-compliance must be clearly defined in the construction agreements. The agreement must be the ultimate map of the parties in order to see what is the very crux of the matter—that the project owner take possession of his new building or edifice, and that the contractor be fairly paid for work done within the work specifications and plans.

Uncertainty is lessened as the parties navigate the choppy seas of construction agreements and disputes when a contract specifically defines the metes and bounds of their specific undertaking. Clarity and an unequivocal understanding of obligations are the *je ne se quoi* of a vibrant commercial world.

Bibliography

Statutes

Civil Code, § 1320, Rep. Act 386, as amended (Phil.).

Civil Code, § 1356, Rep. Act 386, as amended (Phil.).

Civil Code, § 1357, 1358, 1316, Rep. Act 386, as amended (Phil.).

Jurisprudence

Diesel Construction Co., Inc. v. UPSI Property Holdings, Inc, G.R. No. 154885, (24 March 2008) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/44845>

New Kanlaon Construction, Inc. v. First Global BYO Corporation, G.R. No. 231467-68 (16 October 2019) (Phil.), <https://legaldex.com/jurisprudence/new-kanlaon-construction-inc-v-first-global-byo-corp>

R.V. Santos Company, Inc. v. Belle Corporation, G.R. No. 159561-62, (3 October 2012) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/55286>

Transcept Construction and Management Professionals, Inc. v. Aguilar, G.R. No. 117556, (December 8, 2010) (Phil.).

Werr Corp. International v. Highlands Prime, Inc., G.R. No. 187543, (28 February 2017) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/62803>

William Golangco Construction Corporation v. Philippine Commercial International Bank, G.R. No. 142830, (24 March, 2006) (Phil.), <https://elibrary.judiciary.gov.ph/thebookshelf/showdocs/1/40703>

Online Sources

2019 Proposed Amendments to the Revised Rules on Evidence, <https://sc.judiciary.gov.ph/2019-amendments-to-the-1989-revised-rules-on-evidence/>

Arcadis, 2021 Global Construction Disputes Report (2021), <https://www.arcadis.com/en/knowledge-hub/perspectives/global/global-construction-disputes-report>

Black's Law Dictionary.

Construction Industry Authority of the Philippines [CIAP], Mission/Vision,
<http://construction.gov.ph/about-ciap/mission-vision/>

Foreword, Construction Industry Authority of the Philippines [CIAP], Revised CIAP
Document 102 (2022 Ed.),
<http://construction.gov.ph/wp-content/uploads/2023/02/Revised-CIAP-Document-102-2022-Ed.pdf>

Jus Mundi, Construction Arbitration Report 2021, available at
<https://dailyjus.com/wp-content/uploads/2021/10/Construction-Arbitration-Report-20-Oct-2021.pdf>



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